STOCHASTIC POLYAK STEP-SIZES AND MOMENTUM: CONVERGENCE GUARANTEES AND PRACTICAL PERFORMANCE

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ABSTRACT

Stochastic gradient descent with momentum, also known as Stochastic Heavy Ball method (SHB), is one of the most popular algorithms for solving large-scale stochastic optimization problems in various machine learning tasks. In practical scenarios, tuning the step-size and momentum parameters of the method is a prohibitively expensive and time-consuming process. In this work, inspired by the recent advantages of stochastic Polyak step-size in the performance of stochastic gradient descent (SGD), we propose and explore new Polyak-type variants suitable for the update rule of the SHB method. In particular, using the Iterate Moving Average (IMA) viewpoint of SHB, we propose and analyze three novel step-size selections: $\text{MomSPS}_{\max}$, MomDecSPS, and MomAdaSPS. For $\text{MomSPS}_{\max}$, we provide convergence guarantees for SHB to a neighborhood of the solution for convex and smooth problems (without assuming interpolation). If interpolation is also satisfied, then using $\text{MomSPS}_{\max}$, SHB converges to the true solution at a fast rate matching the deterministic HB. The other two variants, MomDecSPS and MomAdaSPS, are the first adaptive step-size for SHB that guarantee convergence to the exact minimizer - without a priori knowledge of the problem parameters and without assuming interpolation. Our convergence analysis of SHB is tight and obtains the convergence guarantees of stochastic Polyak step-size for SGD as a special case. We supplement our analysis with experiments validating our theory and demonstrating the effectiveness and robustness of our algorithms.

1 INTRODUCTION

We consider the unconstrained finite-sum optimization problem,

$$\min_{x\in\mathbb{R}^d}\left[f(x)=\frac{1}{n}\sum_{i=1}^{n}f_i(x)\right], \tag{1}$$

where each $f_i:\mathbb{R}^d\to\mathbb{R}$ is convex, smooth, and lower bounded by ℓ_i^*. Let X^* be the set of minimizers of (1). We assume that $X^*\neq\emptyset$ and we fix $x^*\in X^*$. This problem is the cornerstone of machine learning tasks, (Hastie et al., 2009), where x corresponds to the model parameters, $f_i(x)$ represents the loss on the training point i, and the aim is to minimize the average loss $f(x)$ across training points.

When n is large, stochastic gradient methods are the preferred methods for solving (1) mainly because of their cheap per iteration cost. One of the most popular stochastic algorithms for solving such large-scale machine learning optimization problems is stochastic gradient descent (SGD) with momentum, (Sutskever et al., 2013), a.k.a. stochastic heavy ball method (SHB) given by:

$$x^{t+1}=x^t-\gamma_t\nabla f_{S_t}(x^t)+\beta_t(x^t-x^{t-1}). \tag{SHB}$$

where $S_t\subseteq[n]$ a random subset of data-points (mini-batch) with cardinality B sampled independently at each iteration t, and $\nabla f_{S_t}(x^t)=\frac{1}{B}\sum_{i\in S_t}\nabla f_i(x^t)$ is the mini-batch gradient. Here $\gamma_t>0$ is the step-size/learning rate at iteration t while $\beta_t\geq 0$ represents the momentum parameter.

随机Polyak步长与动量：收敛保证与实践性能

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摘要

具有动量的随机梯度下降，也称为随机重球法（SHB），是解决各种机器学习任务中大规模随机优化问题最流行的算法之一。在实际场景中，调整该方法的步长和动量参数是一个极其昂贵且耗时过程。在这项工作中，受随机Polyak步长在随机梯度下降（SGD）性能方面近期优势的启发，我们提出并探索了适用于SHB方法更新规则的新型Polyak类型变体。特别是，利用SHB的迭代移动平均（IMA）视角，我们提出了并分析了三种新型步长选择：$\text{MomSPS}_{\max}$、MomDecSPS和MomAdaSPS。对于$\text{MomSPS}_{\max}$，我们为凸和光滑问题（不假设插值）中的SHB到解邻域提供了收敛保证。如果也满足插值条件，那么使用$\text{MomSPS}_{\max}$，SHB将以与确定性HB相同的快速速率收敛到真实解。其他两种变体MomDecSPS和MomAdaSPS是SHB的第一个保证收敛到精确最小值的自适应步长——无需问题参数的先验知识，也不需要假设插值。我们对SHB的收敛分析是紧密的，并将随机Polyak步长的收敛保证作为SGD的一个特例获得。我们用实验补充了我们的分析，验证了我们的理论，并展示了我们算法的有效性和鲁棒性。

1 简介

我们考虑无约束有限和优化问题，

$$\min_{x\in\mathbb{R}^d}\left[f(x)=\frac{1}{n}\sum_{i=1}^{n}f_i(x)\right], \tag{1}$$

其中每个 $f_i:\mathbb{R}^d\to\mathbb{R}$ 是凸的、光滑的，并且下界为 ℓ_i^*。令 X^* 为(1)的最小化者的集合。我们假设 $X^*\neq\emptyset$ 并固定 $x^*\in X^*$。这个问题是机器学习任务的基石，(Hastie等人。，2009)，其中 x 对应于模型参数，$f_i(x)$表示在训练点 i上的损失，并且目标是最小化训练点上的平均损失 $f(x)$ 。

当 n 较大时，随机梯度方法是解决（1）的首选方法，主要是因为它们每次迭代的成本较低。用于解决此类大规模机器学习优化问题的一种最受欢迎的随机算法是具有动量的随机梯度下降（SGD），（Sutskever 等人。， 2013），也称为随机重球法（SHB），其定义为：

$$x^{t+1}=x^t-\gamma_t\nabla f_{S_t}(x^t)+\beta_t(x^t-x^{t-1}). \tag{SHB}$$

其中 $S_t\subseteq[n]$ 每次迭代独立采样一个包含 B 个数据点的随机子集（小批量），而 $\nabla f_{S_t}(x^t)=\frac{1}{B}\sum_{i\in S_t}\nabla f_i(x^t)$ 是小批量梯度。这里 $\gamma_t>0$ 是迭代 t 的步长/学习率，而 $\beta_t\geq 0$ 代表动量参数。

When the momentum parameter $\beta_t = 0, \forall t \geq 0$, then the update rule SHB is equivalent to the well-studied mini-batch SGD, $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t)$, (Robbins and Monro, 1951), which has been efficiently analyzed under different properties of problem (1) and different step-size selections γ_t (Nemirovski and Yudin, 1983; Nemirovski et al., 2009; Hardt et al., 2016; Needell et al., 2016; Nguyen et al., 2018; Gower et al., 2019; 2021). Additionally, when the cardinality of the random subset S_t is $B = n$, then the update rule of SHB is equivalent to the deterministic heavy ball method (HB) proposed by Polyak (1964), as a way to improve the convergence behavior of deterministic Gradient Descent (GD).

There is a rich literature on the convergence of SHB and HB in different scenarios. In Polyak (1964), it was proved that for a specific choice of the step-size γ and the momentum parameter β, the HB method enjoys an accelerated linear convergence when minimizing strongly convex quadratic functions while more recently, Ghadimi et al. (2015) proved a global sublinear convergence guarantee for HB for convex and smooth functions. In the stochastic setting, several works focus on convergence guarantees of SHB under constant step-size and momentum parameters (Ma and Yarats, 2019; Kidambi et al., 2018; Yan et al., 2018; Gitman et al., 2019; Liu et al., 2020). However, in practical scenarios, these choices suffer from a prohibitively expensive and time-consuming hyper-parameter tuning process. This has motivated a large body of research on the development of adaptive SHB - a method that adapts their parameters using information collected during the iterative process. Such analysis is challenging, and the current adaptive versions of SHB either focus on the full batch setting (deterministic) (Barré et al., 2020; Saab Jr et al., 2022; Wang et al., 2023), or assume that an interpolation condition is satisfied (Schaipp et al., 2023b) or focus on moving averaged gradient (a different form of momentum) (Wang et al., 2023).

Previous studies in the fully stochastic (non-interpolated) scenario have predominantly concentrated on analyzing adaptive versions of SGD, with limited attention given to developing adaptive variants for the SHB. In this work, we take inspiration from the recently introduced and highly efficient Polyak-type adaptive step-sizes for SGD and investigate its applicability and extension to SHB.

1.1 Main Contributions

Our main contributions are summarized below.

◇ **Efficient Polyak Step-sizes via IMA viewpoint.** We explain and illustrate by experiment (see Figure 1) why naively using $\text{SPS}_{\max}$ of Loizou et al. (2021) as a step-size γ_t in the update rule of SHB is not robust, leading to divergence even in simple problems. To resolve this issue, we provide an alternative way of selecting Polyak-type step-sizes for SHB via the Iterate Moving Average viewpoint from Sebbouh et al. (2021). Through our approach, we propose three novel adaptive step-size selections, namely $\text{MomSPS}_{\max}$, MomDecSPS, MomAdaSPS. Each of the proposed step-sizes depends on the choice of the momentum parameter β adding further stability to SHB and comes with specific benefits over their constant step-size counterparts or other adaptive variants of SHB.

◇ **$\text{MomSPS}_{\max}$: Convergence of SHB in non-interpolated setting.** Our first step-size selection of SHB is $\text{MomSPS}_{\max}$, which has a similar structure to $\text{SPS}_{\max}$ of Loizou et al. (2021) but includes also $(1 - \beta)$ in its expression. For this choice, we provide convergence guarantees for SHB to a neighborhood of the solution for convex and smooth problems. Our analysis provides the first convergence guarantees of adaptive SHB using Polyak-type step-size. Previous works on Polyak step-size with momentum in the stochastic setting have guarantees either only under the interpolation setting (Schaipp et al., 2023b) or for a moving averaged gradient momentum (Wang et al., 2023). In addition, as a corollary of our main theoretical results, we show that $\text{MomSPS}_{\max}$ under the interpolation setting and in deterministic scenarios (full batch) converges to the true solution at a fast rate matching the deterministic HB.

◇ **Convergence of SHB to exact solution via MomDecSPS and MomAdaSPS.** Inspired by two recent Polyak-type step-size selections for SGD, the DecSPS in Orvieto et al. (2022) and AdaSPS in Jiang and Stich (2023), we propose two new ways for tuning the step-size for SHB. These are MomDecSPS and MomAdaSPS. Our analysis provides the first $O(1/\sqrt{T})$ convergence guarantees to the exact solution in the non-interpolated regime for a Polyak-type adaptive variant of SHB. Our proposed update rules converge for any choice of momentum parameter $\beta \in [0, 1)$, which makes them particularly useful in practical scenarios.

当动量参数 $\beta_t = 0 \forall t \geq 0$ 时，更新规则 SHB 等价于研究已久的基于小批量的随机梯度下降（SGD），$x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t)$（Robbins and Monro, 1951），该算法在不同问题 (1) 的不同性质和不同步长选择下已被高效分析γ_t （Nemirovski and Yudin, 1983; Nemirovski et al., 2009; Hardt et al., 2016; Needell et al., 2016; Nguyen et al., 2018; Gower et al., 2019; 2021）。此外，当随机子集的基数 $S_t = B = n$ 时，SHB 的更新规则等价于 Polyak (1964) 提出的确定性哈密顿方法（HB），作为一种改进确定性梯度下降（GD）收敛性的方法。

关于 SHB 和 HB 在不同场景下的收敛性研究非常丰富。在 Polyak (1964) 中，已经证明对于特定的步长 γ 和动量参数 β 选择，当最小化强凸二次函数时，HB 方法具有加速线性收敛性；而最近，Ghadimi 等人 (2015) 证明了对于凸和光滑函数，HB 方法具有全局次线性收敛保证。在随机设置下，一些工作关注在恒定步长和动量参数下的 SHB 收敛性保证 (Ma and Yarats, 2019; Kidambi et al., 2018; Yan et al., 2018; Gitman et al., 2019; Liu et al., 2020)。然而，在实际场景中，这些选择需要经过非常昂贵且耗时的超参数调整过程。这促使了大量关于自适应 SHB 的研究——一种使用迭代过程中收集的信息来调整其参数的方法。这种分析具有挑战性，当前的 SHB 自适应版本要么关注于完整批次设置（确定性）(Barr´e et al., 2020; Saab Jr et al., 2022; Wang et al., 2023)，要么假设满足插值条件 (Schaipp et al., 2023b)，要么关注于移动平均梯度（一种不同的动量形式）(Wang et al., 2023)。

以往在完全随机（非插值）场景下的研究主要集中于分析随机梯度下降（SGD）的自适应版本，而对开发自适应的 SHB 变体关注有限。在本工作中，我们从最近提出的、高效的 Polyak类型自适应步长方法中获得灵感，并研究其应用于 SHB 的适用性和扩展性。

1.1 主要贡献

我们的主要贡献总结如下 w.

◇ **基于IMA观点的高效Polyak步长**。我们通过实验（见图 1）解释并说明，为何简单地使用 $\max$ Loizou等人((2021)的SPS γ_t 作为 SHB 更新规则中的步长是不稳定的，即使在简单问题中也可能导致发散。为解决此问题，我们通过 Sebbouh等人(Sebbouh等人(2021)提出的迭代移动平均观点，为 SHB提供了一种选择 Polyak类型步长的新方法。通过我们的方法，我们提出了三种新的自适应步长选择方法，即$\text{MomSPS}_{\max}$、MomDecSPS、MomAdaSPS。每种提出的步长都依赖于动量参数 β 的选择，为 SHB 增加了额外的稳定性，并且相对于其常数步长对应物或其他 SHB的自适应变体具有特定的优势。

◇ **$\text{MomSPS}_{\max}$：SHB在非插值设置中的收敛性**。我们首先提出的SHB步长选择是SHB ，它是 $\text{MomSPS}_{\max}$，其结构与$\text{SPS}_{\max}$ 的Loizou等人(2021)类似，但表达式中还包括$(1 - \beta)$ 。对于这个选择，我们为凸和光滑问题中的SHB到解的邻域提供了收敛保证。我们的分析首次为使用 Polyak类型步长的自适应SHB提供了收敛保证。之前关于随机设置中带有动量的Polyak步长的工作，其保证要么仅在插值设置下（Schaipp等人， 2023b）成立，要么针对移动平均梯度动量（Wang等人， 2023）。此外，作为我们主要理论结果的推论，我们证明了$\text{MomSPS}_{\max}$ 在插值设置和确定性场景（完整批次）下以与确定性HB相同的快速速率收敛到真实解。

◇ **SHB通过MomDecSPS和MomAdaSPS收敛到精确解**。 受两种最近为SGD选择的Polyak类型步长算法的启发，Orvieto等人 （2022）中的DecSPS和Jiang和Stich （2023）中的 AdaSPS，我们提出了两种新的 SHB步长调整方法。这些方法是MomDecSPS 和 MomAdaSPS。我们的分析首次为 $O(1/\sqrt{T})$ Polyak类型自适应 SHB的非插值模式下的精确解提供了收敛保证。我们提出的更新规则对于任何动量参数 $\beta \in [0, 1)$都收敛，这使得它们在实际场景中特别有用。

Step-size	Assumptions / Setting	Adaptive	Exact Convergence	Rate
Constant (Liu et al., 2020)	Knowledge of L	✗	✗	$O(\frac{1}{T}+\hat{\sigma}^2)$
IMA (Sebbouh et al., 2021)	Knowledge of L	✗	✓	$O(\frac{1}{T}+\hat{\sigma}^2)$
ALR-SMAG (Wang et al., 2023)	Moving Averaged Gradient	✓	✗	$O(\frac{1}{T}+\sigma^2)$
MomSPS$_{\max}$ (Thm 3.2)	Restriction on β	✓	✗	$O(\frac{1}{T}+\sigma^2)$
MomDecSPS (Thm 3.6)	Bounded Iterates	✓	✓	$O(\frac{1}{\sqrt{T}})$
MomAdaSPS (Thm 3.7)	Bounded Iterates	✓	✓	$O(\frac{1}{T}+\frac{\sigma}{\sqrt{T}})$

Table 1: Summary of the considered step-sizes and the corresponding theoretical results in the stochastic setting. All the rates are given for convex and smooth functions. The quantity of convergence in our rates is $\mathbb{E}[f(\overline{x}^T)-f(x^*)]$, where $\overline{x}^T=\frac{1}{T}\sum_{t=0}^{T}x^t$. Here $\hat{\sigma}^2=\mathbb{E}\|\nabla f_i(x^*)\|^2<\infty$ and $\sigma^2=\mathbb{E}[f_i(x^*)-\ell_i^*]<\infty$. The "Exact Convergence" column refers to convergence to the exact solution x^* with no interpolation assumption.

◇ **Tight Convergence Guarantees.** All of our convergence guarantees are *true* generalizations of the theoretical analysis of SGD using SPS$_{\max}$, DecSPS, and AdaSPS. That is, if $\beta=0$ (no momentum) in the update rule of SHB, our theorems obtain as a special case the best-known convergence rates of Polyak-type step-size for SGD, highlighting the tightness of our analysis. See also Table 1 for a summary of our main complexity results and a comparison with closely related works.

◇ **Further Convergence Results.** As a byproduct of our theoretical analysis, we provide two interesting corollaries: a *novel analysis of constant step-size SHB* (as a corollary of our Theorem on MomSPS$_{\max}$) and the first *robust convergence of SHB* via our theorem on MomAdaSPS. For a constant step-size, the Corollary 3.5 of our Theorem 3.2 allows larger step-sizes than the analysis of SHB in Liu et al. (2020) and provides convergence without assuming the restrictive bounded variance condition (there exist $q>0$ such that $\mathbb{E}\|\nabla f_i(x)-\nabla f(x)\|^2<q$). In addition, via Theorem 3.7 we provide the first robust convergence of adaptive SHB that guarantees convergence to the exact solution and automatically adapts to whether our problem is interpolated or not. That is, if interpolation is assumed, then the rate of SHB with MomAdaSPS is the same as the rate of SHB with MomSPS$_{\max}$ (or constant step-size), and if no interpolation is assumed, then it matches the rate of SHB with MomDecSPS. The analysis achieves the best-known rates in both settings.

◇ **Numerical Evaluation.** In Section 4, we verify our theoretical results via numerical experiments on various problems, demonstrating the effectiveness and practicality of our approach. An open-source implementation of our method is available at https://github.com/dimitris-oik/MomSPS.

2 Exploring the Interplay of SPS and Heavy Ball Momentum

In this section, we present the expression of Stochastic Polyak Step-size (SPS) and its different variants. We illustrate that a naive combination of SPS with momentum is not robust, leading to divergence in simple problems. We explain how we resolve this issue using the Iterate Moving Average viewpoint of SHB and propose three adaptive Polyak-type step-sizes for SHB.

2.1 Background on Stochastic Polyak Step-size

The deterministic Polyak step-size (PS) was first introduced by Polyak (1987) as an efficient step-size selection for GD for solving convex optimization problems. It has the following expression: $\gamma_t=\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2}$ which is obtained by minimizing an upper bound of the quantity $\|x^{t+1}-x^*\|^2$ in the analysis of GD. Since its original proposal, the PS has been successfully used in the analysis of deterministic subgradient methods in different settings with favorable convergence guarantees (Boyd et al., 2003; Davis et al., 2018; Hazan and Kakade, 2019). PS requires the prior knowledge of $f(x^*)$, which might look like a strong assumption. However, as shown in Boyd et al. (2003), this is known in several applications, including finding a point in the intersection of convex sets and positive semi-definite matrix completion.

Inspired by the convergence of PS in the deterministic setting, Loizou et al. (2021) has effectively modified the Polyak step-size for the stochastic setting, achieving convergence rates comparable to those of finetuned SGD. The proposed stochastic Polyak step-size (SPS) has several benefits,

步长	假设 / 设置	自适应	精确收敛率	
常数（刘等人，2020）	对 L 的了解	✗	✗	$O(\frac{1}{T}+\hat{\sigma}^2)$
IMA (Sebbouh等人, 2021)	对 L的了解	✗	✓	$O(\frac{1}{T}+\hat{\sigma}^2)$
ALR-SMAG (Wang等人, 2023)	移动平均梯度	✓	✗	$O(\frac{1}{T}+\sigma^2)$
MomSPS$_{\max}$ (Thm 3.2)	对 β的限制	✓	✗	$O(\frac{1}{T}+\sigma^2)$
MomDecSPS (Thm 3.6)	有界迭代	✓	✓	$O(\frac{1}{\sqrt{T}})$
MomAdaSPS (Thm 3.7)	有界迭代	✓	✓	$O(\frac{1}{T}+\frac{\sigma}{\sqrt{T}})$

表1：考虑的步长算法及其在随机设置中的理论结果摘要。所有速率均针对凸和光滑函数给出。我们的收敛速率是 $\mathbb{E}[f(\overline{x}^T)-f(x^*)]$，，其中 $\overline{x}^T=\frac{1}{T}\sum_{t=0}^{T}x^t$。这里 $\hat{\sigma}^2=\mathbb{E}\|\nabla f_i(x^*)\|^2<\infty$和 $\sigma^2=\mathbb{E}[f_i(x^*)-\ell_i^*]<\infty$。 "精确收敛"一列指的是无插值假设的精确解收敛 x^*。

◇ **严格的收敛保证。** 我们所有的收敛保证都是 对使用*SPS*、*DecSPS*和*AdaSPS*进行*SGD*的理论分析的推广$_{\max}$，即，如果 $\beta=0$(无动量)在 SHB的更新规则中，我们的定理作为Polyak类型步长SGD的最佳已知收敛速度的一个特例得到，突显了我们分析的紧密性。另见表 1，总结我们主要复杂度结果并与密切相关的工作进行比较。

◇进一步收敛结果。作为我们理论分析的副产品，我们提供了两个有趣的推论：常数步长SHB的新分析（作为我们关于MomSPS$_{\max}$的定理的推论）以及我们关于MomAdaSPS的定理通过SHB的第一个鲁棒收敛。对于常数步长，我们定理3.2的推论3.5允许比Liu等人（2020）中SHB的分析更大的步长，并且提供收敛而不需要假设限制性的有界方差条件（存在 $q>0$ 使得 $\mathbb{E}\|\nabla f_i(x)-\nabla f(x)\|^2<q$）。此外，通过定理3.7，我们提供了自适应SHB的第一个鲁棒收敛，保证收敛到精确解并自动适应我们的问题是插值还是不插值。也就是说，如果假设插值，则MomAdaSPS的SHB速度与MomSPS$_{\max}$ (或常数步长)的SHB速度相同，如果不假设插值，则它与MomDecSPS的SHB速度匹配。该分析在两种情况下都实现了最佳已知速度。

◇ **数值评估。** 在第4章，我们通过在各种问题上进行的数值实验验证了我们的理论结果，展示了我们方法的有效性和实用性。我们方法的开放源代码实现可在 https://github.com/ dimitris-oik/MomSPS。

2 探索SPS与重球动量的相互作用

在本节中，我们介绍了随机Polyak步长（SPS）及其不同变体的表达式。我们说明SPS与动量的简单组合并不稳健，会导致简单问题中的发散。我们解释了如何利用SHB 的迭代移动平均观点来解决这个问题，并针对SHB提出了三种自适应Polyak类型步长。

2.1 随机Polyak步长背景

确定性Polyak步长（PS）最早由Polyak（1987）提出，作为用于解决凸优化问题的梯度下降（GD）的有效步长选择。其表达式如下：$\gamma_t=\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2}$，这是通过最小化GD分析中量 $\|x^{t+1}-x^*\|^2$的上界得到的。自最初提出以来，PS已在不同设置中成功应用于确定性次梯度法分析，并具有有利的收敛保证（Boyd等人，2003；Davis等人，2018；Hazan和Kakade，2019）。PS需要预先知道 $f(x^*)$，这看起来可能是一个强假设。然而，正如Boyd等人（2003）所示，这在多种应用中是已知的，包括在凸集交集和正半定矩阵补全中找到点。

受确定性设置中PS收敛的启发，Loizou等人(2021) 已有效修改了随机设置中的Polyak步长，实现了与微调SGD相当的收敛速度。所提出的最优随机Polyak步长（SPS）具有多项优势

including independence on parameters of the problem (e.g., L-smoothness or μ strong convexity) and competitive performance in over-parametrized models. More specifically, Loizou et al. (2021) proposed the $\text{SPS}_{\max}$ given below[1]:

$$\gamma_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}. \tag{SPS$_{\max}$}$$

Here, $\gamma_b > 0$ is a bound that restricts SPS from being very large and is essential to ensure convergence to a small neighborhood around the solution and $c > 0$ is a positive constant that depends on the function class the objective f belongs to.

As mentioned in Orvieto et al. (2022), the $\text{SPS}_{\max}$ comes with strong convergence guarantees and competitive performance; however, it has one main drawback when used in non-over-parameterized regimes: It can guarantee convergence only to a neighborhood of the solution. For this reason, Orvieto et al. (2022) suggests a decreasing variant of the original $\text{SPS}_{\max}$ named DecSPS, given by:

$$\gamma_t = \frac{1}{c_t}\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{\|\nabla f_{S_t}(x^t)\|^2}, c_{t-1}\gamma_{t-1}\right\}, \tag{DecSPS}$$

where c_t is an increasing sequence of positive real numbers and $c_{-1} := c_0$ and $\gamma_{-1} = \gamma_b > 0$. The authors proved that SGD with DecSPS and $c_t = \sqrt{t+1}$, converges with a sublinear rate $O(1/\sqrt{T})$ for convex and smooth functions with bounded iterates (i.e., $D^2 := \max_{t\in[T-1]}\|x^t - x^*\|^2 < \infty$).

More recently, in Jiang and Stich (2023), another decreasing variant of SPS was introduced, named AdaSPS:

$$\gamma_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}\frac{1}{\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}, \tag{AdaSPS}$$

where $\gamma_{-1} = +\infty$ and $c > 0$. For convex and L-smooth functions with bounded iterates, it can be shown that SGD with AdaSPS converges to an exact solution with a rate $O\left(\frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}}\right)$, where $\tau = 2cLD^2 + \frac{1}{c}$. The interesting aspect of the convergence of AdaSPS is that it provides a robust result for SGD, meaning that the method recovers the best bounds for both the interpolated ($\sigma = 0$) and non-interpolated regimes.

2.2 Naive SPS in SHB

All of the above variants of SPS were proposed and analyzed for SGD (SHB with no momentum). With the increased popularity of momentum in machine learning, one can naturally ask the following question: *Is it possible to combine SPS with momentum?* This question was partially answered in Wang et al. (2023) when they proposed a Polyak-type step-size for a moving averaged gradient (MAG) momentum but not for the SHB. Their MAG framework is not equivalent to SHB when their stepsizes are adaptive and while their proposed stepsize reduces to the original $\text{SPS}_{\max}$ stepsize when $\beta = 0$, their guarantees do not reduce to the original $\text{SPS}_{\max}$ guarantees for convex functions.

The most straightforward approach is to directly apply the $\text{SPS}_{\max}$ in the SHB update rule. Let us call this update rule *$\text{SPS}_{\max}$ with naive momentum*. Unfortunately, this approach does not necessarily lead to convergence for natural choices of momentum parameter $\beta \in (0, 1)$, as shown in Figure 1.[2] In this experiment, even for simple convex and smooth problems like a logistic regression with synthetic data, the naive rule fails to converge when β gets larger (a typical choice for momentum parameter is $\beta = 0.9$). This indicates that a more careful step-size selection is needed, which may also depend

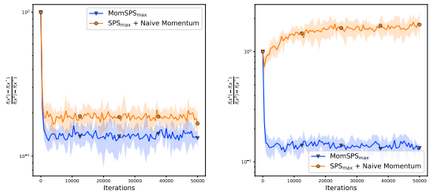


Figure 1: Comparison of $\text{MomSPS}_{\max}$ versus $\text{SPS}_{\max}$ with naive momentum for different momentum parameters β on a logistic regression problem. Left: $\beta = 0.2$, Right: $\beta = 0.5$

[1] Originally Loizou et al. (2021) use $f^*_{S_t}$ instead of the lower bound $\ell^*_{S_t}$. The lower bound is due to Orvieto et al. (2022), which proves that the more relaxed lower bound can still lead to the same convergence guarantees.

[2] See also Figure 9 in Appendix for more values of β.

包括对问题参数的独立性（例如 L-光滑性或 μ 强凸性）以及在过参数化模型中的优异性能。更具体地说，Loizou等人(2021) 提出了如下$\text{SPS}_{\max}$： [1]:

$$\gamma_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}. \tag{SPS$_{\max}$}$$

这里， $\gamma_b > 0$ 是一个限制SPS不会过大的界限，对于确保收敛到解附近的小邻域至关重要，而 $c > 0$ 是一个取决于目标 f 所属函数类的正常数。

如前所述Orvieto等人(2022)，$\text{SPS}_{\max}$ 具有强收敛保证和具有竞争力的性能；然而，在非过参数化条件下使用时，它有一个主要缺点：只能保证收敛到解的邻域。因此，Orvieto等人(2022)提出了一种原始SPS的递减变体$_{\max}$，称为DecSPS，其定义为：

$$\gamma_t = \frac{1}{c_t}\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{\|\nabla f_{S_t}(x^t)\|^2}, c_{t-1}\gamma_{t-1}\right\}, \tag{DecSPS}$$

其中 c_t 是一个正实数的递增序列， $c_{-1}: = c_0$ 和 $\gamma_{-1} = \gamma_b > 0$。作者证明了SGD与DecSPS 和 $c_t = \sqrt{t+1}$，对于具有有界迭代（即 $D^2: = \max_{t\in[T-1]}\|x^t - x^*\|^2 < \infty$）的凸和光滑函数，收敛速度为次线性 $O(1/\sqrt{T})$。

最近，在Jiang和Stich (2023)，又引入了一种递减的SPS变体，命名为AdaSPS：

$$\gamma_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}\frac{1}{\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}, \tag{AdaSPS}$$

其中 $\gamma_{-1} = +\infty$ 和 $c > 0$。对于凸且 L-光滑的函数，当迭代有界时，可以证明使用AdaSPS的SGD以速率 $O\left(\frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}}\right)$收敛到精确解，其中$\tau = 2cLD^2 + \frac{1}{c}$。~~AdaSPS~~收敛的有趣之处在于它为SGD提供了一个稳健的结果，这意味着该方法在插值($\sigma = 0$)和非插值情况下都能恢复最佳界限。

2.2 NAIVE SPS INSHB

上述所有SPS变体都是为SGD（SHB且无动量）提出的，并进行了分析。随着动量在机器学习中的日益普及，人们自然会提出以下问题：是否可以将*SPS*与动量结合？这个问题在Wang等人 (2023)的研究中得到了部分解答，当时他们提出了一种用于移动平均梯度（MAG）动量的Polyak类型步长，但未用于SHB。当他们的步长是自适应时，其MAG框架与SHB 并不等价，而他们提出的步长在特定条件下会简化为原始的

$\text{SPS}_{\max}$ 当步长时 $\beta = 0$，它们的保证不会降低到原始$\text{SPS}_{\max}$ 对凸函数的保证。

最直接的方法是直接应用 $\text{SPS}_{\max}$ 在 SHB 更新规则中。我们称这个更新规则为*$\text{SPS}_{\max}$* 带朴素动量。不幸的是，这种方法对于自然选择的动量参数 $\beta \in (0,1)$，并不一定收敛，如图1所示。[2] 在本实验中，即使是像逻辑回归这样的简单凸和光滑问题（使用合成数据），当 β 增大时（动量参数的典型选择是 $\beta = 0.9$），朴素规则也无法收敛。这表明需要更仔细的步长选择，这可能还取决于

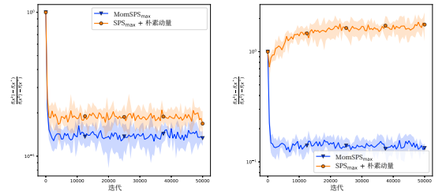


图1：比较$\text{MomSPS}_{\max}$ 与$\text{SPS}_{\max}$结合朴素动量对不同动量参数 β 在逻辑回归问题上的表现 $\beta = 0.2$，右图： $\beta = 0.5$

[1] 最初Loizou等人(2021)使用 $f^*_{S_t}$ 而不是下界 $\ell^*_{S_t}$。下界归因于Orvieto等人(2022)，该研究证明更宽松的下界仍可保证收敛性。[2] 另见图9 附录中的更多 β。

on the momentum parameter. In Figure 1, we compare the SPSmax with naive momentum with one of our proposed and analyzed step-size selection MomSPS$_{\max}$. As seen in Figure 1, when β is small, then both SPS$_{\max}$ with naive momentum and SHB with MomSPS$_{\max}$ have similar behavior. When $\beta = 0$, the two methods have identical performance as both are reduced to the same method: SGD with SPS$_{\max}$. However, as β gets larger, the performance of SPS$_{\max}$ with naive momentum gets worse and less stable, and at some point, it diverges. On the other hand, in this example, SHB with MomSPS$_{\max}$ converges for any of the selected β.

2.3 Iterative Moving Average: Balance between SPS and Momentum

Having explained how naively combining SPS$_{\max}$ with heavy ball momentum can lead to divergence, in this section, we leverage the iterate moving-average (IMA) viewpoint of SHB from Sebbouh et al. (2021) to propose SPS-type adaptations that depend on momentum parameter as well for SHB. As we will see later, this viewpoint provides stability and robustness to the proposed update rules.

Iterate Moving Average (IMA). Sebbouh et al. (2021) provide the IMA as an alternative way of expressing the update rule of SHB and explain how the IMA formulation is crucial in comparing SHB and SGD as it allows to establish connections between the step-sizes of the two methods. The Iterate Moving Average method is given by the following update rule:

$$z^{t+1} = z^t - \eta_t \nabla f_{S_t}(x^t), \quad x^{t+1} = \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1},$$

where $z^0 = x^0$, $\eta_t > 0$ and $\lambda_t \geq 0$. As proved in Sebbouh et al. (2021), if for any $t \in \mathbb{N}$, holds $1 + \lambda_{t+1} = \frac{\lambda_t}{\beta_t}$ and $\eta_t = (1 + \lambda_{t+1})\gamma_t$ then the x_t iterates of the IMA method are equal to the x_t iterates produced by the SHB method. For completeness, we include the proof of this statement in Appendix C.3.

In our proposed methods, we select the most common setting of constant momentum ($\beta_t = \beta$). Considering the equivalence between IMA and SHB under the correct parameter selection, let us present the following proposition that will allow us to obtain convergence guarantees for SHB via a convergence analysis of IMA.

Proposition 2.1. *If $\lambda_t = \lambda \geq 0$ then assuming that $\beta = \frac{\lambda}{1+\lambda}$ and $\gamma_t = (1-\beta)\eta_t$ the x_t iterates of the IMA method are equal to the x_t iterates produced by the SHB method.*

Using the above proposition, let us provide the following corollary that explains the derivation of our proposed step-size selections for SHB.

Corollary 2.2. *Let the step-size η_t in IMA be one of the previously proposed Polyak step-sizes: SPS$_{\max}$, DecSPS or AdaSPS and let $\lambda_t = \lambda$. Then via Proposition 2.1, the SHB has constant momentum parameter $\beta_t = \beta$ and the step-sizes are MomSPS$_{\max}$, MomDecSPS or MomAdaSPS respectively.*

Three New Adaptive Step-sizes for SHB. Using the IMA viewpoint and Proposition 2.1 and Corollary 2.2 let us present three Polyak-type step-size for SHB.

(i) MomSPS$_{\max}$. Let us start with the variant associated with the SPS$_{\max}$. That is, via Corollary 2.2 if we use SPS$_{\max}$ as a step-size for IMA, then this is equivalent to using the following step-size in the update rule of SHB:

$$\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}. \qquad \text{(MomSPS}_{\max}\text{)}$$

Here, the parameter $\gamma_b > 0$ has the same purpose as in the original SPS$_{\max}$, and it is a bound that restricts MomSPS$_{\max}$ from being very large and is essential to ensure convergence to a neighborhood around the solution. It is clear from its expression that MomSPS$_{\max}$ has the same form as SPS$_{\max}$ but multiplied by a *correcting momentum factor* $1-\beta$. This follows from Corollary 2.2. In practice, this small change allows the SHB to be more "stable" for different momentum parameters, as we show in Figure 1. Using MomSPS$_{\max}$ and a restriction on the momentum parameter β in Section 3, we establish $O(1/T)$ convergence for SHB up to a neighborhood of the solution for convex and smooth functions.

动量参数。在图1中，我们比较了带朴素动量的 SPSmax 和我们提出并分析的步长选择 MomSPS$_{\max}$。如图1所示，当 β 较小时，带朴素动量的 SPS 和SHB带MomSPS$_{\max}$ 的行为相似。当 $\beta = 0$时，两种方法的性能相同，因为它们都简化为相同的算法：带 SPS 的 SGD$_{\max}$。然而，当 β 增大时，带朴素动量的 SPS 性能变差且稳定性降低，在某个时刻会发散。另一方面，在本例中，SHB带MomSPS$_{\max}$ 对任何选定的 β都会收敛。

2.3 迭代移动平均：SPS 与动量的平衡

在解释了 SPS$_{\max}$ 与重球动量（HB）的简单组合可能导致发散之后，在本节中，我们利用 SHB从Sebbouh 等人(2021) 提出的迭代移动平均（IMA）观点，提出了依赖于动量参数的 SPS 类型自适应方法，用于SHB。正如我们稍后将看到的，这种观点为所提出的更新规则提供了稳定性和鲁棒性。

迭代移动平均（IMA）。Sebbouh 等人(2021) 将 IMA 作为表达 SHB 更新规则的一种替代方法，并解释了 IMA 公式在比较SHB和随机梯度下降（SGD）时的关键作用，因为它允许在两种方法的步长之间建立连接。迭代移动平均方法由以下更新规则给出：

$$z^{t+1} = z^t - \eta_t \nabla f_{S_t}(x^t), \quad x^{t+1} = \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1},$$

其中 $z^0 = x^0$, $\eta_t > 0$ 和 $\lambda_t \geq 0$。如 Sebbouh 等人(2021) 所证明的，如果对于任何 $t \in \mathbb{N}$, 满足$1 + \lambda_{t+1} = \frac{\lambda_t}{\beta_t}$ 和 $\eta_t = (1 + \lambda_{t+1})\gamma_t$，则 IMA 方法的迭代值等于 x_t SHB方法产生的迭代值。为了完整性，我们在附录C.3中包含了该命题的证明。

在我们的方法中，我们选择最常见的恒定动量设置（$\beta_t = \beta$）。考虑到IMA和SHB 在正确参数选择下的等价性，让我们提出以下命题，它将允许我们通过IMA的收敛性分析来获得SHB的收敛保证。

命题2.1。 如果 $\lambda_t = \lambda \geq 0$，则假设 $\beta = \frac{\lambda}{1+\lambda}$ 和 $\gamma_t = (1-\beta)\eta_t$，那么 x_t *IMA*方法的迭代值等于 x_t 由*SHB* 方法产生的迭代值。

利用上述命题，让我们提供以下推论，它解释了我们为SHB提出的步长选择的推导过程。

推论2.2。 设*IMA*中的步长 η_t 是先前提出的*Polyak*步长之一：*SPS*$_{\max}$、*DecSPS*或*AdaSPS*，且设 $\lambda_t = \lambda$。那么通过命题2.1，*SHB* 具有恒定动量参数 $\beta_t = \beta$，且步长分别为*MomSPS*$_{\max}$、*MomDecSPS* 或*MomAdaSPS*。

SHB的三个新自适应步长。采用IMA视角和命题2.1 以及推论2.2，让我们介绍三个Polyak类型步长SHB。

(i) MomSPS$_{\max}$。 让我们从与SPS相关的变体开始$_{\max}$。也就是说，通过推论2.2，如果我们使用SPS$_{\max}$ 作为IMA的步长，那么这相当于在SHB的更新规则中使用以下步长：

$$\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}. \qquad \text{(MomSPS}_{\max}\text{)}$$

这里，参数 $\gamma_b > 0$ 与原始SPS中的参数具有相同的作用$_{\max}$，它是一个限制MomSPS$_{\max}$ 不要过大并且对于确保收敛到解的邻域至关重要。从其表达式可以看出MomSPS$_{\max}$ 与SPS具有相同的形式$_{\max}$但乘以一个校正动量因子 $1-\beta$。这源于引理2.2。在实践中，这种小的变化使得SHB 对于不同的动量参数更加“稳定”，如图1所示。使用MomSPS$_{\max}$ 和动量参数的限制 β 在第三节中，我们建立了 $O(1/T)$ 对于凸函数和光滑函数的SHB 在解的邻域内的收敛性。

Our following two proposed step-size selections are variants of the two decreasing variants of Polyak step-sizes DecSPS and AdaSPS. As such, we can prove SHB's convergence to the exact solution instead of a neighborhood. More importantly, using MomDecSPS or MomAdaSPS, we can provide convergence guarantees for any choice of the momentum coefficients $\beta \in [0, 1)$ making SHB fully adaptive (no tuning necessary).

(ii) MomDecSPS. Firstly, we propose an adaptation of DecSPS with momentum. We call this step-size selection MomDecSPS, and it is given by:

$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}c_{t-1}}{c_t}\right\}, \tag{MomDecSPS}$$

where $\gamma_{-1} := \gamma_b > 0$ is a step-size bound and $c_t = c\sqrt{t+1}$ for $t \geq 0$ with $c_{-1} := c_0 = c > 0$ be a constant to regulate the step-size. Note that γ_t is indeed decreasing. This step-size is adaptive and, in its definition, does not require knowledge of any function properties (e.g., smoothness/strong convexity constants). In Section 3 we prove that SHB with MomDecSPS and constant momentum $\beta \in [0, 1)$ converges to exact solution with a rate $O(1/\sqrt{T})$.

(iii) MomAdaSPS. Similar to the previous two steps, let us propose the following adaptation of AdaSPS with momentum. We call this MomAdaSPS, and is given by:

$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}. \tag{MomAdaSPS}$$

By definition, this is a decreasing step-size ($\gamma_t \leq \gamma_{t-1}$). In Section 3, we prove that SHB with MomAdaSPS also converges to the exact solution for any choice of $\beta \in [0, 1)$. More specifically, we show that MomAdaSPS converges to the exact solution with a rate $O\left(\frac{1}{T} + \frac{\sigma}{\sqrt{T}}\right)$. Thus, when we are in the interpolation regime (i.e., $\sigma^2 = 0$), the convergence of SHB with MomAdaSPS matches SHB with $\text{MomSPS}_{\max}$ while when the setting does not satisfy interpolation condition, this became equivalent to the $O(1/\sqrt{T})$ of MomDecSPS. Following the terminology of Jiang and Stich (2023), our results are the first *robust* adaptive step-size selection for SHB, as it can automatically adapt to the optimization setting (interpolation vs. non-interpolation).

Let us close this section by mentioning two remarks related to the above three step-size selections:

Remark 2.3. The "correcting factor" $1 - \beta$ is outside the minimum in the $\text{MomSPS}_{\max}$ while for the decreasing variants MomDecSPS and MomAdaSPS, it only appears in the first term of the minimum. This follows from Proposition 2.1 and the derivation of Corollary 2.2 and is explained in detail in Appendix C.3. In Appendix F.3, via experiments, we also test other variants of the above step-sizes with the correcting factor outside of the minimum. We observe that the other choices do not have good practical performance.

Remark 2.4. For no momentum, i.e., $\beta = 0$, all three proposed step-size choices, $\text{MomSPS}_{\max}$, MomDecSPS and MomAdaSPS are reduced to $\text{SPS}_{\max}$, DecSPS and AdaSPS and our convergence analysis recovers the convergence guarantees for SGD showing the tightness of our approach.

3 Convergence Analysis

In this section, we present the convergence results for SHB with all of the proposed Polyak step-sizes. For the formal definitions and helpful lemmas, see Appendix C. The proofs can be found in Appendix D. For all of our results, we make the following assumption:

Assumption 3.1 (Finite optimal objective difference)**.**

$$\sigma^2 := \mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] = f(x^*) - \mathbb{E}_{S_t}[\ell^*_{S_t}] < \infty$$

This assumption was first introduced in Loizou et al. (2021) and Orvieto et al. (2022). Note that $\sigma^2 < \infty$ when all f_i are lower bounded like we have assumed. We say that problem (1) is *interpolated* if $\sigma^2 = 0$. If the interpolation condition is satisfied then there exists $x^* \in X^*$ such

我们提出的以下两个步长选择方法是Polyak步长DecSPS 和AdaSPS的两个减少变体的变体。因此，我们可以证明SHB收敛到精确解而不是邻域。更重要的是，使用MomDecSPS或MomAdaSPS，我们可以为任何动量系数 $\beta \in [0, 1$的选择提供收敛保证，使得SHB 完全自适应（无需调整）。

(ii) MomDecSPS.首先，我们提出了一种带有动量的DecSPS自适应方法。我们称这种方法为MomDecSPS，其表达式为：

$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}c_{t-1}}{c_t}\right\}, \tag{MomDecSPS}$$

其中 $\gamma_{-1} := \gamma_b > 0$ 是一个步长界限， $c_t = c\sqrt{t+1}$ 对于 $t \geq 0$ ， $c_{-1} := c_0 = c > 0$ 是一个用于调节步长的常数。请注意， γ_t 确实在递减。这个步长是自适应的，在其定义中不需要任何函数属性的知识（例如，光滑性/强凸性常数）。在第3 节中，我们证明了SHB 在MomDecSPS 和常数动量$\beta \in [0, 1)$ 下收敛到精确解，其速率为 $O(1/\sqrt{T})$。

(iii) MomAdaSPS. 类似于前两个步骤，让我们提出以下带有动量的AdaSPS自适应方法。我们称这种方法为MomAdaSPS，其表达式为：

$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}. \tag{MomAdaSPS}$$

根据定义，这是一个递减的步长 ($\gamma_t \leq \gamma_{t-1}$)。在第3节中，我们证明了对于任何选择的 $\beta \in [0, 1)$，SHB结合MomAdaSPS也会收敛到精确解。更具体地说，我们展示了MomAdaSPS以速率 $O\left(\frac{1}{T} + \frac{\sigma}{\sqrt{T}}\right)$ 收敛到精确解。因此，当我们在插值状态（即 $\sigma^2 = 0$）时，~~SHB~~结合MomAdaSPS的收敛性与SHB结合$\text{MomSPS}_{\max}$ 相匹配；而当设置不满足插值条件时，这相当于MomDecSPS的 $O(1/\sqrt{T})$。遵循Jiang和Stich（2023）的术语，我们的结果是首个针对SHB的鲁棒自适应步长选择方法，因为它可以自动适应优化设置（插值与非插值）。

让我们通过提及两点与上述三种步长选择相关的内容来结束本节：

备注 2.3。 "修正因子" $1 - \beta$ 位于 $\text{MomSPS}_{\max}$ 的最小值范围之外，而对于递减变体 MomDecSPS 和 MomAdaSPS，它仅出现在最小值的第一项中。这源于命题 2.1 以及推论2.2 的推导，并在附录 C.3中有详细说明。在附录 F.3中，我们通过实验还测试了其他具有修正因子位于最小值范围之外的上述步长变体。我们观察到其他选择在实际性能上并不理想。

备注 2.4。 对于无动量的情况，即 $\beta = 0$，三种提出的步长选择 $\text{MomSPS}_{\max}$、MomDecSPS和 MomAdaSPS 均简化为 $\text{SPS}_{\max}$、DecSPS 和 AdaSPS ，我们的收敛性分析恢复了 SGD 的收敛性保证，这表明了我们方法的有效性。

3 收敛性分析

在本节中，我们介绍了 SHB 在所有提出的Polyak步长下的收敛性结果。关于正式定义和有用引理，请参见附录 C。证明可以在附录D中找到。对于所有我们的结果，我们做出以下假设：

假设3.1(有限最优目标差)。

$$\sigma^2 := \mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] = f(x^*) - \mathbb{E}_{S_t}[\ell^*_{S_t}] < \infty$$

这一假设最早由Loizou等人(2021)提出，并由Orvieto等人(2022)进一步研究。需要注意的是$\sigma^2 < \infty$ 当所有 f_i 都像我们假设的那样存在下界时。我们称问题（1）是插值的，如果 $\sigma^2 = 0$。如果插值条件成立，那么存在 $x^* \in X^*$ 。

that $f(x^*) = f_{S_t}(x^*) = \ell^*_{S_t}$ for all $S_t \subseteq [n]$. Many modern machine learning models satisfy this condition. Examples include non-parametric regression (Liang and Rakhlin, 2020) and over-parameterized deep neural networks (Zhang et al., 2021; Ma et al., 2018).

3.1 Convergence to a neighborhood of the solution

We start with the analysis of SHB with $\text{MomSPS}_{\max}$.

Theorem 3.2. *Assume that each f_i is convex and L_i-smooth. Then, the iterates of SHB with $\text{MomSPS}_{\max}$ with $c = 1$ and $\beta \in \left[0, \frac{\alpha}{2\gamma_b - \alpha}\right)$ where $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ and $L_{\max} = \max_i\{L_i\}$, converge as*

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{C_1\|x^0 - x^*\|^2}{T} + C_2\sigma^2,$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$ and the constants $C_1 = \frac{1-\beta}{\alpha\beta+\alpha-2\beta\gamma_b}$ and $C_2 = \frac{2\gamma_b-\alpha\beta-\alpha}{\alpha\beta+\alpha-2\beta\gamma_b}$.

Firstly, let us note that both constants C_1 and C_2 in Theorem 3.2 are positive since the denominator of both is positive because $\beta \in [0, 1)$ and the numerator of C_2 is positive because $\gamma_b \geq \alpha > 0$. The above result shows that $\text{MomSPS}_{\max}$ has a sublinear convergence to a neighborhood matching the best-known rate for SHB in the convex setting, see Liu et al. (2020). Moreover, when $\beta = 0$, the update rule of SHB with $\text{MomSPS}_{\max}$ becomes equivalent to SGD with $\text{SPS}_{\max}$ (no momentum) and the result of Theorem 3.2 reduces to convergence guarantees provided in Loizou et al. (2021), only with a slightly better neighborhood of convergence (here when $\beta = 0$ we have $C_2 = (2\gamma_b - \alpha)/\alpha$ while in the original it is $C_2 = 2\gamma_b/\alpha$).

In addition, note that in Theorem 3.2, there is a restriction in the momentum coefficient. This stems from the technique used in the proof (which forces C_1 and C_2 to be positive). However, this restriction is not vital as in practical scenarios, the method converges even when β is selected outside the given interval. Moreover, let us highlight that the constants C_1 and C_2 are increasing when viewed as functions of β in the given interval (see also Figure 14). This means that our theory suggests that $\beta = 0$ (no momentum) is the best theoretical choice, which is typical in many works on stochastic methods with momentum (Wang et al., 2023; Loizou and Richtárik, 2020). This is not ideal but does not undermine the importance of Theorem 3.2, as this is the first result showing the convergence of SHB with SPS in a non-interpolated setting.

From Theorem 3.2, we can also deduce rates for the deterministic and interpolated regimes.

Corollary 3.3 (Deterministic Heavy Ball Method)**.** *Assume that f is convex and L-smooth. Then HB with $\beta \in [0, \alpha/(2\gamma_b - \alpha))$ and*

$$\gamma_t = (1-\beta)\min\left\{\frac{f(x^t) - f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\}, \qquad (\text{MomPS}_{\max})$$

where $\alpha = \min\left\{\frac{1}{2L}, \gamma_b\right\}$, converges as $\min_{t\in[T]}\{f(x^t) - f(x^*)\} \leq \frac{(1-\beta)\|x^0-x^*\|^2}{(\alpha\beta+\alpha-2\beta\gamma_b)T}$.

Note that in the setting of Corollary 3.3, we have used $f(x^*)$ instead of a lower bound ℓ^*. This is in accordance with previous work in the deterministic setting (Polyak, 1987). We highlight that in Hazan and Kakade (2019) (PS) and Wang et al. (2023) (ALR-MAG), there is no upper bound in the step-size for the deterministic setting. This is due to the fact that in these works, the step-size is derived as a minimizer of an upper bound of the quantity $\|x^t - x^*\|^2$. Our step-size $\text{MomPS}_{\max}$ is *not* a minimizer of such bound when $\beta > 0$. Nevertheless, when $\beta = 0$ (no momentum), by setting $\gamma_b = \infty$ we recover the corresponding result in Hazan and Kakade (2019). In our setting, we cannot set $\gamma_b = \infty$ when $\beta > 0$ because the term $\beta\gamma_b$ appears in the denominator of our convergence result.

Corollary 3.4 (Interpolation)**.** *Assume interpolation ($\sigma^2 = 0$) and let all assumptions of Theorem 3.2 be satisfied. Then SHB with $\text{MomSPS}_{\max}$ and $\beta \in [0, \alpha/(2\gamma_b - \alpha))$ where $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ converges as $\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{(1-\beta)\|x^0-x^*\|^2}{(\alpha\beta+\alpha-2\beta\gamma_b)T}$, where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$.*

这意味着 $f(x^*) = f_{S_t}(x^*) = \ell^*_{S_t}$ 对于所有 $S_t \subseteq [n]$。许多现代机器学习模型满足这一条件。例如非参数回归（Liang和Rakhlin，2020）和过参数化的深度神经网络（Zhang等人,2021； Ma等人， 2018）。

3.1 收敛到解的邻域

我们从SHB的分析开始与$\text{MomSPS}_{\max}$。

定理3.2。假设每个 f_i 都是凸的且 L_i-光滑的。那么，*SHB* 与*MomSPS*$_{\max}$ 的迭代和 $c = 1$，其中 $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ 和 $L_{\max} = \max_i\{L_i\}$，收敛为

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{C_1\|x^0 - x^*\|^2}{T} + C_2\sigma^2,$$

其中 $x^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$ 和常数 $C_1 = \frac{1-\beta}{\alpha\beta+\alpha-2\beta\gamma_b}$ 和 $C_2 = \frac{2\gamma_b-\alpha\beta-\alpha}{\alpha\beta+\alpha-2\beta\gamma_b}$。

首先，让我们注意到定理3.2中的两个常数 C_1 和 C_2 都是正的，因为它们的分母都是正的，因为 $\beta \in [0, 1)$，而 C_2 的分子是正的，因为 $\gamma_b \geq \alpha > 0$。上述结果表明$\text{MomSPS}_{\max}$ 在凸设置中具有次线性收敛到与SHB最佳已知速率匹配的邻域，参见刘等人(2020)。此外，当 $\beta = 0$时，SHB与$\text{MomSPS}_{\max}$ 的更新规则等效于具有$\text{SPS}_{\max}$ (无动量)的SGD，定理3.2的结果简化为Loizou等人(2021)中提供的收敛保证，只是收敛邻域略好（当 $\beta = 0$ 时，我们有 $C_2 = (2\gamma_b - \alpha)/\alpha$，而在原始版本中它是 $C_2 = 2\gamma_b/\alpha$）。

此外，请注意，在定理3.2中，动量系数存在一个限制。这源于证明中使用的技巧（该技巧迫使 C_1 和 C_2 为正）。然而，这个限制并非关键，因为在实际场景中，即使当 β 选择在给定区间之外时，该方法仍然收敛。此外，让我们强调的是，当作为 β 的函数在给定区间内观察时，常数 C_1 和 C_2 是递增的（另见图14）。这意味着我们的理论表明 $\beta = 0$ (无动量)是最佳的理论选择，这在许多关于随机动量方法的文献中很常见（Wang等人,2023; Loizou和Richt´arik, 2020）。这不是理想的，但并不削弱定理3.2的重要性，因为这是第一个证明SHB 在非插值设置下与SPS收敛的结果。

从定理3.2中，我们还可以推导出确定性和插值情况下的收敛速度。

推论 3.3 (确定性哈密顿方法)**.** 假设 f 是凸的且 L-光滑的。那么，具有 $\beta \in [0, \alpha/(2\gamma_b - \alpha))$ 和

$$\gamma_t = (1-\beta)\min\left\{\frac{f(x^t) - f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\}, \qquad (\text{MomPS}_{\max})$$

其中 $\alpha = $最小值 $\left\{\frac{1}{2L}, \gamma_b\right\}$, 收敛到 最小值$_{t\in[T]}\{f(x^t) - f(x^*)\} \leq \frac{(1-\beta)\|x^0-x^*\|^2}{(\alpha\beta+\alpha-2\beta\gamma_b)T}$.

注意，在推论 3.3 的设置中，我们使用了 $f(x^*)$ 而不是下界 ℓ^*。这符合先前在确定性设置（Polyak，1987）中的工作。我们强调，在 Hazan 和 Kakade (2019) (PS) 以及 Wang 等人 (2023) (ALR-MAG) 中，确定性设置中的步长没有上界。这是由于在这些工作中，步长被推导为某个量 $\|x^t - x^*\|^2$ 上界的极小值。我们的步长 $\text{MomPS}_{\max}$ 在 $\beta > 0$ 时并不是该上界的极小值。然而，当 $\beta = 0$ （无动量）时，通过设置 $\gamma_b = \infty$，我们重新获得了 Hazan 和 Kakade (2019) 中的相应结果。在我们的设置中，当 $\beta > 0$ 时我们不能设置 $\gamma_b = \infty$，因为我们的收敛结果分母中出现了项 $\beta\gamma_b$。

推论 3.4 (插值)**.** 假设插值 ($\sigma^2 = 0$)，且满足定理 *3.2* 的所有条件。那么 *SHB* 与 *MomSPS*$_{\max}$ 和 $\beta \in [0, \alpha/(2\gamma_b - \alpha))$ 其中 $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ 收敛为 $\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{(1-\beta)\|x^0-x^*\|^2}{(\alpha\beta+\alpha-2\beta\gamma_b)T}$，其中 $x^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$。

In the closely related works Loizou et al. (2021) ($\text{SPS}_{\max}$) and Wang et al. (2023) (ALR-SMAG), there is no upper bound in the proposed step-sizes for the interpolated setting. In our analysis, we focus on the fully stochastic setting (interpolation is only a corollary), and the result of Corollary 3.4 shows a saddle connection between the momentum parameter and the parameter γ_b (we cannot select $\gamma_b = \infty$ when momentum is used). Nevertheless, via our corollary, when $\beta = 0$ by setting $\gamma_b = \infty$, we can recover the corresponding result in Loizou et al. (2021).

Another Corollary of Theorem 3.2 is a novel analysis for SHB with constant step-size.

Corollary 3.5 (Constant Step-size)**.** *Let all assumptions of Theorem 3.2 be satisfied. If* $\gamma_b \leq \frac{1}{2L_{\max}}$*, then SHB with* $\gamma \leq \frac{1-\beta}{2L_{\max}}$ *and* $\beta \in [0,1)$ *converges as* $\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\|x^0 - x^*\|^2}{T\gamma} + \sigma^2$*, where* $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$*.*

Note that in Corollary 3.5, there is no restriction that depends on the smoothness parameter in the momentum parameter β. We allow to have $\beta \in [0,1)$. A similar result is obtained in Liu et al. (2020), where the authors establish the convergence rate $O\left(\frac{f(x^0)-f(x^*)}{T} + \frac{L\gamma\hat{\sigma}^2}{1-\beta}\right)$ for the quantity $\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla f_{S_t}(x^t)\|^2]$ when $\gamma \leq \frac{(1-\beta)^2}{L}\min\left\{\frac{1}{4-\beta+\beta^2}, \frac{1}{2\sqrt{2\beta+2\beta^2}}\right\}$. Here $\hat{\sigma}^2$ is assumed to be a bound of the variance $\mathbb{E}_t\|\nabla f_{S_t}(x^t) - \nabla f(x^t)\|^2 \leq \hat{\sigma}^2$. In comparison, in our analysis, we provide the same asymptotic rate $O(1/T)$ for the common in the convex setting quantity $\mathbb{E}[f(\overline{x}^T) - f(x^*)]$. By comparing the two results, our step-size is larger when $\beta \geq \sqrt{5} - 2 \approx 0.236$. However, in our result, the neighborhood of convergence is constant and does not depend on the step-size γ or the momentum coefficient. For a numerical comparison of SHB with the two different constant step-sizes, see Appendix F.2.

3.2 Convergence to the exact solution

In this section, we provide theoretical results for the adaptive decreasing step-sizes, MomDecSPS and MomAdaSPS. The main advantage of these decreasing step-sizes is that we can guarantee convergence to the exact solution while keeping the main adaptiveness properties. Due to the nature of the decreasing step-sizes, there is no restriction in the momentum parameter, as was the case with Theorem 3.2. For the results of this section, we make the extra assumption of *bounded iterates*, i.e., we assume that $D^2 = \max_{t\in[T]}\|x^t - x^*\|^2$ is finite. This is a standard assumption for several adaptive step-sizes, see: (Reddi et al., 2018; Ward et al., 2020; Orvieto et al., 2022; Jiang and Stich, 2023).

Theorem 3.6. *Assume that each* f_i *is convex and* L_i*-smooth. Let* $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$*,* $\alpha = \min\left\{\frac{1}{2L}, \gamma_b\right\}$ *and* $D^2 = \max_{t\in[T]}\|x^t - x^*\|^2$*. Then, the iterates of SHB with MomDecSPS with* $c = 1$ *and* $\beta \in [0,1)$ *converge as:*

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{2\beta[f(x^0) - f(x^*)]}{(1-\beta)T} + \frac{(1+\beta)D^2}{(1-\beta)\alpha\sqrt{T}} + \frac{2\sigma^2}{\sqrt{T}}.$$

Notice that when there is no momentum, i.e., for $\beta = 0$, Theorem 3.6 recovers the original result of SGD with DecSPS from Orvieto et al. (2022). To our knowledge, the only other result that guarantees convergence to the exact solution for SHB in the non-interpolated regime appears in Sebbouh et al. (2021). In particular, Sebbouh et al. (2021) proves that for $\gamma_t = \frac{\eta_t}{1+\lambda_{t+1}}$ and $\beta_t = \frac{\lambda_t}{1+\lambda_t}$ where $\eta_t = \frac{\eta}{\sqrt{t+1}}$ with $0 \leq \eta \leq \frac{1}{4L_{\max}}$ and $\lambda_t = \frac{1}{2}\sum_{i=0}^{t-1}\sqrt{\frac{t+1}{i+1}}$ with $\lambda_0 = 0$, SHB converges as $\mathbb{E}[f(x^t) - f(x^*)] \leq O(\log t/\sqrt{t})$. This result only holds for a specific choice of β_t, it needs knowledge of $L_{\max}$, and only guarantees $O(\log t/\sqrt{t})$ rate. However, it does not make a bounded iteration assumption, and it shows convergence of the last iterate. In contrast, our convergence of SHB with MomDecSPS holds for any choice of $\beta \in [0,1)$, the step-size is adaptive and does not depend on $L_{\max}$, and it achieves a rate of $O(1/\sqrt{k})$.

Next, we present the theoretical guarantees for MomAdaSPS.

在密切相关的研究Loizou等人 (2021) ($\text{SPS}_{\max}$) 和 Wang等人(2023) (ALR-SMAG) 中，所提出的插值设置步长没有上界。在我们的分析中，我们关注完全随机设置（插值只是一个推论），而推论 3.4 显示动量参数与参数 γ_b (我们无法选择$\gamma_b = \infty$ 当使用动量时)。尽管如此，通过我们的推论，当 $\beta = 0$ 通过设置 $\gamma_b = \infty$，我们可以恢复 Loizou等人(2021) 中的相应结果。

定理的另一个推论 3.2 是针对SHB 的常步长新分析。

推论 3.5 (常步长)。 设定理 *3.2* 的所有假设成立。如果 $\gamma_b \leq \frac{1}{2L_{\max}}$，则 *SHB* 具有 $\gamma \leq \frac{1-\beta}{2L_{\max}}$ 和 $\beta \in [0,1)$ 收敛为 $\mathbb{E}[f(x^T) - f(x^*)] \leq \frac{\|x^0 - x^*\|^2}{T\gamma} + \sigma^2$，其中 $x^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$。

注意，在推论 3.5 中，对于动量参数 β 的光滑度参数没有限制。我们允许存在 $\beta \in [0,1)$。在 Liu 等人 (2020) 中获得了类似的结果，其中作者建立了当 $\gamma \leq \frac{(1-\beta)^2}{L}\min\left\{\frac{1}{4-\beta+\beta^2}, \frac{1}{2\sqrt{2\beta+2\beta^2}}\right\}$ 时，量 $\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[\|\nabla f_{S_t}(x^t)\|^2]$ 的收敛速度 $O\left(\frac{f(x^0)-f(x^*)}{T} + L\gamma\frac{\sigma^2}{1-\beta}\right)$。相比之下，在我们的分析中，我们为凸设置中的公共量 $\mathbb{E}[f(x^T) - f(x^*)]$ 提供了相同的渐近速度 $O(1/T)$。通过比较这两个结果，当 $\beta \geq \sqrt{5} - 2 \approx 0.236$ 时，我们的步长更大。然而，在我们的结果中，收敛的邻域是常数，并且不依赖于步长 γ 或动量系数。关于 SHB 与两种不同常步长的数值比较，参见附录 F.2。

3.2 收敛到精确解

在本节中，我们为自适应减小步长，MomDecSPS和MomAdaSPS提供了理论结果。这些减小步长的主要优点在于，我们可以在保持主要自适应特性的同时保证收敛到精确解。由于减小步长的性质，动量参数没有限制，就像定理3.2的情况一样。对于本节的这些结果，我们额外假设了有界迭代，即我们假设 $D^2 = \max_{t\in[T]}\|x^t - x^*\|^2$ 是有限的。这是几种自适应步长的一个标准假设，参见：(Reddi等人，2018；Ward等人，2020；Orvieto等人，2022；Jiang和Stich，2023)。

定理3.6。 假设每个 f_i 是凸的且 L_i-光滑的。令$x^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$，$\alpha = \min\left\{\frac{1}{2L}, \gamma_b\right\}$ 和 $D^2 = \max_{t\in[T]}\|x^t - x^*\|^2$。那么，*SHB* 使用*MomDecSPS*和 $c = 1$ 以及 $\beta \in [0,1)$的迭代收敛如下：

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{2\beta[f(x^0) - f(x^*)]}{(1-\beta)T} + \frac{(1+\beta)D^2}{(1-\beta)\alpha\sqrt{T}} + \frac{2\sigma^2}{\sqrt{T}}.$$

请注意，当没有动量时，即对于 $\beta = 0$，定理3.6 恢复了 Orvieto等人 (2022) 中 SGD 与 DecSPS 的原始结果。据我们所知，在非插值情况下保证 SHB 收敛到精确解的另一个结果出现在 Sebbouh等人 (2021) 中。特别是，Sebbouh等人 (2021) 证明了对于 $\gamma_t = \frac{\eta_t}{1+\lambda_{t+1}}$ 和 $\beta_t = \frac{\lambda_t}{1+\lambda_t}$，其中 $\eta_t = \frac{\eta}{\sqrt{t+1}}$ 满足 $0 \leq \eta \leq \frac{1}{4L_{\max}}$ 且 $\lambda_t = \frac{1}{2}\sum_{i=0}^{t-1}\sqrt{\frac{t+1}{i+1}}$ 满足 $\lambda_0 = 0$，SHB 收敛为 $\mathbb{E}[f(x^t) - f(x^*)] \leq O(\log t/\sqrt{t})$。该结果仅适用于特定的 β_t 选择，需要 $L_{\max}$ 的知识，并且仅保证 $O(\log t/\sqrt{t})$ 的速率。然而，它不进行有界迭代假设，并且显示了最后迭代点的收敛性。相比之下，我们 MomDecSPS 的 SHB 收敛适用于任何 $\beta \in [0,1$ 选择)，步长是自适应的且不依赖于 $L_{\max}$，并实现了 $O(1/\sqrt{k})$ 的速率。

接下来，我们介绍MomAdaSPS的理论保证。

Theorem 3.7. *Assume that each f_i is convex and L_i-smooth. Then, the iterates of SHB with MomAdaSPS and $\beta \in [0,1)$ converge as*

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\tau = \frac{\beta\sqrt{f(x^0)-f(x^*)}}{1-\beta} + \frac{(1+\beta)cLD^2}{2(1-\beta)} + \frac{1}{2c}$ *and* $D^2 = \max_{t\in[T]} \|x^t - x^*\|$.

As in our previous theorems, when there is no momentum ($\beta = 0$), Theorem 3.7 recovers the convergence guarantees of AdaSPS from Jiang and Stich (2023). Let us highlight that our convergence guarantees of SHB with MomAdaSPS offers the first *robust* step-size selection for SHB, in the sense that it can automatically adapt to the optimization setting. More specifically, for no interpolation, our result has a rate of $O(1/\sqrt{T})$, which matches the best-known rate for SHB while if we assume interpolation ($\sigma^2 = 0$), then we are able to achieve a rate of $O(1/T)$ which matches the best-known rate for SHB in the convex setting (see Liu et al. (2020) and Theorem 3.2). Furthermore, under interpolation, Theorem 3.7 improves the convergence of Corollary 3.4. In particular, when $\sigma^2 = 0$, Theorem 3.7 reaches the rate $O(1/T)$ for *any* $\beta \in [0,1)$, while in Corollary 3.4 this is possible only under tighter restrictions on β.

4 Numerical Experiments

In this section, we test our proposed algorithms in deterministic and stochastic convex problems as well as in training popular deep neural networks (DNNs). Our experiments are designed to highlight the benefits of our momentum variants over the vanilla (no-momentum) SPS.

Deterministic Setting. For the deterministic setting, we focus on the least squares problem. The loss function in this case is given by $f(x) = \frac{1}{2}\|Ax - b\|^2$, where $A \in \mathbb{R}^{n\times d}$ and $b \in \mathbb{R}^n$. In our experiments, we follow the setting of Wang et al. (2023) and choose $n = d = 1000$ while the matrix A has been generated according to Lenard and Minkoff (1984) such that the condition number of A^TA is 10^4. We test the following algorithms: Gradient Descent (GD), Nesterov's Accelerated Gradient Descent (AGD) (Nesterov, 2018), Gradient Descent with Polyak's step-size (Polyak, 1987), Adaptive Heavy Ball (AHB) (Saab Jr et al., 2022), Polyak's Heavy Ball (HB) (Polyak, 1964), Accelerated Gradient Method (AGM) (Barré et al., 2020), ALR-MAG (Wang et al., 2023) and finally our step-size from Corollary 3.3, called (MomPS$_{\max}$), which guarantees $O(1/T)$ rate. For the least squares problem, there is an established theory from Polyak (1964) that guarantees acceleration of HB with optimal step-size and momentum coefficient given by $\beta^* = (\sqrt{L} - \sqrt{\mu})^2/(\sqrt{L} + \sqrt{\mu})^2$ and $\gamma^* = (1 + \sqrt{\beta^*})/L$. For ALR-MAG and SHB with MomPS$_{\max}$, we have used the optimal momentum β^* for direct comparison with optimal HB. Moreover, for GD Polyak, ALR-MAG, and MomPS$_{\max}$, we have used the precomputed $f(x^*)$ (via GD), and for MomPS$_{\max}$, we have chosen $\gamma_b = 100$ (for other choices, see Appendix F.4). The results in Figure 2 show that MomPS$_{\max}$ has performance comparable to AGM and ALR-MAG which are faster than vanilla Polyak step-size after the initial iterations. Furthermore, to explore the behavior of the various adaptive step-sizes, we plot the Polyak, the ALR-SMAG, the MomPS$_{\max}$, and the optimal HB step-sizes. We can see that the original Polyak step-size oscillates around the optimal HB step-size while MomPS$_{\max}$ converges from below to the optimal.

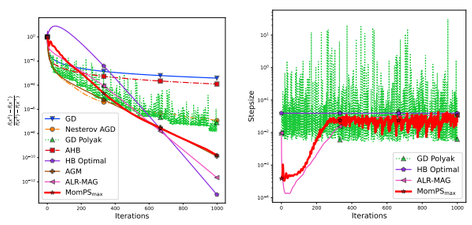


Figure 2: Comparison of deterministic algorithms for least squares. Left: Relative error, Right: Step-sizes.

Stochastic Setting: Convex Problems. Here, we compare our step-sizes with previous works in the stochastic setting. As noticed initially for SPS$_{\max}$ (Loizou et al., 2021), the value of the upper bound γ_b that results in good convergence depends on the problem and requires careful parameter

定理3.7。 假设每个 f_i 是凸的并且 L_i-光滑的。那么，*SHB* 使用*MomAdaSPS* 和 $\beta \in [0,1)$ 的迭代将收敛为

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\tau = \frac{\beta\sqrt{f(x^0)-f(x^*)}}{1-\beta} + \frac{(1+\beta)cLD^2}{2(1-\beta)} + \frac{1}{2c}$ *and* $D^2 = \max_{t\in[T]} \|x^t - x^*\|$.

如我们之前的定理一样，当没有动量($\beta = 0$)时，定理3.7重新获得了Jiang和Stich（2023）提出的AdaSPS的收敛保证。让我们强调的是，我们提出的SHB与MomAdaSPS的收敛保证为SHB提供了首个鲁棒的步长选择方法，因为它可以自动适应优化设置。更具体地说，对于无插值的情况，我们的结果具有 $O(1/\sqrt{T})$的速率，这匹配了SHB目前最已知的速率；如果我们假设插值($\sigma^2 = 0$)，那么我们能够达到 $O(1/T)$的速率，这匹配了SHB在凸设置下的最已知的速率（参见刘等人（2020）和定理3.2）。此外，在插值情况下，定理3.7改进了推论3.4的收敛性。特别是，当 $\sigma^2 = 0$时，定理3.7对任何 $\beta \in [0,1)$都达到了 $O(1/T)$的速率，而在推论3.4中，这只有在更严格的 β限制下才可能实现。

4 数值实验

在本节中，我们在确定性凸问题和训练流行深度神经网络（DNNs）中测试我们提出的算法。我们的实验旨在突出我们动量变体相对于无动量的SPS的优势。

确定性设置。 对于确定性设置，我们关注最小二乘问题。在这种情况下，损失函数为 $f(x) = \frac{1}{2}\|Ax - b\|^2$，其中 $A \in \mathbb{R}^{n\times d}$ 和 $b \in \mathbb{R}^n$。在我们的实验中，我们遵循 Wang等人(2023)的设置，并选择 $n = d = 1000$，而矩阵 A 已根据 Lenard和Minkoff(1984)生成，使得 A^TA 的条件数为 10^4。我们测试以下算法：梯度下降（GD）、Nesterov加速梯度下降（AGD）（Nesterov, 2018）、带Polyak步长的梯度下降

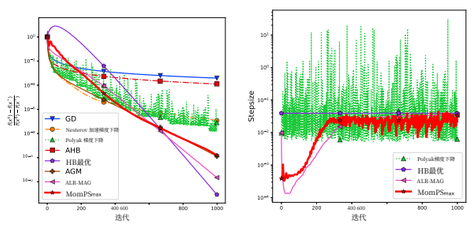


图2：确定性算法的最小二乘问题比较。左：相对误差，右：步长。

(Polyak, 1987), 自适应哈密顿方法 (AHB) (Saab Jr等人, 2022), Polyak的哈密顿方法 (HB) (Polyak, 1964), 加速梯度方法 (AGM) (Barr´e等人, 2020), ALR-MAG (Wang等人, 2023) 以及我们来自Corollary 3.3的步长，称为 (MomPS$_{\max}$)，它保证了$O(1/T)$速率。对于最小二乘问题，Polyak (1964) 建立了理论，保证了HB在最优步长和动量系数$\beta^* = (\sqrt{L} - \sqrt{\mu})^2/(\sqrt{L} + \sqrt{\mu})^2$ 以及 $\gamma^* = (1 + \sqrt{\beta^*})/L$下的加速。对于ALR-MAG和SHB使用MomPS$_{\max}$，我们使用了最优动量 β^*，以便与最优HB直接比较。此外，对于GD Polyak、ALR-MAG和MomPS$_{\max}$，我们使用了预先计算的 $f(x^*)$ (通过GD)，而对于MomPS$_{\max}$，我们选择了 $\gamma_b = 100$ (其他选择，参见附录F.4)。图2中的结果表明，MomPS$_{\max}$ 的性能与AGM和ALR-MAG相当，后者在初始迭代后比vanilla Polyak步长更快。此外，为了探索各种自适应步长的行为，我们绘制了Polyak、ALR-SMAG、MomPS$_{\max}$以及最优HB步长。我们可以看到，原始Polyak步长在最优HB步长周围振荡，而MomPS$_{\max}$ 从下方收敛到最优值。

随机设置：凸问题。 在此，我们将我们的步长与先前在随机设置中的工作进行比较。最初注意到SPS$_{\max}$ (Loizou等人， 2021)，导致良好收敛的上限 γ_b 的值取决于问题，需要仔细参数

tuning. To alleviate this problem, Loizou et al. (2021) uses a smoothing procedure that prevents large fluctuations in the step-size across iterations. That is, $\gamma_b^t = \tau^{b/n}\gamma_{t-1}$ for each iteration t where $\tau = 2$, b is the batch-size, and n is the number of examples. We use the same smoothing trick for γ_b in the implementation of the proposed methods. Similar smoothing procedures have been used in Tan et al. (2016); Vaswani et al. (2019b).

We considered multi-class logistic regression applied to commonly used benchmark datasets from the LIBSVM repository (Chang and Lin, 2011). We separate our experiments between two classes of step-sizes: non-decreasing step-sizes (which might guarantee convergence to a neighborhood of the solution) and decreasing step-sizes (which guarantee convergence to the exact solution). For the first class of step-sizes we test the following algorithms: SGD (Nesterov, 2018), SHB with the constant step-size and $\beta = 0.9$, ADAM as described in Kingma and Ba (2015), the warm-up version of ALR-SMAG with the hyper-parameters suggested in Wang et al. (2023), the $\text{SPS}_{\max}$, the $\text{SPS}_{\max}$ with naive momentum as described in Section 2.2 and our proposed $\text{MomSPS}_{\max}$. For the Polyak-based step-sizes such as ALR-SMAG, $\text{SPS}_{\max}$, and $\text{MomSPS}_{\max}$, we select $\ell_i^* = 0$ and $c = 1$. For SHB, ALR-SMAG, $\text{MomSPS}_{\max}$, $\text{SPS}_{\max}$, with naive momentum, we use momentum $\beta = 0.9$. All the convex experiments are run for 100 epochs and for 5 trials. We plot the average of the trials and the standard deviation. Since there are no standard train/test splits, and due to the small sizes of the datasets, we present training loss and accuracy curves only. We present the comparison of these methods in Figure 3. In all experiments, we observe that the performance of our proposed $\text{MomSPS}_{\max}$ significantly outperforms the other methods in both training loss and test accuracy. For the decreasing variants, we test AdaGrad-Norm (Duchi et al., 2011), SGD with DecSPS, SGD with AdaSPS, and our proposed SHB with MomDecSPS and SHB with MomAdaSPS. We present the outcome of this comparison in Figure 4. MomDecSPS and MomAdaSPS outperform their no-momentum counterparts in both training loss and test accuracy.

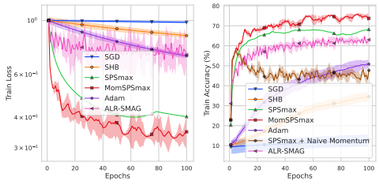


Figure 3: LibSVM dataset: vowel, Batch size: 52

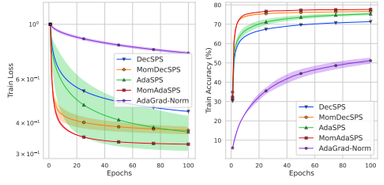


Figure 4: LibSVM dataset: letter, Batch size: 1500

MomSPS for DNNs. In our final experiment, we go beyond the convex setting of our convergence guarantees, and we test $\text{MomSPS}_{\max}$ on the training of DNNs. We consider non-convex minimization for multi-class classification using deep network models on the CIFAR 10 and CIFAR 100 datasets (Krizhevsky et al., 2009). We use two standard image-classification architectures: ResNet18 and ResNet34, (He et al., 2016). For space concerns, we report only the ResNet34 experiments in the main paper and relegate the ResNet18 to the Appendix. In these experiments we run $\text{MomSPS}_{\max}$ with $c = 0.4$. We present the results for CIFAR 10 in Figure 5 and for CIFAR 100 in Figure 6. We observe that $\text{MomSPS}_{\max}$ consistently outperforms its no-momentum counterpart $\text{SPS}_{\max}$ and has competitive generalization performance compared to other popular optimizers.

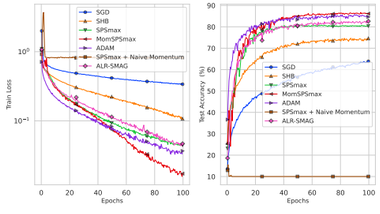


Figure 5: Resnet 34 on CIFAR 10

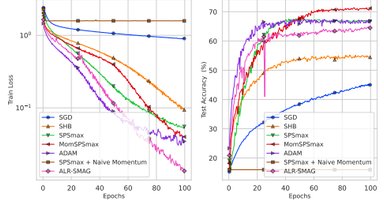


Figure 6: ResNet 34 on CIFAR 100

调整。为了缓解这个问题， Loizou等人(2021)使用了一种平滑程序，防止步长在迭代之间出现大幅波动。也就是说， $\gamma_b^t = \tau^{b/n}\gamma_{t-1}$ 对于每次迭代 t 其中$\tau = 2$, b 是批次大小，而 n 是示例数量。我们为γ_b 在所提出方法的实现中使用了相同的平滑技巧。类似的平滑程序已在Tan等人(2016)； Vaswani等人(2019b)中使用。

我们考虑了多类逻辑回归应用于LIBSVM库（Chang和Lin，2011年）中常用的基准数据集。我们将实验分为两类步长：非递减步长（可能保证收敛到解的邻域）和递减步长（保证收敛到精确解）。对于第一类步长，我们测试以下算法：SGD（Nesterov，2018年）、常步长的SHB和 $\beta = 0.9$、Kingma和Ba（2015年）中描述的ADAM、Wang等人（2023年）中建议超参数的ALR-SMAG的预热版本、$\text{SPS}_{\max}$、$\text{SPS}_{\max}$（如第2.2节中描述的朴素动量）以及我们提出的$\text{MomSPS}_{\max}$。对于基于Polyak的步长，如ALR-SMAG、$\text{SPS}_{\max}$和$\text{MomSPS}_{\max}$，我们选择 $\ell_i^* = 0$ 和 $c = 1$。对于SHB、ALR-SMAG、$\text{MomSPS}_{\max}$、$\text{SPS}_{\max}$（带朴素动量），我们使用动量 $\beta = 0.9$。所有凸性实验均运行100个周期和5次试验。我们绘制了试验的平均值和标准差。由于没有标准的训练/测试拆分，并且由于数据集规模较小，我们仅呈现训练损失和准确率曲线。我们在图3中展示了这些方法的比较。在所有实验中，我们观察到我们提出的$\text{MomSPS}_{\max}$ 在训练损失和测试准确率方面均显著优于其他方法。对于递减变体，我们测试了AdaGrad-Norm（Duchi等人，2011年）、带DecSPS的SGD、带AdaSPS的SGD，以及我们提出的带MomDecSPS的SHB和带MomAdaSPS的SHB。我们在图4中展示了此比较的结果。MomDecSPS和MomAdaSPS在训练损失和测试准确率方面均优于其无动量的对应方法。



图3：Libsvm数据集：元音，批大小：52

图4：Libsvm数据集：字母，批大小：1500

深度神经网络的MomSPS。 在我们的最终实验中，我们超越了收敛保证的凸设置，并测试 $\text{MomSPS}_{\max}$ 在深度神经网络（DNNs）的训练上。我们考虑在CIFAR 10和CIFAR 100数据集（Krizhevsky等人， 2009）上使用深度网络模型进行多类分类的非凸最小化。我们使用两种标准的图像分类架构：ResNet18和ResNet34，(He等人,2016）。由于篇幅限制，我们在正文中仅报告ResNet34实验，并将ResNet18移至附录。在这些实验中我们运行$\text{MomSPS}_{\max}$ 与 $c = 0.4$。我们展示了CIFAR 10的结果（图5 ）和CIFAR 100的结果（图6）。我们观察到$\text{MomSPS}_{\max}$ 始终优于其无动量对应方法$\text{SPS}_{\max}$，并且与其他流行优化器相比具有有竞争力的泛化性能。

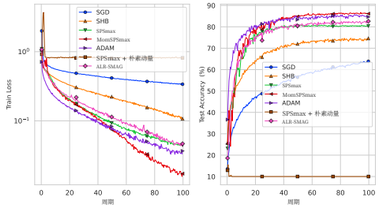


图5：Resnet 34 在 CIFAR 10 上

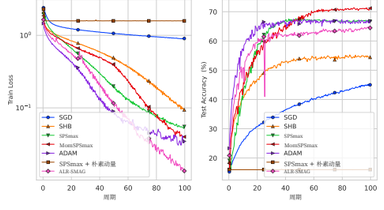


图6：ResNet 34在CIFAR 100上

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Supplementary Material

The Supplementary Material is organized as follows: In Appendix A, we have more details on related work on adaptive and momentum methods. In Appendix B, we include the main pseudo-codes for our algorithms. In Appendix C, we give the basic definitions and lemmas as well as the basic theory of IMA. Appendix D presents the proofs of the theoretical guarantees from the main paper. In Appendix F, we describe in detail our experimental setup and provide additional experiments.

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补充材料

补充材料组织如下：在附录 A，我们提供了关于自适应和动量方法相关工作的更多细节。在附录 B，我们包含了我们算法的主要伪代码。在附录 C，我们给出了基本定义、引理以及IMA的基本理论。附录 D 展示了主论文中理论保证的证明。附录F，我们详细描述了我们的实验设置并提供了额外的实验。

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A FURTHER RELATED WORK

Adaptive methods. Two of the first adaptive algorithms are AdaGrad (Duchi et al., 2011) and RMSProp (Hinton et al., 2012). The very popular algorithm Adam (Kingma and Ba, 2015) was introduced as a momentum extension of RMSProp, and its more recent weight-decay variant AdamW (Loshchilov and Hutter, 2019) is the de facto optimizer for deep neural networks. Investigating the convergence guarantees of adaptive methods across various settings continues to be an active area of research (Vaswani et al., 2020; Ward et al., 2020; Li and Orabona, 2019; Shi et al., 2022; Défossez et al., 2022; Choudhury et al., 2024b; Defazio and Jelassi, 2022).

More recently, a new line of work for adaptive algorithms has appeared, inspired by Polyak step-sizes. Some attempts for efficiently generalizing the Polyak step-size from the deterministic setting to the stochastic were made in Rolinek and Martius (2018); Prazeres and Oberman (2021); Berrada et al. (2020). Loizou et al. (2021) was the first work proposing $\text{SPS}_{\max}$ and providing strong convergence guarantees in different settings, including strongly convex, convex, and non-convex functions. Further extensions with decreasing variants were proposed in Orvieto et al. (2022); Jiang and Stich (2023). Many recent works propose SPS-type update rules for solving optimization problems in different settings. For example, extensions of SPS to proximal setting analyzed in Schaipp et al. (2023a), SPS variants for mirror descent presented in D'Orazio et al. (2023), and SPS for federated learning proposed in Mukherjee et al. (2023). There are also strong connections between SPS and Model-Based approaches (Asi and Duchi, 2019a;b; Chadha et al., 2022). For further results related to SPS, see Gower et al. (2022); Li et al. (2023); Garrigos et al. (2023) and Abdukhakimov et al. (2023).

Momentum with constant step-size. Polyak (1964) introduced the concept of momentum for the Gradient Descent (GD) method. He showed that for strongly convex quadratic problems, the momentum method provably accelerates GD with optimal momentum coefficient $\beta^* = (\sqrt{L} - \sqrt{\mu})^2/(\sqrt{L} + \sqrt{\mu})^2$ and $\gamma^* = (1 + \sqrt{\beta^*})^2/L$ where μ is the strong convexity constant and L the smoothness constant. Ghadimi et al. (2015) proved a global sublinear convergence guarantee for HB for convex and smooth functions and global linear convergence when the function is also smooth and strongly convex. Moreover, Nesterov (1983) introduced Nesterov's Accelerated Gradient Descent (AGD) method, where he shows acceleration over GD for strongly convex and smooth as well as convex and smooth objectives. Note that both of these methods require knowledge of both μ and L. In the stochastic setting, Liu et al. (2020) provides new convergence guarantees for SHB with constant step-size. In particular, it shows that for both strongly convex and non-convex objectives, SHB enjoys the same convergence bound as SGD. Sebbouh et al. (2021), among other things, shows that in the smooth and convex setting SHB converges in expectation at the last iterate with a rate of $O(1/T)$ to a neighborhood of the minimum and at a $O(\log T/\sqrt{T})$ rate to the minimum exactly. For these guarantees, one needs knowledge of the problem parameters (such as the smoothness constant), and their results hold for a specific selection of the momentum parameter. Momentum has also been successfully applied in linear systems, see Loizou and Richtárik (2020) where it provides convergence guarantees of SGD, stochastic Newton, stochastic proximal point, and stochastic dual subspace ascent with momentum for consistent linear systems.

Adaptive Momentum. In this paragraph, we review the bibliography for adaptive methods with momentum. Momentum can be either SHB or Nesterov's acceleration, and in this case, the adaptivity can be either the step-size or the momentum coefficient. In the deterministic setting, Barré et al. (2020) proposed the Accelerated Gradient Method (AGM), an adaptive algorithm built upon AGD that guarantees acceleration over GD. AGM approximates the strong convexity constant μ by the inverse of the classical Polyak Step-size. However, their algorithm still requires the knowledge of L (but not μ). Furthermore, Saab Jr et al. (2022) suggests an Adaptive Heavy Ball (AHB) algorithm where it approximates the constants μ and L iteratively in each step based on the formula of the optimal constants of HB, but it does not show any acceleration over HB or GD. In the stochastic setting, Wang et al. (2023) propose an adaptive algorithm for a moving averaged gradient momentum, named ALR-SMAG. It shows that ALR-SMAG enjoys a linear convergence rate for semi-strongly convex and smooth functions. Schaipp et al. (2023b) proposes a new adaptive learning rate that can be combined with any momentum-based method. Finally, Zeng et al. (2023) proposes adaptive variants for SHB in linear systems solvers.

进一步相关工作

自适应方法。最早的两种自适应算法是 AdaGrad（Duchi 等人，2011 年）和 RMSProp（Hinton 等人，2012 年）。非常流行的算法 Adam（Kingma 和 Ba，2015 年）被引入为 RMSProp 的动量扩展，而其较新的权重衰减变体 AdamW（Loshchilov 和 Hutter，2019 年）已成为深度神经网络的实际优化器。在各种设置下研究自适应方法的收敛保证仍然是活跃的研究领域（Vaswani 等人，2020 年；Ward 等人，2020 年；Li 和 Orabona，2019 年；Shi 等人，2022 年；D´efossez 等人，2022 年；Choudhury 等人，2024b；Defazio 和 Jelassi，2022 年）。

最近，出现了一项受Polyak步长启发的新工作，用于自适应算法。在Rolinek和Martius (2018)；Prazeres和Oberman (2021)；Berrada等人 (2020) 中，曾尝试高效地将Polyak步长从确定性设置推广到随机性设置。Loizou等人 (2021) 是首个提出$\text{SPS}_{\max}$并针对不同设置（包括强凸、凸和非凸函数）提供强收敛保证的工作。Orvieto等人 (2022)；Jiang和Stich (2023) 进一步提出了具有减小变体的扩展。许多近期工作提出了SPS型更新规则，用于在不同设置中解决优化问题。例如，Schaipp等人 (2023a) 分析了SPS在近端设置中的扩展，D'Orazio等人 (2023) 提出了用于镜像下降的SPS变体，Mukherjee等人 (2023) 提出了用于联邦学习的SPS。SPS与基于模型的方法之间也存在强连接 (Asi和Duchi, 2019a;b; Chadha等人, 2022)。有关SPS的进一步结果，参见Gower等人 (2022)；Li等人 (2023)；Garrigos等人 (2023) 和Abdukhakimov等人 (2023)。

具有固定步长的动量。Polyak (1964) 为梯度下降 (GD) 方法引入了动量概念。他表明，对于强凸二次问题，动量方法在最优动量系数 $\beta^* = (\sqrt{L} - \sqrt{\mu})^2/(\sqrt{L} + \sqrt{\mu})^2$ 和 $\gamma^* = (1 \pm \sqrt{\beta^*})^2/L$ 下，能够可证明地加速GD，其中 μ 是强凸常数，L 是光滑常数。Ghadimi 等人 (2015) 证明了对于凸和光滑函数，哈密顿方法 (HB) 具有全局次线性收敛保证，当函数也光滑且强凸时，则具有全局线性收敛保证。此外，Nesterov (1983) 引入了 Nesterov 加速梯度下降 (AGD) 方法，他表明该方法对于强凸和光滑以及凸和光滑目标都优于 GD。请注意，这两种方法都需要知道 μ 和 L。在随机设置中，刘等人 (2020) 为具有固定步长的 SHB 提供了新的收敛保证。特别是，它表明对于强凸和非凸目标，SHB 享有与随机梯度下降 (SGD) 相同的收敛界。Sebbouh 等人 (2021) 证明了在光滑和凸设置中，SHB 以 $O(1/T)$ 的速率在期望意义上在最后一个迭代点收敛到最小值附近的邻域，并以 $O(\log T/\sqrt{T})$ 的速率精确收敛到最小值。为了获得这些保证，需要知道问题参数（例如光滑常数），并且他们的结果适用于动量参数的特定选择。动量也已成功应用于线性系统，参见 Loizou 和 Richt´arik (2020)，其中它提供了 SGD、随机牛顿、随机近端点法和随机对偶子空间上升（带动量）对于一致线性系统的收敛保证。

自适应动量。在本段中，我们回顾了具有动量的自适应方法的文献。动量可以是SHB或Nesterov加速，在这种情况下，自适应性可以是步长或动量系数。在确定性设置中，Barr´e等人。(2020) 提出了加速梯度方法（AGM），这是一种基于AGD的自适应算法，它保证了比GD的加速。AGM通过经典Polyak步长的倒数来近似强凸常数 μ，但是他们的算法仍然需要知道 L(但不是 μ)。此外，Saab Jr等人。(2022) 提出了一种自适应重球（AHB）算法，它在每一步中根据HB最优常数的公式迭代地近似常数 μ 和 L，但它没有显示出比HB或GD的加速。在随机设置中，Wang等人。(2023) 提出了一种用于移动平均梯度动量的自适应算法，命名为ALR-SMAG。它表明ALR-SMAG对于半强凸和光滑函数具有线性收敛速度。Schaipp等人。(2023b) 提出了一种新的自适应学习率，它可以与任何基于动量的方法结合使用。最后，Zeng等人。(2023) 提出了SHB 在线性系统求解器中的自适应变体。

On Technical Assumptions for Convergence. In the literature of stochastic optimization problems, there are several assumptions on the noise of the stochastic estimators that typically are made on top of smoothness and convexity to prove the convergence of stochastic optimization algorithms.

For example, many works (Recht et al., 2011; Rakhlin et al., 2012; Shamir and Zhang, 2013; Nguyen et al., 2018) assume bounded gradients, i.e., that is, there exists a $M \in \mathbb{R}$ such that $\mathbb{E}\left\|\nabla f_i(x)\right\|^2 \leq M$. While this might look like a natural assumption, in the unconstrained setting, it contradicts the assumption of strong convexity leading to convergence guarantees that hold for an empty set of problems (Nguyen et al., 2018; Gower et al., 2019; 2021). A much more relaxed assumption used in the literature is the growth condition on the stochastic gradients, (Bertsekas and Tsitsiklis, 1996; Schmidt and Roux, 2013; Vaswani et al., 2019a). It states that there exist constants $\rho \in \mathbb{R}$ and $\delta \in \mathbb{R}$ such that $\mathbb{E}\left\|\nabla f_i(x)\right\|^2 \leq \rho\|\nabla f(x)\|^2 + \delta$. Based on a strong growth condition ($\delta = 0$), Schmidt and Roux (2013) were the first to establish linear convergence of SGD, with Vaswani et al. (2019a) later showing that SGD can find a first-order stationary point as efficiently as full gradient descent in non-convex settings. Similar conditions have also been proved and used in the analysis of decentralized variants of SGD (Koloskova et al., 2020; Assran et al., 2019). More recently, a line of work that uses smoothness (via expected smoothness/residual conditions) to provide closed-form expressions for the values of ρ and δ of growth condition was able to provide tight convergence analysis of several stochastic algorithms, including SGD (Gower et al., 2019; 2021), variance reduced methods (Khaled et al., 2023), stochastic algorithms for min-max optimization (Loizou et al., 2021; 2020; Gorbunov et al., 2022; Choudhury et al., 2024a).

In the convex regime, the original analysis of SGD with SPS of Loizou et al. (2021) was one of the first papers that did not require any additional assumptions to guarantee convergence for SGD. Following the convergence guarantees of Loizou et al. (2021), we highlight that our proposed analysis of SHB with Polyak step-size does not require any additional assumptions for guaranteeing convergence. To the best of our knowledge, as mentioned in the main paper, our approach provides the first analysis of SHB without the restrictive bounded variance and growth conditions.

For proving convergence of SHB, Liu et al. (2020) makes the strong assumption of bounded variance and assumes knowledge of the smoothness L for tuning the parameters. Another work on theoretical guarantees for SHB is Sebbouh et al. (2021), where a last iterate convergence is proved for a specific choice of γ_t and β_t, but also requires knowledge of L and only achieves rate $O(\log T/\sqrt{T})$ for the decreasing step-size variant. Furthermore, the authors of Wang et al. (2022) propose a Polyak-based stepsize for a moving averaged gradient momentum variant, different from SHB, and provide guarantees for the deterministic and stochastic setting for strongly convex objectives. Finally, Schaipp et al. (2023b) assumes a stochastic convex and *interpolated* regime for their results.

关于收敛性的技术假设。在随机优化问题的文献中，存在几种关于随机估计器噪声的假设，这些假设通常在光滑性和凸性的基础上提出，以证明随机优化算法的收敛性。

例如，许多工作（Recht等人，2011年；Rakhlin等人，2012年；Shamir和Zhang，2013年；Nguyen等人，2018年）假设梯度有界，即存在一个 $M \in \mathbb{R}$，使得 $\mathbb{E}\left\|\nabla f_i(x)\right\|^2 \leq M$。虽然这看起来像是一个自然的假设，但在无约束设置下，它违背了强凸性的假设，导致仅适用于空问题集的收敛保证（Nguyen等人，2018年；Gower等人，2019年；2021年）。文献中使用的更宽松的假设是随机梯度的增长条件（Bertsekas和Tsitsiklis，1996年；Schmidt和Roux，2013年；Vaswani等人，2019a）。它指出存在常数 $\rho \in \mathbb{R}$ 和 $\delta \in \mathbb{R}$，使得 $\mathbb{E}\left\|\nabla f_i(x)\right\|^2 \leq \rho\|\nabla f(x)\|^2 + \delta$。基于一个强增长条件（$\delta = 0$），Schmidt和Roux（2013年）首次建立了SGD的线性收敛性，Vaswani等人（2019a）后来表明，在非凸设置中，SGD找到一阶驻点的效率与全梯度下降相同。类似的条件也已被证明并在SGD的去中心化变体的分析中使用（Koloskova等人，2020年；Assran等人，2019年）。最近，一系列使用平滑性（通过期望平滑性/残差条件）来为增长条件的 ρ 和 δ 值提供封闭形式表达式的工作，能够为包括SGD（Gower等人，2019年；2021年）、方差减少方法（Khaled等人，2023年）、用于最小-最大优化的随机算法（Loizou等人，2021年；2020年；Gorbunov等人，2022年；Choudhury等人，2024a）在内的几种随机算法提供紧密的收敛分析。

在凸区域内，Loizou等人⁽²⁰²¹⁾的原始分析是首批无需任何额外假设即可保证随机梯度下降(SGD)收敛的论文之一。在Loizou等人⁽²⁰²¹⁾的收敛保证基础上，我们强调，我们提出的Polyak步长SHB分析无需额外假设即可保证收敛。据我们所知，正如正文中所述，我们的方法首次对无限制方差增长条件的SHB进行了分析。

为证明收敛性，SHB，Liu等人(2020)做出了强方差有界的假设，并假设已知光滑性 L 以调整参数。另一项关于SHB 理论保证的工作是Sebbouh等人(2021)，其中对于特定的 γ_t 和 β_t 证明了最后迭代收敛，但也需要知道 L，并且仅对递减步长变体实现了 $O(\log T/\sqrt{T})$ 的收敛速度。此外，Wang等人(2022)提出了一种基于Polyak的步长用于移动平均梯度动量变体，不同于SHB，并为强凸目标在确定性和随机设置下提供了保证。最后，Schaipp等人(2023b)假设了随机凸和插值的机制以支持他们的结果。

B PSEUDO-CODES

In this section, we include the pseudo-codes of the SHB with the proposed stochastic Polyak step-sizes for a better understanding of the methods and easier comparison with other algorithms.

SHB with MomSPSmax. We start with the pseudo-code for SHB with MomSPS$_{\max}$. In Theorem 3.2 we established convergence to a neighborhood for $\beta \in \left[0, \frac{\alpha}{2\gamma_b - \alpha}\right)$ where $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ and for $c = 1$. However, in practice, it seems to have convergence even for different choices of c or even if β is chosen outside of this bound, so we provide the pseudo-code for a general $\beta \in [0, 1)$ and c. Thus, the user must provide the momentum coefficient β, the upper bound γ_b, the constant c, and the lower bounds $\ell^*_{S_t}$. In practice, the usual choices are $\beta = 0.9$, $c = 1$, and $\ell^*_{S_t} = 0$.

Algorithm 1 SHB with MomSPS$_{\max}$

Parameters: $\beta \in [0, 1)$, $\gamma_b > 0$, $c > 0$, $\ell^*_{S_t}$ lower bounds
Initialization: $x^0 \in \mathbb{R}^d$, $x^{-1} = x^0$
for $t = 0, 1, 2, \ldots$ **do**
  Choose uniformly at random $S_t \subseteq \{1, \ldots, n\}$
  $\gamma_t = (1 - \beta) \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}$
  $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$
end for

SHB with MomDecSPS. For the decreasing step-sizes, Theorems 3.6 and 3.7 make no assumption on $\beta \in [0, 1)$. For SHB with MomDecSPS, the user needs to provide the momentum coefficient β, the upper bound γ_b, the constant c and the lower bounds ℓ_{S_t}, like the previous algorithm. In practice, the usual choices are $c = 1$ and $\ell^*_{S_t} = 0$. Note that when $t = 0$, the step-size is equal to

$$\gamma_0 = \min\left\{\frac{(1-\beta)[f_{S_0}(x^0) - \ell^*_{S_0}]}{c\|\nabla f_{S_0}(x^0)\|^2}, \frac{\gamma_b c}{c}\right\} = (1-\beta)\min\left\{\frac{f_{S_0}(x^0) - \ell^*_{S_0}}{c\|\nabla f_{S_0}(x^0)\|^2}, \gamma_b\right\},$$

which is equal to MomSPS$_{\max}$ for $t = 0$.

Algorithm 2 SHB with MomDecSPS

Parameters: $\beta \in [0, 1)$, $\gamma_{-1} = \gamma_b > 0$, $c > 0$, $c_{-1} = c > 0$, $c_t = c\sqrt{t+1}$ for $t \geq 0$, $\ell^*_{S_t}$ lower bounds
Initialization: $x^0 \in \mathbb{R}^d$, $x^{-1} = x^0$
for $t = 0, 1, 2, \ldots$ **do**
  Choose uniformly at random $S_t \subseteq \{1, \ldots, n\}$
  $\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}c_{t-1}}{c_t}\right\}$
  $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$
end for

SHB with MomAdaSPS. Finally, for SHB with MomAdaSPS the requirements are less since the user only needs to provide the momentum coefficient β, the constant c and the lower bounds $\ell^*_{S_t}$. Furthermore, according to Jiang and Stich (2023) one can set $c = \frac{1}{\sqrt{f_{S_0}(x^0) - \ell^*_{S_0}}}$ after randomly choosing $S_0 \subseteq [n]$ in the first iteration. Moreover, the minimum with respect to γ_{t-1} is to ensure that the step-size is decreasing.

B 伪代码

在本节中，我们包含SHB 的伪代码，其中提出了随机Polyak步长，以便更好地理解这些方法，并与其他算法进行更简单的比较。

SHB 与 MomSPSmax。我们从具有 MomSPS$_{\max}$ 的 SHB 伪代码开始。在定理 3.2 中，我们证明了 $\beta \in \left[0, \frac{\alpha}{2\gamma_b - \alpha}\right)$ 收敛到 $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ 以及 $c = 1$ 的邻域。然而，在实践中，即使对于不同的 c 选择，或者即使 β 被选择在此界限之外，似乎也有收敛，因此我们提供了针对一般 $\beta \in [0, 1)$ 和 c 的伪代码。因此，用户必须提供动量系数 β、上界 γ_b、常数 c 以及下界 $\ell^*_{S_t}$。在实践中，通常的选择是 $\beta = 0.9$、$c = 1$ 和 $\ell^*_{S_t} = 0$。

算法1SHB 与 MomSPS$_{\max}$

1: 参数： $\beta \in [0, 1)$, $\gamma_b > 0$, $c > 0$, $\ell^*_{S_t}$ **下界 2: 初始化：** $x^0 \in \mathbb{R}^d$, $x^{-1} = x^0$ **3: 对于** $t = 0, 1, 2, \ldots$ **执行 4: 随机均匀选择** $S_t \subseteq \{1, \ldots, n\}$ **5:** $\gamma_t = (1 - \beta)$ **min**$\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}$ **6:** $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$ **7: 结束循环**

SHB与 MomDecSPS. 对于递减步长，定理 3.6 和 3.7 不做任何假设关于 $\beta \in [0, 1)$。对于 SHB 与 MomDecSPS，用户需要提供动量系数 β，上界 γ_b，常数 c 和下界 ℓ_{S_t}，与之前的算法类似。在实践中，常见的选择是 $c = 1$和 $\ell^*_{S_t} = 0$。注意当 $t = 0$，步长等于

$$\gamma_0 = \min\left\{\frac{(1-\beta)[f_{S_0}(x^0) - \ell^*_{S_0}]}{c\|\nabla f_{S_0}(x^0)\|^2}, \frac{\gamma_b c}{c}\right\} = (1-\beta)\min\left\{\frac{f_{S_0}(x^0) - \ell^*_{S_0}}{c\|\nabla f_{S_0}(x^0)\|^2}, \gamma_b\right\},$$

等于MomSPS$_{\max}$ 对于 $t = 0$。

算法2SHB 与MomDecSPS

1: 参数： $\beta \in [0, 1)$, $\gamma_{-1} = \gamma_b > 0$, $c > 0$, $c_{-1} = c > 0$, $c_t = c\sqrt{t+1}$ **对于** $t \geq 0$, $\ell^*_{S_t}$ **下界 2: 初始化：** $x^0 \in \mathbb{R}^d$, $x^{-1} = x^0$ **3: 对于** $t = 0, 1, 2, \ldots$ **执行 4: 在** $S_t \subseteq \{1, \ldots, n\}$**中均匀随机选择 5:** $\gamma_t =$ **min**$\left\{(1-\beta)\frac{[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}c_{t-1}}{c_t}\right\}$ **6:** $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$ **7: 结束循环**

SHB与MomAdaSPS。 最后，对于SHB 与 MomAdaSPS 来说，要求较低，因为用户只需提供动量系数 β、常数 c 以及下界 $\ell^*_{S_t}$。此外，根据Jiang和Stich(2023) 的研究，可以在第一次迭代中随机选择 $S_0 \subseteq [n]$ 后设置 $c = \frac{1}{\sqrt{f_{S_0}(x^0) - \ell^*_{S_0}}}$。再者，关于 γ_{t-1} 的最小值是为了确保步长在减小。

Algorithm 3 SHB with MomAdaSPS

Parameters: $\beta \in [0,1)$, $\gamma_{-1} = +\infty$, $c > 0$, $\ell^*_{S_t}$ lower bounds
Initialization: $x^0 \in \mathbb{R}^d$, $x^{-1} = x^0$
for $t = 0, 1, 2, \ldots$ **do**
  Choose uniformly at random $S_t \subseteq \{1, \ldots, n\}$
  $\gamma_t = \min\left\{ \frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2 \sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1} \right\}$
  $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$
end for

Note that for all the above algorithms, one needs only lower bounds $\ell^*_{S_t}$, which can be chosen equal to $\ell^*_{S_t} = 0$ in practice. Of course, if the true infima $f^*_{S_t} = \inf_{x \in \mathbb{R}^d} f_{S_t}$ are known, then one should use them because it will lead to better convergence in terms of the neighborhood. Furthermore, notice that we start by initializing $x^{-1} = x^0$, where x^0 is the given starting point. This is because the first step of SHB (i.e. for $t = 0$) is just an iteration of SGD because we have no previous iterate x^{-1}, that is $x^1 = x^0 - \gamma_0 \nabla f_{S_0}(x^0) + \beta(x^0 - x^{-1}) = x^0 - \gamma_0 \nabla f_{S_0}(x^0)$.

算法3SHB 与MomAdaSPS

1: 参数： $\beta \in [0,1)$, $\gamma_{-1} = +\infty$, $c > 0$, $\ell^*_{S_t}$ **下界 2: 初始化：** $x^0 \in \mathbb{R}^d$, $x^{-1} = x^0$ **3: for** $t = 0$, **1, 2, .. . do 4: 在** $S_t \subseteq \{1, \ldots, n\}$**上均匀随机选择 5:** $\gamma_t =$ **min** $\left\{ \right.$ **(**

$$\frac{1-\beta\ [f_{S_t}\ x^t\ - \ell^*_{S_t}]}{c\|\nabla f_{S_t}\ x^t\ \|^2 \sqrt{\sum_{s=0}^t f_{S_s}}}, \gamma_{t-1} \left. \right\}$$

6: $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$ **7: end for**

请注意，对于所有上述算法，只需要下界 $\ell^*_{S_t}$，在实践中可以将其设为 $\ell^*_{S_t} = 0$。当然，如果已知真实下确界 $f^*_{S_t} = \inf_{x \in \mathbb{R}^d} f_{S_t}$，则应使用它们，因为这将在邻域方面带来更好的收敛性。此外，请注意我们首先通过初始化 $x^{-1} = x^0$，其中 x^0 是给定的起始点。这是因为SHB (即对于 $t = 0$) 的第一步只是SGD的迭代，因为我们没有之前的迭代x^{-1}，也就是说 $x^1 = x^0 - \gamma_0 \nabla f_{S_0}(x^0) + \beta(x^0 - x^{-1}) = x^0 - \gamma_0 \nabla f_{S_0}(x^0)$.

C TECHNICAL PRELIMINARIES

C.1 BASIC DEFINITIONS

In this section, we present some basic definitions we use throughout the paper.

Definition C.1 (Convexity)**.** A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is convex if

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle,$$

for all $x, y \in \mathbb{R}^d$.

Definition C.2 (L-smooth)**.** A differentiable function $f : \mathbb{R}^d \to \mathbb{R}$ is L-smooth if there exists a constant $L > 0$ such that

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|,$$

or equivalently

$$|f(x) - f(y) - \langle \nabla f(y), x - y \rangle| \leq \frac{L}{2}\|x - y\|^2$$

for all $x, y \in \mathbb{R}^d$.

C.2 BASIC LEMMAS

Here we have gathered useful lemmas for various Polyak related step-sizes. These lemmas are used repeatedly in the proofs of the main results.

Lemma C.3 ((Loizou et al., 2021))**.** *Suppose each f_i is L_i-smooth, then the step-size of SPS$_{\max}$ satisfies:*

$$\alpha = \min\left\{\frac{1}{2cL_{\max}}, \gamma_b\right\} \leq \gamma_t \leq \gamma_b \text{ and } \gamma_t^2\|\nabla f_{S_t}(x^t)\|^2 \leq \gamma_t[f_{S_t}(x^t) - \ell_{S_t}^*], \tag{2}$$

where $L_{\max} = \max_{i\in[n]}\{L_i\}$.

Lemma C.4 (Lemma 1, (Orvieto et al., 2022))**.** *Suppose each f_i is L_i-smooth, then the step-size of DecSPS satisfies*

$$\min\left\{\frac{1}{2c_tL_{\max}}, \frac{c_0\gamma_b}{c_t}\right\} \leq \gamma_t \leq \frac{c_0\gamma_b}{c_t} \text{ and } \gamma_{t-1} \leq \gamma_t \text{ and } \gamma_t^2\|\nabla f_{S_t}(x^t)\|^2 \leq \frac{\gamma_t}{c_t}[f_{S_t}(x^t) - \ell_{S_t}^*], \tag{3}$$

where $L_{\max} = \max_{i\in[n]}\{L_i\}$.

Lemma C.5 (Lemma 12, (Jiang and Stich, 2023))**.** *For any non-negative sequence $a_0, \ldots, a_T$, the following holds:*

$$\sqrt{\sum_{t=0}^{T} a_t} \leq \sum_{t=0}^{T} \frac{a_t}{\sqrt{\sum_{s=0}^{t} a_s}} \leq 2\sqrt{\sum_{t=0}^{T} a_t}. \tag{4}$$

If $a_0 \geq 1$, then the following holds:

$$\sum_{t=0}^{T} \frac{a_t}{\sqrt{\sum_{s=0}^{t} a_s}} \leq \log\left(\sum_{t=0}^{T} a_t\right) + 1. \tag{5}$$

C 技术预备知识

C.1 基本定义

在本节中，我们介绍论文中始终使用的一些基本定义。

定义 C.1 (凸性)。可微函数 $f: \mathbb{R}^d \to \mathbb{R}$ 是凸的，如果

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle,$$

对于所有 $x, y \in \mathbb{R}^d$。

定义 C.2 (L-光滑)。可微函数 $f: \mathbb{R}^d \to \mathbb{R}$ 是 L-光滑的，如果存在一个常数 $L > 0$ 使得

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|,$$

或者等价地

$$|f(x) - f(y) - \langle \nabla f(y), x - y \rangle| \leq \frac{L}{2}\|x - y\|^2$$

对于所有 $x, y \in \mathbb{R}^d$。

C.2 基本引理

在这里，我们收集了与各种Polyak相关步长相关的有用引理。这些引理在主要结果的证明中反复使用。

引理 C.3((Loizou等人, 2021))。假设每个 f_i 都是 L_i光滑的$_{\max}$，则*SPS*的步长满足：

$$\alpha = \min\left\{\frac{1}{2cL_{\max}}, \gamma_b\right\} \leq \gamma_t \leq \gamma_b \text{ and } \gamma_t^2\|\nabla f_{S_t}(x^t)\|^2 \leq \gamma_t[f_{S_t}(x^t) - \ell_{S_t}^*], \tag{2}$$

其中 $L_{\max} = \max_{i\in[n]}\{L_i\}$。

引理 C.4 (引理 1, (Orvieto等人,2022))**.** 假设每个 f_i 都是 L_i光滑的，则*DecSPS*的步长满足

$$\min\left\{\frac{1}{2c_tL_{\max}}, \frac{c_0\gamma_b}{c_t}\right\} \leq \gamma_t \leq \frac{c_0\gamma_b}{c_t} \text{ and } \gamma_{t-1} \leq \gamma_t \text{ and } \gamma_t^2\|\nabla f_{S_t}(x^t)\|^2 \leq \frac{\gamma_t}{c_t}[f_{S_t}(x^t) - \ell_{S_t}^*], \tag{3}$$

其中 $L_{\max} = \max_{i\in[n]}\{L_i\}$.

引理 C.5 (引理 12, (Jiang和Stich, 2023))**.** 对于任何非负序列 $a_0, \ldots, a_T$，如下成立：

$$\sqrt{\sum_{t=0}^{T} a_t} \leq \sum_{t=0}^{T} \frac{a_t}{\sqrt{\sum_{s=0}^{t} a_s}} \leq 2\sqrt{\sum_{t=0}^{T} a_t}. \tag{4}$$

如果 $a_0 \geq 1$，那么以下成立：

$$\sum_{t=0}^{T} \frac{a_t}{\sqrt{\sum_{s=0}^{t} a_s}} \leq \log\left(\sum_{t=0}^{T} a_t\right) + 1. \tag{5}$$

Lemma C.6 (Lemma 13, (Jiang and Stich, 2023)). *If $x^2 \leq a(x+b)$ for $a \geq 0$ and $b \geq 0$, then it holds that $x \leq a + \sqrt{ab}$.*

Lemma C.7 (Lemma 16, (Jiang and Stich, 2023)). *Suppose each f_i is L_i-smooth, then the step-size of AdaSPS satisfies*

$$\frac{1}{2cL_{\max}} \frac{1}{\sqrt{\sum_{s=0}^{t} f_{S_s}(x^s) - \ell^*_{S_s}}} \leq \eta_t \leq \frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2} \frac{1}{\sqrt{\sum_{s=0}^{t} f_{S_s}(x^s) - \ell^*_{S_s}}}, \tag{6}$$

where $L_{\max} = \max_{i\in[n]}\{L_i\}$.

C.3 Iterate Moving Average

As mentioned in the main paper, the iterates of the SHB method are equal to the 2.3 method. The following proposition is from Sebbouh et al. (2021). We include the proof for completeness.

Proposition C.8 (Proposition 1.6, (Sebbouh et al., 2021)). *Let $\eta_t > 0$ and $\lambda_t \geq 0$. Consider the Iterate Moving Average (IMA) method given by the following update rule:*

$$z^{t+1} = z^t - \eta_t \nabla f_{S_t}(x^t) \tag{7}$$

$$x^{t+1} = \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1}, \tag{8}$$

where $z^0 = x^0$. If the following equations hold for any $t \in \mathbb{N}$

$$1 + \lambda_{t+1} = \frac{\lambda_t}{\beta_t} \tag{9}$$

$$\eta_t = (1+\lambda_{t+1})\gamma_t, \tag{10}$$

then the x_t iterates of the 2.3 method are equal to the x_t iterates produced by the SHB method.

Proof. We will start from the 2.3 update rule and we will prove that its (x^t) iterates satisfy the SHB update rule. Suppose that we have the (z^{t+1}) and (x^{t+1}) iterates as well as the (λ_t) and (η_t) sequences. Define the sequences (β_t) and (γ_t) using eqs. (9) and (10). We will show that (x^t) iterates satisfy the SHB update rule. Solving for z^{t+1} in eq. (8) we get

$$z^{t+1} := x^{t+1} + \lambda_{t+1}(x^{t+1} - x^t) \tag{11}$$

Now substituting eqs. (7) and (11) in eq. (8) we have

$$\begin{aligned}
x^{t+1} &= \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1} \\
&\overset{(7)}{=} \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} \left(z^t - \eta_t \nabla f_{S_t}(x^t)\right) \\
&\overset{(11)}{=} \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} \left(x^t + \lambda_t(x^t - x^{t-1}) - \eta_t \nabla f_{S_t}(x^t)\right) \\
&= x^t - \frac{\eta_t}{\lambda_{t+1}+1} \nabla f_{S_t}(x^t) + \frac{\lambda_t}{\lambda_{+t+1}+1}(x^t - x^{t-1}) \\
&\overset{(9,10)}{=} x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta_t (x^t - x^{t-1}),
\end{aligned}$$

as wanted. □

The above proposition essentially states that momentum is a convex combination of SGD under a certain transformation. In this paper, we work only with a constant momentum coefficient, i.e., $\beta_t = \beta$. For this case, we also assume constant λ_t, and we have Proposition 2.1 and Corollary 2.2.

引理 C.6 (引理 13, (Jiang和Stich, 2023)). *If $x^2 \leq a(x+b)$ for $a \geq 0$ and $b \geq 0$, then itholds that $x \leq a + \sqrt{ab}$.*

引理 C.7 (引理 16, (Jiang和Stich,2023)). 假设每个 f_i 都是 L_i-光滑的，则*AdaSPS*的步长满足

$$\frac{1}{2cL_{\max}} \frac{1}{\sqrt{\sum_{s=0}^{t} f_{S_s}(x^s) - \ell^*_{S_s}}} \leq \eta_t \leq \frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2} \frac{1}{\sqrt{\sum_{s=0}^{t} f_{S_s}(x^s) - \ell^*_{S_s}}}, \tag{6}$$

其中 $L_{\max} = \max_{i\in[n]}\{L_i\}$.

C.3 迭代移动平均

如主论文所述，SHB方法的迭代等于 2.3 方法。以下命题来自 Sebbouh等人(2021)。我们包含证明以供完整性。

命题 C.8 (命题 1.6, (Sebbouh等人, 2021)). 令 $\eta_t > 0$ 和 $\lambda_t \geq 0$。考虑由以下更新规则给出的迭代移动平均（*IMA*）方法:

$$z^{t+1} = z^t - \eta_t \nabla f_{S_t}(x^t) \tag{7}$$

$$x^{t+1} = \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1}, \tag{8}$$

在 $z^0 = x^0$ 处。如果对于任意 $t \in \mathbb{N}$，以下等式成立

$$1 + \lambda_{t+1} = \frac{\lambda_t}{\beta_t} \tag{9}$$

$$\eta_t = (1+\lambda_{t+1})\gamma_t, \tag{10}$$

则 x_t 的*2.3* 方法迭代等于 x_t 由*SHB* 方法产生的迭代 .

证明。我们将从 2.3 更新规则开始，并证明其 (x^t) 迭代满足 SHB 更新规则。假设我们已经有了 (z^{t+1}) 和 (x^{t+1}) 迭代以及 (λ_t) 和 (η_t) 序列。使用等式(9)和(10)定义 (β_t) 和 (γ_t) 序列。我们将证明 (x^t) 迭代满足 SHB 更新规则。在等式(8)中求解 z^{t+1}，我们得到

$$z^{t+1} := x^{t+1} + \lambda_{t+1}(x^{t+1} - x^t) \tag{11}$$

现在将等式(7) 和 (11) 代入等式(8) 中，我们有

$$\begin{aligned}
x^{t+1} &= \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1} \\
&\overset{(7)}{=} \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} \left(z^t - \eta_t \nabla f_{S_t}(x^t)\right) \\
&\overset{(11)}{=} \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} \left(x^t + \lambda_t(x^t - x^{t-1}) - \eta_t \nabla f_{S_t}(x^t)\right) \\
&= x^t - \frac{\eta_t}{\lambda_{t+1}+1} \nabla f_{S_t}(x^t) + \frac{\lambda_t}{\lambda_{+t+1}+1}(x^t - x^{t-1}) \\
&\overset{(9,10)}{=} x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta_t (x^t - x^{t-1}),
\end{aligned}$$

如所求。 □

上述命题本质上表明，动量是在某种变换下随机梯度下降的凸组合。在本文中，我们仅使用常数动量系数，即$\beta_t = \beta$。对于这种情况，我们还假设常数 λ_t，并得到命题2.1 和推论2.2。

Proof of Proposition 2.1. Setting $\lambda_t = \lambda$ in Equation (9) we have

$$1 + \lambda = \frac{\lambda}{\beta_t} \Rightarrow \beta_t = \frac{\lambda}{1+\lambda} =: \beta.$$

Then solving for λ we get $\lambda = \frac{\beta}{1-\beta}$. Hence

$$\begin{aligned}\gamma_t &= \frac{\eta_t}{1+\lambda} \\ &= \frac{\eta_t}{1+\frac{\beta}{1-\beta}} \\ &= (1-\beta)\eta_t\end{aligned}$$

□

Proof of Corollary 2.2.

- Let $\eta_t = \text{SPS}_{\max}$. Then

$$\begin{aligned}\gamma_t &= (1-\beta)\eta_t \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \eta_b\right\} \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}.\end{aligned}$$

  Here $\gamma_b = \eta_b$.

- Let $\eta_t = \text{DecSPS}$. Then

$$\begin{aligned}\gamma_t &= (1-\beta)\eta_t \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\eta_{t-1}c_{t-1}}{c_t}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{(1-\beta)\eta_{t-1}c_{t-1}}{c_t}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}c_{t-1}}{c_t}\right\},\end{aligned}$$

  since $\gamma_{t-1} = (1-\beta)\eta_{t-1}$ from Proposition 2.1.

- Let $\eta_t = \text{AdaSPS}$. Then

$$\begin{aligned}\gamma_t &= (1-\beta)\eta_t \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \eta_{t-1}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, (1-\beta)\eta_{t-1}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}.\end{aligned}$$

□

C.4 Connection of MomSPS$_{\max}$ and SPS$_{\max}$

In this section, we explore the connection between the MomSPS$_{\max}$ and the SPS$_{\max}$ step-sizes. In particular, one can view the MomSPS$_{\max}$ step-size as the SPS$_{\max}$ step-size with $c = 1/(1-$

命题2.1的证明。 将 $\lambda_t = \lambda$ 代入方程（9）中，我们得到

$$1 + \lambda = \frac{\lambda}{\beta_t} \Rightarrow \beta_t = \frac{\lambda}{1+\lambda} =: \beta.$$

然后求解 λ，我们得到 $\lambda = \frac{\beta}{1-\beta}$。因此

$$\begin{aligned}\gamma_t &= \frac{\eta_t}{1+\lambda} \\ &= \frac{\eta_t}{1+\frac{\beta}{1-\beta}} \\ &= (1-\beta)\eta_t\end{aligned}$$

□

推论 2.2 的证明2.2。

- 令 $\eta_t = \text{SPS}_{\max}$。然后

$$\begin{aligned}\gamma_t &= (1-\beta)\eta_t \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \eta_b\right\} \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}.\end{aligned}$$

  这里 $\gamma_b = \eta_b$。

- 令 $\eta_t = \text{DecSPS}$。然后

$$\begin{aligned}\gamma_t &= (1-\beta)\eta_t \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\eta_{t-1}c_{t-1}}{c_t}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{(1-\beta)\eta_{t-1}c_{t-1}}{c_t}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c_t\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}c_{t-1}}{c_t}\right\},\end{aligned}$$

  由于 $\gamma_{t-1} = (1-\beta)\eta_{t-1}$ 根据命题 2.1。

- 令 $\eta_t = \text{AdaSPS}$。然后

$$\begin{aligned}\gamma_t &= (1-\beta)\eta_t \\ &= (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \eta_{t-1}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, (1-\beta)\eta_{t-1}\right\} \\ &= \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}.\end{aligned}$$

□

C.4 MomSPS$_{\max}$ 与 SPS$_{\max}$的连接

在本节中，我们探讨了 MomSPS$_{\max}$ 与 SPS$_{\max}$ 步长的关系。特别是，可以将 MomSPS$_{\max}$ 步长视为 SPS$_{\max}$ 步长，其具有 $c = 1/(1-$

β). Here, we explain why these are two different step-sizes. Recall from Proposition 2.1 and Corollary 2.2 that our proposed step-sizes for the SHB setting are scaled versions of Polyak step-sizes for SGD. In particular, let's assume that

$$\gamma_t^{SPSmax} = \min\left\{\frac{f_{S_t}(x^t) - f_{S_t}^*}{c_{SPSmax}\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b^{SPSmax}\right\}$$

and

$$\gamma_t^{MomSPSmax} = (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - f_{S_t}^*}{c_{MomSPSmax}\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b^{MomSPSmax}\right\}$$

so we have $\gamma_{SPSmax}^t = \gamma_{MomSPSmax}^t$ if and only if

$$\frac{1-\beta}{c_{MomSPSmax}} = \frac{1}{c_{SPSmax}} \text{ and } \gamma_b^{SPSmax} = (1-\beta)\gamma_b^{MomSPSmax}.$$

Without loss of generalization, we can assume that $c_{MomSPSmax} = 1$ (this is also what Theorem 3.2 guarantees). In this case we get $\beta = 1 - \frac{1}{c_{SPSmax}}$. Hence, we can restate Theorem 3.2 as follows:

Theorem C.9. *Assume that each f_i is convex and L_i-smooth. Then, the iterates of SHB with $SPS_{\max}$ with $c \in \left[1, \frac{2\gamma_b - 2\alpha}{2\gamma_b - \alpha}\right)$ and $\beta = 1 - \frac{1}{c}$ where $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ and $L_{\max} = \max_{i\in[n]}\{L_i\}$, converge as*

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{C_1\|x^0 - x^*\|^2}{T} + C_2\sigma^2,$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$ is the Cesaro average and the constants $C_1 = \frac{1-\beta}{\alpha\beta+\alpha-2\beta\gamma_b}$ and $C_2 = \frac{2\gamma_b - \alpha\beta - \alpha}{\alpha\beta+\alpha-2\beta\gamma_b}$.

This means that $\text{MomSPS}_{\max}$ can be viewed as a special case of $\text{SPS}_{\max}$. However, note that the update rules are different since in SHB we have the extra term $+\beta(x^t - x^{t-1})$ so a different analysis is needed.

The same line of thought can be applied to other works related to HB. For example, in Ghadimi et al. (2015), the first global guarantees for HB were established. The authors show that for a convex and L-smooth function f HB with any $\beta \in [0,1)$ and $0 < \gamma < \frac{2(1-\beta)}{L}$ converges with a rate of $O(1/T)$. Also, recall that GD for convex and L-smooth function converges for $0 < \gamma < \frac{2}{L}$ with rate $O(1/T)$. Hence, we can restate the result of Ghadimi et al. (2015) as follows: Let f be a convex and L-smooth function. Then HB with $\gamma = \frac{c}{L}$, where $c \in (0,2)$, and $\beta = 1 - \frac{c}{2}$ converges with rate $O(1/T)$.

Similarly, for SGD in the convex and smooth setting, we are guaranteed convergence $O(1/T)$ to a neighborhood of the solution when $\gamma \leq \frac{1}{L}$. In Liu et al. (2020), the authors guarantee that SHB with any $\beta \in (0,1)$ and $\gamma \leq \frac{(1-\beta)^2}{L}\min\left\{\frac{1}{4-\beta+\beta^2}, \frac{1}{2\sqrt{2\beta^2+2\beta}}\right\}$ converges with rate $O(1/T)$ to a neighborhood. This can be restated as follows: SHB with $\gamma = \frac{c}{L}$, where $c \in (0,1)$, and β given by the solution of $c = \min\left\{\frac{(1-\beta)^2}{4-\beta+\beta^2}, \frac{(1-\beta)^2}{2\sqrt{2\beta^2+2\beta}}\right\}$, converges with rate $O(1/T)$ to a neighborhood.

β)。在这里，我们解释为什么这些是两种不同的步长。回顾命题 2.1 和推论 2.2 ，我们为SHB设置提出的步长是 SGD 的 Polyak 步长的缩放版本。特别是，让我们假设

$$\gamma_t^{SPSmax} = \min\left\{\frac{f_{S_t}(x^t) - f_{S_t}^*}{c_{SPSmax}\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b^{SPSmax}\right\}$$

and

$$\gamma_t^{MomSPSmax} = (1-\beta)\min\left\{\frac{f_{S_t}(x^t) - f_{S_t}^*}{c_{MomSPSmax}\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b^{MomSPSmax}\right\}$$

所以我们有 $\gamma_{SPSmax}^t = \gamma_{MomSPSmax}^t$ 当且仅当

$$\frac{1-\beta}{c_{MomSPSmax}} = \frac{1}{c_{SPSmax}} \text{ and } \gamma_b^{SPSmax} = (1-\beta)\gamma_b^{MomSPSmax}.$$

无需丧失一般性，我们可以假设 $c_{MomSPSmax} = 1$ (这也是定理3.2 所保证的)。在这种情况下我们得到 $\beta = 1 - \frac{1}{c_{SPSmax}}$。因此，我们可以重新表述定理3.2 如下：

定理 C.9。 假设每个 f_i 是凸的并且 L_i-光滑的。那么，*SHB* 的迭代点$_{\max}$ 与 $c \in \left[1, \frac{2\gamma_b - 2\alpha}{2\gamma_b - \alpha}\right)$ 和 $\beta = 1 - \frac{1}{c}$，其中 $\alpha = \min\left\{\frac{1}{2L_{\max}}, \gamma_b\right\}$ 和 $L_{\max} = \max_{i\in[n]}\{L_i\}$，收敛为

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{C_1\|x^0 - x^*\|^2}{T} + C_2\sigma^2,$$

其中 $x^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$ 是*Cesaro*平均值，而常数是 $C_1 = \frac{1-\beta}{\alpha\beta+\alpha-2\beta\gamma_b}$ 和 $C_2 = \frac{2\gamma_b - \alpha\beta - \alpha}{\alpha\beta+\alpha-2\beta\gamma_b}$。

这意味着 $\text{MomSPS}_{\max}$ 可以看作是SPS的一种特殊情况$_{\max}$。但是请注意，更新规则是不同的，因为在 SHB 中我们有额外的项 $+\beta(x^t - x^{t-1})$，因此需要不同的分析。

同样的思路可以应用于其他与哈密顿方法（HB）相关的研究。例如，在Ghadimi等人（2015年）的研究中，首次建立了HB的全局收敛保证。作者表明，对于凸且 L-光滑的函数 f HB，在任意 $\beta \in [0,1)$和 $0 < \gamma < {}^{2}(1-\beta)$下，$L$ 以速率$O(1/T)$收敛。此外，请回忆，对于凸且 L-光滑函数，梯度下降（GD）在 $0 < \gamma < \frac{2}{L}$ 下以速率$O(1/T)$收敛。因此，我们可以将Ghadimi等人（2015年）的结果重新表述如下：设 f 是一个凸且 L-光滑的函数。那么，哈密顿方法（HB）在 $\gamma = \frac{c}{L}$ 下收敛，其中 $c \in (0,2)$，并且 $\beta = 1 - \frac{c}{2}$ 以速率$O(1/T)$收敛。

类似地，对于凸和光滑设置中的随机梯度下降，当满足 $\gamma \leq \frac{1}{L}$ 时，我们保证收敛到解的邻域 $O(1/T)$。在<样式 id='4'>刘等人</样式>(<样式 id='6'>2020</样式>)中，作者保证<样式 id='8'>SHB</样式>对于任何 $\beta \in$ (0<样式 id='11'>, </样式>1)和 $\gamma \leq$ <样式 id='14'>(</样式>$_{1-\beta}$<样式 id='17'>)</样式>$_{L}$ $^{2}\min\left\{\frac{1}{4-\beta+\beta^2}\right.$ <样式 id='21'>,</样式> $\left.\frac{1}{2\sqrt{2\beta^2+2\beta}}\right\}$ 以速率 $O(1/T)$收敛到邻域。这可以重新表述如下：<样式 id='26'>SHB </样式>对于 $\gamma = \frac{c}{L}$，其中 $c \in$ (0<样式 id='30'>, </样式>1)，并且 β 由 $c = \min\left\{\right.$ <样式 id='35'>(</样式> $\frac{1-\beta^2}{4-\beta+\beta^2}$ <样式 id='38'>)</样式><样式 id='40'>, </样式><样式 id='42'>(</样式> $\left.\frac{1-\beta^2}{2\sqrt{2\beta^2+2\beta}}\right\}$ <样式 id='45'>)</样式>的解给出，以速率 $O(1/T)$收敛到邻域。

D PROOFS OF THE MAIN RESULTS

In this section we present the proofs of the main theoretical results presented in the main paper, i.e., the convergence analysis of SHB with $\text{MomSPS}_{\max}$, MomDecSPS and MomAdaSPS for convex and smooth functions f_i and f of Problem 1. The idea for all the proofs is that we firstly show the corresponding result in the 2.3 setting, and then we transfer it to the SHB setting with the help of Proposition 2.1.

Compared to the analysis of SGD with SPS, the SHB update rule requires taking into consideration extra terms related to the previous iterate x^{t-1}. This needs new algebraic tricks in order to use convexity, and after that, one needs to deal with the two quantities $f(x^t)$ and $f(x^{t-1})$ at the same time. We deal with this using the 2.3 framework, similarly with Sebbouh et al. (2021). However, we highlight that we finish our proofs with a different approach. More specifically, in the proof of Theorem G.1 in Sebbouh et al. (2021) the authors choose the momentum coefficients λ_t in such a way that the main sum telescopes and the final result only has the expectation of the last iterate. In our proof, we make no such simplifications, and instead, we use Jensen's inequality to finish. This approach allows us to have more freedom in the selection of λ_t. This is more prominent in the proofs of the decreasing variants.

Compared to the classical analysis of constant step-size SHB, the use of SPS requires an adaptive step-size that uses the loss and stochastic gradient estimates at an iterate, resulting in correlations of the step-size with the gradient, i.e. we might have $\mathbb{E}[\gamma_t \nabla f_i(x^t)] \neq \gamma_t \mathbb{E}[\nabla f_i(x^t)]$. One of the technical challenges in the proofs is to carefully analyze the SHB iterates, taking these correlations into account. Moreover, since we try to be adaptive to the Lipschitz constant, we can not use any standard descent lemmas (implied by the smoothness and the SHB update) that are used in the classical analysis of SGD and constant step-size SHB. This makes the convex proof more challenging than the standard analysis of SHB.

D.1 PROOF OF THEOREM 3.2

First, we state and prove the corresponding theorem for $\text{SPS}_{\max}$ in the 2.3 setting.

Theorem D.1. *Assume that each f_i is convex and L_i-smooth. Then, the iterates of 2.3 with*

$$\eta_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{\|\nabla f_{S_t}(x^t)\|^2}, \eta_b\right\} \text{ and } \lambda_t = \lambda \in \left[0, \frac{\alpha}{2(\eta_b - \alpha)}\right),$$

where $\alpha = \min\left\{\frac{1}{2L_{\max}}, \eta_b\right\}$ and $L_{\max} = \max_{i\in[n]}\{L_i\}$, converge as

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\|x^0 - x^*\|^2}{(2\alpha\lambda + \alpha - 2\eta_b\lambda)T} + \frac{(2\eta_b + 2\eta_b\lambda - 2\alpha\lambda - \alpha)\sigma^2}{2\alpha\lambda + \alpha - 2\eta_b\lambda},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$ and $\sigma^2 = \mathbb{E}[f_i(x^*) - f_i^*]$.

Proof. We have

$$\begin{aligned}
\|z^{t+1} - x^*\|^2 &= \|z^t - x^*\|^2 - 2\eta_t\langle\nabla f_{S_t}(x^t), z^t - x^*\rangle + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2 \\
&\overset{(11)}{=} \|z^t - x^*\|^2 - 2\eta_t\langle\nabla f_{S_t}(x^t), x^t - x^*\rangle - 2\eta_t\lambda_t\langle\nabla f_{S_t}(x^t), x^t - x^{t-1}\rangle \\
&\quad + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2 \\
&\overset{\text{convexity}}{\leq} \|z^t - x^*\|^2 - 2\eta_t[f_{S_t}(x^t) - f_{S_t}(x^*)] - 2\eta_t\lambda_t[f_{S_t}(x^t) - f_{S_t}(x^{t-1})] \\
&\quad + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2 \\
&\overset{(2)}{\leq} \|z^t - x^*\|^2 - 2\eta_t[f_{S_t}(x^t) - f_{S_t}(x^*)] - 2\eta_t\lambda_t[f_{S_t}(x^t) - f_{S_t}(x^{t-1})] \\
&\quad + \eta_t[f_{S_t}(x^t) - \ell^*_{S_t}] \\
&= \|z^t - x^*\|^2 - 2\eta_t[f_{S_t}(x^t) - \ell^*_{S_t}] + 2\eta_t[f_{S_t}(x^*) - \ell^*_{S_t}] - 2\eta_t\lambda_t[f_{S_t}(x^t) - \ell^*_{S_t}]
\end{aligned}$$

D 主要结果的证明

<样式 id='1'>在本节中，我们介绍了主要论文中提出的主要理论结果的证明，即凸和光滑函数</样式><样式 id='3'>SHB </样式><样式 id='5'>与</样式><样式 id='7'>MomSPS</样式>$_{\max}$<样式 id='10'>，</样式><样式 id='12'>MomDecSPS </样式><样式 id='14'>和</样式><样式 id='16'>MomAdaSPS</样式><样式 id='18'>对于问题</样式><样式 id='20'>1</样式><样式 id='22'>的收敛性分析。所有证明的想法是，我们首先在</样式><样式 id='24'>2.3</样式><样式 id='26'>设置中展示相应结果，然后借助命题</样式><样式 id='28'>2.1</样式><样式 id='30'>将其转移到</样式><样式 id='32'>SHB</样式><样式 id='34'>设置中。

与SGD结合SPS的分析相比，SHB 更新规则需要考虑与先前迭代相关的额外项 x^{t-1}。这需要新的代数技巧才能利用凸性，之后，人们需要同时处理两个量 $f(x^t)$ 和 $f(x^{t-1})$ 。我们使用2.3 框架来处理这个问题，与Sebbouh等人(2021)类似。然而，我们强调我们用不同的方法完成了证明。更具体地说，在Sebbouh等人(2021)中的定理G.1的证明中，作者选择动量系数 λ_t ，使得主和式望远镜，最终结果只有最后迭代的期望。在我们的证明中，我们没有进行这种简化，而是使用Jensen不等式来完成。这种方法使我们能够在选择 λ_t方面拥有更大的自由度。这在递减变体的证明中更为明显。

与经典固定步长分析相比SHB，使用SPS需要自适应步长，该步长利用迭代处的损失和随机梯度估计，导致步长与梯度存在相关性，即我们可能有 $\mathbb{E}[\gamma_t \nabla f_i(x^t)] \neq \gamma_t \mathbb{E}[\nabla f_i(x^t)]$。在证明中的一个技术挑战是仔细分析SHB迭代，同时考虑这些相关性。此外，由于我们试图适应利普希茨常数，因此不能使用任何标准下降引理（这些引理由光滑性和SHB更新所隐含），这些引理在经典随机梯度下降和固定步长SHB分析中使用。这使得凸证明比标准SHB分析更具挑战性。

D.1 证明THEOREM 3.2

首先，我们陈述并证明SPS在$_{\max}$ 2.3 设置下的相应定理。

定理 D.1。 假设每个 f_i 是凸的并且 L_i-光滑的。那么，*2.3* 的迭代

$$\eta_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{\|\nabla f_{S_t}(x^t)\|^2}, \eta_b\right\} \text{ and } \lambda_t = \lambda \in \left[0, \frac{\alpha}{2(\eta_b - \alpha)}\right),$$

其中 $\alpha =$最小$\left\{\frac{1}{2L_{\max}}, \eta_b\right\}$和 $L_{\max} =$最大$_{i\in[n]}\{L_i\}$收敛为

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\|x^0 - x^*\|^2}{(2\alpha\lambda + \alpha - 2\eta_b\lambda)T} + \frac{(2\eta_b + 2\eta_b\lambda - 2\alpha\lambda - \alpha)\sigma^2}{2\alpha\lambda + \alpha - 2\eta_b\lambda},$$

在 $x^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$和 $\sigma^2 = \mathbb{E}[f_i(x^*) - f_i^*]$处。

证明。我们有

$$\begin{aligned}
\|z^{t+1} - x^*\|^2 &= \|z^t - x^*\|^2 - 2\eta_t\langle\nabla f_{S_t}(x^t), z^t - x^*\rangle + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2 \\
&\overset{(11)}{=} \|z^t - x^*\|^2 - 2\eta_t\langle\nabla f_{S_t}(x^t), x^t - x^*\rangle - 2\eta_t\lambda_t\langle\nabla f_{S_t}(x^t), x^t - x^{t-1}\rangle \\
&\quad + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2 \\
&\overset{\text{convexity}}{\leq} \|z^t - x^*\|^2 - 2\eta_t[f_{S_t}(x^t) - f_{S_t}(x^*)] - 2\eta_t\lambda_t[f_{S_t}(x^t) - f_{S_t}(x^{t-1})] \\
&\quad + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2 \\
&\overset{(2)}{\leq} \|z^t - x^*\|^2 - 2\eta_t[f_{S_t}(x^t) - f_{S_t}(x^*)] - 2\eta_t\lambda_t[f_{S_t}(x^t) - f_{S_t}(x^{t-1})] \\
&\quad + \eta_t[f_{S_t}(x^t) - \ell^*_{S_t}] \\
&= \|z^t - x^*\|^2 - 2\eta_t[f_{S_t}(x^t) - \ell^*_{S_t}] + 2\eta_t[f_{S_t}(x^*) - \ell^*_{S_t}] - 2\eta_t\lambda_t[f_{S_t}(x^t) - \ell^*_{S_t}]
\end{aligned}$$

$$
\begin{aligned}
&+ 2\eta_t\lambda_t[f_{S_t}(x^{t-1}) - \ell^*_{S_t}] + \eta_t[f_{S_t}(x^t) - \ell^*_{S_t}] \\
&= \|z^t - x^*\|^2 - 2\eta_t\left(1 + \lambda_t - \frac{1}{2}\right)[f_{S_t}(x^t) - \ell^*_{S_t}] + 2\eta_t\lambda_t[f_{S_t}(x^{t-1}) - \ell^*_{S_t}] \\
&\quad + 2\eta_t[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&\overset{(2)}{\leq} \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f_{S_t}(x^t) - \ell^*_{S_t}] + 2\eta_b\lambda_t[f_{S_t}(x^{t-1}) - \ell^*_{S_t}] \\
&\quad + 2\eta_b[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&\leq \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f_{S_t}(x^t) - f_{S_t}(x^*)] - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&\quad + 2\eta_b\lambda_t[f_{S_t}(x^{t-1}) - f_{S_t}(x^*)] + 2\eta_b\lambda_t[f_{S_t}(x^*) - \ell^*_{S_t}] + 2\eta_b[f_{S_t}(x^*) - \ell^*_{S_t}].
\end{aligned}
$$

Now take expectation conditional on x^t to get

$$
\begin{aligned}
\mathbb{E}_{S_t}\|z^{t+1} - x^*\|^2 &\leq \mathbb{E}_{S_t}\|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}_{S_t}[f_{S_t}(x^t) - f_{S_t}(x^*)] \\
&\quad - 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] + 2\eta_b\lambda_t\,\mathbb{E}_{S_t}[f_{S_t}(x^{t-1}) - f_{S_t}(x^*)] \\
&\quad + 2\eta_b\lambda_t\,\mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] + 2\eta_b\,\mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&= \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f(x^t) - f(x^*)] - 2\alpha\left(\lambda_t + \frac{1}{2}\right)\sigma^2 \\
&\quad + 2\eta_b\lambda_t[f(x^{t-1}) - f(x^*)] + 2\eta_b\lambda_t\sigma^2 + 2\eta_b\sigma^2 \\
&= \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f(x^t) - f(x^*)] + 2\eta_b\lambda_t[f(x^{t-1}) - f(x^*)] \\
&\quad + 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (12)
\end{aligned}
$$

Note that in eq. (12) the coefficient $2\left(\eta_b\left(1+\lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)$ of σ^2 is positive since $\eta_b \geq \alpha$. Taking expectation again and using the tower property we have

$$
\begin{aligned}
&2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)] \\
&\leq \mathbb{E}\|z^t - x^*\|^2 - \mathbb{E}\|z^{t+1} - x^*\|^2 + 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (13)
\end{aligned}
$$

Summing eq. (13) for $t = 0, \ldots, T-1$ and telescoping we get

$$
\begin{aligned}
&\sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)]\right] \\
&\leq \sum_{t=0}^{T-1}\left[\mathbb{E}\|z^t - x^*\|^2 - \mathbb{E}\|z^{t+1} - x^*\|^2\right] + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2 \\
&= \mathbb{E}\|z^0 - x^*\|^2 - \mathbb{E}\|z^T - x^*\|^2 + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2 \\
&\overset{z^0 = x^0}{\leq} \|x^0 - x^*\|^2 + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (14)
\end{aligned}
$$

The LHS of eq. (14) can be bounded as follows (using $x^{-1} = x^0$)

$$
\sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)]\right]
$$

$$
\begin{aligned}
&+ 2\eta_t\lambda_t[f_{S_t}(x^{t-1}) - \ell^*_{S_t}] + \eta_t[f_{S_t}(x^t) - \ell^*_{S_t}] \\
&= \|z^t - x^*\|^2 - 2\eta_t\left(1 + \lambda_t - \frac{1}{2}\right)[f_{S_t}(x^t) - \ell^*_{S_t}] + 2\eta_t\lambda_t[f_{S_t}(x^{t-1}) - \ell^*_{S_t}] \\
&\quad + 2\eta_t[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&\overset{(2)}{\leq} \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f_{S_t}(x^t) - \ell^*_{S_t}] + 2\eta_b\lambda_t[f_{S_t}(x^{t-1}) - \ell^*_{S_t}] \\
&\quad + 2\eta_b[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&\leq \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f_{S_t}(x^t) - f_{S_t}(x^*)] - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&\quad + 2\eta_b\lambda_t[f_{S_t}(x^{t-1}) - f_{S_t}(x^*)] + 2\eta_b\lambda_t[f_{S_t}(x^*) - \ell^*_{S_t}] + 2\eta_b[f_{S_t}(x^*) - \ell^*_{S_t}].
\end{aligned}
$$

现在在 x^t 的条件下取期望得到

$$
\begin{aligned}
\mathbb{E}_{S_t}\|z^{t+1} - x^*\|^2 &\leq \mathbb{E}_{S_t}\|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}_{S_t}[f_{S_t}(x^t) - f_{S_t}(x^*)] \\
&\quad - 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] + 2\eta_b\lambda_t\,\mathbb{E}_{S_t}[f_{S_t}(x^{t-1}) - f_{S_t}(x^*)] \\
&\quad + 2\eta_b\lambda_t\,\mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] + 2\eta_b\,\mathbb{E}_{S_t}[f_{S_t}(x^*) - \ell^*_{S_t}] \\
&= \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f(x^t) - f(x^*)] - 2\alpha\left(\lambda_t + \frac{1}{2}\right)\sigma^2 \\
&\quad + 2\eta_b\lambda_t[f(x^{t-1}) - f(x^*)] + 2\eta_b\lambda_t\sigma^2 + 2\eta_b\sigma^2 \\
&= \|z^t - x^*\|^2 - 2\alpha\left(\lambda_t + \frac{1}{2}\right)[f(x^t) - f(x^*)] + 2\eta_b\lambda_t[f(x^{t-1}) - f(x^*)] \\
&\quad + 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (12)
\end{aligned}
$$

注意在等式（12）中，系数 $2\left(\eta_b\left(1+\lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)$ 的 σ^2 是正的，因为 $\eta_b \geq \alpha$。再次取期望并使用塔性质，我们得到

$$
\begin{aligned}
&2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)] \\
&\leq \mathbb{E}\|z^t - x^*\|^2 - \mathbb{E}\|z^{t+1} - x^*\|^2 + 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (13)
\end{aligned}
$$

对等式（13）从 $t = 0, \ldots, T-1$ 求和并利用望远镜求和公式，我们得到

$$
\begin{aligned}
&\sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)]\right] \\
&\leq \sum_{t=0}^{T-1}\left[\mathbb{E}\|z^t - x^*\|^2 - \mathbb{E}\|z^{t+1} - x^*\|^2\right] + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2 \\
&= \mathbb{E}\|z^0 - x^*\|^2 - \mathbb{E}\|z^T - x^*\|^2 + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2 \\
&\overset{z^0 = x^0}{\leq} \|x^0 - x^*\|^2 + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (14)
\end{aligned}
$$

方程（14）的左侧可以如下界定（使用 $x^{-1} = x^0$）

$$
\sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)]\right]
$$

$$
\begin{aligned}
&= \sum_{t=0}^{T-1} 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=0}^{T-1} 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)] \\
&= \sum_{t=0}^{T-1} 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=0}^{T-2} 2\eta_b\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\alpha\left(\lambda_{T-1} + \frac{1}{2}\right)\mathbb{E}[f(x^{T-1}) - f(x^*)] + \sum_{t=0}^{T-2} 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] \\
&\quad - \sum_{t=0}^{T-2} 2\eta_b\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\alpha\left(\lambda_{T-1} + \frac{1}{2}\right)\mathbb{E}[f(x^{T-1}) - f(x^*)] + \sum_{t=0}^{T-2}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)] \\
&\geq \left[2\alpha\left(\lambda_{T-1} + \frac{1}{2}\right) - 2\eta_b\lambda_T\right]\mathbb{E}[f(x^{T-1}) - f(x^*)] \\
&\quad + \sum_{t=0}^{T-2}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)] \\
&= \sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)]. \qquad (15)
\end{aligned}
$$

Combining eq. (14) and eq. (15) we have

$$
\begin{aligned}
&\sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)] \\
&\leq \|x^0 - x^*\|^2 + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (16)
\end{aligned}
$$

In eq. (16), setting $\lambda_t = \lambda, \forall t > 0$ with $0 \leq \lambda < \frac{\alpha}{2(\eta_b - \alpha)}$ (in order to ensure that quantity $2\alpha\left(\lambda + \frac{1}{2}\right) - 2\eta_b\lambda$ in the LHS is positive) we get

$$
\left[2\alpha\left(\lambda + \frac{1}{2}\right) - 2\eta_b\lambda\right]\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \leq \|x^0 - x^*\|^2 + 2T\left(\eta_b\left(1 + \lambda\right) - \alpha\left(\lambda + \frac{1}{2}\right)\right)\sigma^2. \qquad (17)
$$

Now dividing eq. (17) by $T\left[2\alpha\left(\lambda + \frac{1}{2}\right) - 2\eta_b\lambda\right]$ and using Jensen's inequality we get

$$
\begin{aligned}
\mathbb{E}[f(\overline{x}^T) - f(x^*)] &\leq \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \\
&\leq \frac{\|x^0 - x^*\|^2 + T\left(2\eta_b + 2\eta_b\lambda - 2\alpha\lambda - \alpha\right)\sigma^2}{(2\alpha\lambda + \alpha - 2\eta_b\lambda)T} \\
&= \frac{\|x^0 - x^*\|^2}{(2\alpha\lambda + \alpha - 2\eta_b\lambda)T} + \frac{2\eta_b + 2\eta_b\lambda - 2\alpha\lambda - \alpha}{2\alpha\lambda + \alpha - 2\eta_b\lambda}\sigma^2,
\end{aligned}
$$

as wanted. □

Now the proof of Theorem 3.2 follows immediately from Theorem D.1 setting $\lambda = \frac{\beta}{1-\beta}$ (from Proposition 2.1) and $\gamma_b = \eta_b$ (from Corollary 2.2). The bound on $\beta = \frac{\lambda}{1+\lambda}$ follows from the bound on λ and from the fact that the function $f(x) = \frac{x}{1+x}$ is strictly increasing for $x \geq 0$.

$$
\begin{aligned}
&= \sum_{t=0}^{T-1} 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=0}^{T-1} 2\eta_b\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)] \\
&= \sum_{t=0}^{T-1} 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=0}^{T-2} 2\eta_b\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\alpha\left(\lambda_{T-1} + \frac{1}{2}\right)\mathbb{E}[f(x^{T-1}) - f(x^*)] + \sum_{t=0}^{T-2} 2\alpha\left(\lambda_t + \frac{1}{2}\right)\mathbb{E}[f(x^t) - f(x^*)] \\
&\quad - \sum_{t=0}^{T-2} 2\eta_b\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\alpha\left(\lambda_{T-1} + \frac{1}{2}\right)\mathbb{E}[f(x^{T-1}) - f(x^*)] + \sum_{t=0}^{T-2}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)] \\
&\geq \left[2\alpha\left(\lambda_{T-1} + \frac{1}{2}\right) - 2\eta_b\lambda_T\right]\mathbb{E}[f(x^{T-1}) - f(x^*)] \\
&\quad + \sum_{t=0}^{T-2}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)] \\
&= \sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)]. \qquad (15)
\end{aligned}
$$

结合方程（14）和方程（15），我们得到

$$
\begin{aligned}
&\sum_{t=0}^{T-1}\left[2\alpha\left(\lambda_t + \frac{1}{2}\right) - 2\eta_b\lambda_{t+1}\right]\mathbb{E}[f(x^t) - f(x^*)] \\
&\leq \|x^0 - x^*\|^2 + \sum_{t=0}^{T-1} 2\left(\eta_b\left(1 + \lambda_t\right) - \alpha\left(\lambda_t + \frac{1}{2}\right)\right)\sigma^2. \qquad (16)
\end{aligned}
$$

在公式（16）中，设置 $\lambda_t = \lambda, \forall t > 0$，并使用$0 \leq \lambda < \frac{\alpha}{2(\eta_b - \alpha)}$（为确保左侧的量$2\alpha\left(\lambda + \frac{1}{2}\right) - 2\eta_b\lambda$为正），我们得到

$$
\left[2\alpha\left(\lambda + \frac{1}{2}\right) - 2\eta_b\lambda\right]\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \leq \|x^0 - x^*\|^2 + 2T\left(\eta_b\left(1 + \lambda\right) - \alpha\left(\lambda + \frac{1}{2}\right)\right)\sigma^2. \qquad (17)
$$

现在将公式（17）除以$T\left[2\alpha\left(\lambda + \frac{1}{2}\right) - 2\eta_b\lambda\right]$，并使用Jensen不等式，我们得到

$$
\begin{aligned}
\mathbb{E}[f(\overline{x}^T) - f(x^*)] &\leq \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \\
&\leq \frac{\|x^0 - x^*\|^2 + T\left(2\eta_b + 2\eta_b\lambda - 2\alpha\lambda - \alpha\right)\sigma^2}{(2\alpha\lambda + \alpha - 2\eta_b\lambda)T} \\
&= \frac{\|x^0 - x^*\|^2}{(2\alpha\lambda + \alpha - 2\eta_b\lambda)T} + \frac{2\eta_b + 2\eta_b\lambda - 2\alpha\lambda - \alpha}{2\alpha\lambda + \alpha - 2\eta_b\lambda}\sigma^2,
\end{aligned}
$$

即所求。 □

现在，定理3.2 的证明直接来自定理D.1，设置$\lambda = \frac{\beta}{1-\beta}$ (来自命题2.1) 和 $\gamma_b = \eta_b$ (来自推论 2.2)。对 $\beta = \frac{\lambda}{1+\lambda}$ 的界限来自对 λ 的界限，并且来自函数 $f(x) = \frac{x}{1+x}$ 在 $x \geq 0$上是严格递增的。

D.2 PROOF OF THEOREM 3.6

Here we state and prove the corresponding theorem for DecSPS in the 2.3 setting.

Theorem D.2. *Assume that each f_i is convex and L_i-smooth. Then, the iterates of 2.3 with*

$$\eta_t = \frac{1}{c_t}\min\left\{\frac{f_{S_t}(x^t)-\ell^*_{S_t}}{\|\nabla f_{S_t}(x^t)\|^2}, c_{t-1}\eta_{t-1}\right\} \text{ and } \lambda_t \geq 0,$$

where (c_t) is an increasing sequences with $c_t \geq 1$, $c_{-1}=c_0$, $\eta_{-1}=\eta_b$ and (λ_t) is a decreasing sequence, converge as

$$\mathbb{E}[f(\overline{x}^T)-f(x^*)] \leq \frac{2\lambda_1[f(x^0)-f(x^*)]}{T} + \frac{c_{T-1}D_z^2}{T\alpha} + \frac{\sigma^2}{T}\sum_{t=0}^{T-1}\frac{1}{c_t},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1}x^t$, $\alpha = \min\left\{\frac{1}{2L}, \eta_b c_0\right\}$ and $D_z^2 = \max_{t\in[T]}\|z^t-x^*\|^2$.

Proof. We have

$$\|z^{t+1}-x^*\|^2 \overset{(7)}{=} \|z^t-x^*\|^2 - 2\eta_t\langle\nabla f_{S_t}(x^t), z^t-x^*\rangle + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2.$$

Rearranging and dividing by $\eta_t > 0$ we get

$$2\langle\nabla f_{S_t}(x^t), z^t-x^*\rangle \leq \frac{\|z^t-x^*\|^2}{\eta_t} - \frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \eta_t\|\nabla f_{S_t}(x^t)\|^2.$$

Summing the above for $t=0,\ldots,T-1$ and telescoping we have

$$\begin{aligned}
&2\sum_{t=0}^{T-1}\langle\nabla f_{S_t}(x^t), z^t-x^*\rangle \\
&\leq \sum_{t=0}^{T-1}\left[\frac{\|z^t-x^*\|^2}{\eta_t} - \frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \eta_t\|\nabla f_{S_t}(x^t)\|^2\right] \\
&= \sum_{t=0}^{T-1}\frac{\|z^t-x^*\|^2}{\eta_t} - \sum_{t=0}^{T-1}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=1}^{T-1}\frac{\|z^t-x^*\|^2}{\eta_t} - \sum_{t=0}^{T-1}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=0}^{T-2}\frac{\|z^{t+1}-x^*\|^2}{\eta_{t+1}} - \sum_{t=0}^{T-1}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=0}^{T-2}\frac{\|z^{t+1}-x^*\|^2}{\eta_{t+1}} - \sum_{t=0}^{T-2}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} - \frac{\|z^T-x^*\|^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=0}^{T-2}\left(\frac{1}{\eta_{t+1}}-\frac{1}{\eta_t}\right)\|z^{t+1}-x^*\|^2 - \frac{\|z^T-x^*\|^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&\leq \left(\frac{1}{\eta_0} + \sum_{t=0}^{T-2}\left[\frac{1}{\eta_{t+1}}-\frac{1}{\eta_t}\right]\right)D_z^2 + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2. \qquad (18)
\end{aligned}$$

Taking expectation in eq. (18) we have

$$2\sum_{t=0}^{T-1}\mathbb{E}\left[\langle\nabla f(x^t), z^t-x^*\rangle\right] \leq \mathbb{E}\left[\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2\right]. \qquad (19)$$

D.2 P证明T3.6

在这里，我们陈述并证明适用于DecSPS的相应定理，在 2.3 设置中。

定理D.2. 假设每个 f_i 是凸的且 L_i-光滑的。那么，*2.3* 的迭代 *h*

$$\eta_t = \frac{1}{c_t}\min\left\{\frac{f_{S_t}(x^t)-\ell^*_{S_t}}{\|\nabla f_{S_t}(x^t)\|^2}, c_{t-1}\eta_{t-1}\right\} \text{ and } \lambda_t \geq 0,$$

其中 (c_t) 是一个递增序列，包含 $c_t \geq 1$, $c_{-1}=c_0$, $\eta_{-1}=\eta_b$ ，而 (λ_t) 是一个递减序列，它们收敛为

$$\mathbb{E}[f(\overline{x}^T)-f(x^*)] \leq \frac{2\lambda_1[f(x^0)-f(x^*)]}{T} + \frac{c_{T-1}D_z^2}{T\alpha} + \frac{\sigma^2}{T}\sum_{t=0}^{T-1}\frac{1}{c_t},$$

其中 $\underline{x}^T = \frac{1}{\underline{T}}\sum_{t=0}^{T-1}x^t$, $\alpha = \min\left\{\frac{1}{2\underline{L}}, \eta_b c_0\right\}$ 且 $D_z^2 = \max_{t\in[T]}\|z^t-x^*\|^2$。

证明。 我们有

$$\|z^{t+1}-x^*\|^2 \overset{(7)}{=} \|z^t-x^*\|^2 - 2\eta_t\langle\nabla f_{S_t}(x^t), z^t-x^*\rangle + \eta_t^2\|\nabla f_{S_t}(x^t)\|^2.$$

重新排列并除以 $\eta_t > 0$ ，我们得到

$$2\langle\nabla f_{S_t}(x^t), z^t-x^*\rangle \leq \frac{\|z^t-x^*\|^2}{\eta_t} - \frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \eta_t\|\nabla f_{S_t}(x^t)\|^2.$$

将上述内容对 $t=0$, ..., $T-1$ 求和并利用望远镜求和公式，我们得到

$$\begin{aligned}
&2\sum_{t=0}^{T-1}\langle\nabla f_{S_t}(x^t), z^t-x^*\rangle \\
&\leq \sum_{t=0}^{T-1}\left[\frac{\|z^t-x^*\|^2}{\eta_t} - \frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \eta_t\|\nabla f_{S_t}(x^t)\|^2\right] \\
&= \sum_{t=0}^{T-1}\frac{\|z^t-x^*\|^2}{\eta_t} - \sum_{t=0}^{T-1}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=1}^{T-1}\frac{\|z^t-x^*\|^2}{\eta_t} - \sum_{t=0}^{T-1}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=0}^{T-2}\frac{\|z^{t+1}-x^*\|^2}{\eta_{t+1}} - \sum_{t=0}^{T-1}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=0}^{T-2}\frac{\|z^{t+1}-x^*\|^2}{\eta_{t+1}} - \sum_{t=0}^{T-2}\frac{\|z^{t+1}-x^*\|^2}{\eta_t} - \frac{\|z^T-x^*\|^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{\|z^0-x^*\|^2}{\eta_0} + \sum_{t=0}^{T-2}\left(\frac{1}{\eta_{t+1}}-\frac{1}{\eta_t}\right)\|z^{t+1}-x^*\|^2 - \frac{\|z^T-x^*\|^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&\leq \left(\frac{1}{\eta_0} + \sum_{t=0}^{T-2}\left[\frac{1}{\eta_{t+1}}-\frac{1}{\eta_t}\right]\right)D_z^2 + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \\
&= \frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2. \qquad (18)
\end{aligned}$$

在公式（18）中取期望我们有

$$2\sum_{t=0}^{T-1}\mathbb{E}\left[\langle\nabla f(x^t), z^t-x^*\rangle\right] \leq \mathbb{E}\left[\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2\right]. \qquad (19)$$

Now the LHS of eq. (19) can be bounded as follows

$$
\begin{aligned}
&2\sum_{t=0}^{T-1}\mathbb{E}\left[\langle\nabla f(x^t), z^t - x^*\rangle\right] \\
&\overset{(z^0=x^0)}{=} 2\,\mathbb{E}\left[\langle\nabla f(x^0), x^0 - x^*\rangle\right] + 2\sum_{t=1}^{T-1}\mathbb{E}\left[\langle\nabla f(x^t), z^t - x^*\rangle\right] \\
&\overset{(11)}{=} 2\,\mathbb{E}\left[\langle\nabla f(x^0), x^0 - x^*\rangle\right] + \sum_{t=1}^{T-1}\mathbb{E}\left[2\langle\nabla f(x^t), x^t - x^*\rangle + 2\lambda_t\langle\nabla f(x^t), x^t - x^{t-1}\rangle\right] \\
&\overset{\text{convexity}}{\geq} 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}\mathbb{E}\left[2(f(x^t) - f(x^*)) + 2\lambda_t(f(x^t) - f(x^{t-1}))\right] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}\left[2(1+\lambda_t)\,\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)]\right] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}2(1+\lambda_t)\,\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=1}^{T-1}2\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}2(1+\lambda_t)\,\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=0}^{T-2}2\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + 2\sum_{t=1}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] + \sum_{t=1}^{T-1}2\lambda_t\,\mathbb{E}[f(x^t) - f(x^*)] \\
&\quad - \sum_{t=0}^{T-2}2\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + 2\sum_{t=1}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] + 2\lambda_{T-1}\,\mathbb{E}[f(x^{T-1}) - f(x^*)] \\
&\quad + \sum_{t=1}^{T-2}2\lambda_t\,\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=1}^{T-2}2\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1\,\mathbb{E}[f(x^0) - f(x^*)] \\
&= 2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] + 2\lambda_{T-1}\,\mathbb{E}[f(x^{T-1}) - f(x^*)] \\
&\quad + 2\sum_{t=1}^{T-2}(\lambda_t - \lambda_{t+1})\,\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1\,\mathbb{E}[f(x^0) - f(x^*)] \\
&\geq 2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)], \qquad (20)
\end{aligned}
$$

using the fact that (λ_t) is decreasing and the fact that $\mathbb{E}[f(x^t) - f(x^*)] \geq 0$. Combining eq. (19) and eq. (20) we have

$$
2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \leq \mathbb{E}\left[\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\mathbb{E}\,\eta_t\|\nabla f_{S_t}(x^t)\|^2\right]. \qquad (21)
$$

Now for the RHS of eq. (21) we have

$$
\begin{aligned}
\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 &\overset{(3)}{\leq} \frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^t) - \ell_{S_t}^*] \\
&= \frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^t) - f_{S_t}(x^*)] + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^*) - \ell_{S_t}^*]
\end{aligned}
$$

现在公式（19）的左侧可以如下界定

$$
\begin{aligned}
&2\sum_{t=0}^{T-1}\mathbb{E}\left[\langle\nabla f(x^t), z^t - x^*\rangle\right] \\
&\overset{(z^0=x^0)}{=} 2\,\mathbb{E}\left[\langle\nabla f(x^0), x^0 - x^*\rangle\right] + 2\sum_{t=1}^{T-1}\mathbb{E}\left[\langle\nabla f(x^t), z^t - x^*\rangle\right] \\
&\overset{(11)}{=} 2\,\mathbb{E}\left[\langle\nabla f(x^0), x^0 - x^*\rangle\right] + \sum_{t=1}^{T-1}\mathbb{E}\left[2\langle\nabla f(x^t), x^t - x^*\rangle + 2\lambda_t\langle\nabla f(x^t), x^t - x^{t-1}\rangle\right] \\
&\overset{\text{convexity}}{\geq} 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}\mathbb{E}\left[2(f(x^t) - f(x^*)) + 2\lambda_t(f(x^t) - f(x^{t-1}))\right] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}\left[2(1+\lambda_t)\,\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)]\right] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}2(1+\lambda_t)\,\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=1}^{T-1}2\lambda_t\,\mathbb{E}[f(x^{t-1}) - f(x^*)] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + \sum_{t=1}^{T-1}2(1+\lambda_t)\,\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=0}^{T-2}2\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + 2\sum_{t=1}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] + \sum_{t=1}^{T-1}2\lambda_t\,\mathbb{E}[f(x^t) - f(x^*)] \\
&\quad - \sum_{t=0}^{T-2}2\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] \\
&= 2\,\mathbb{E}\left[f(x^0) - f(x^*)\right] + 2\sum_{t=1}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] + 2\lambda_{T-1}\,\mathbb{E}[f(x^{T-1}) - f(x^*)] \\
&\quad + \sum_{t=1}^{T-2}2\lambda_t\,\mathbb{E}[f(x^t) - f(x^*)] - \sum_{t=1}^{T-2}2\lambda_{t+1}\,\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1\,\mathbb{E}[f(x^0) - f(x^*)] \\
&= 2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] + 2\lambda_{T-1}\,\mathbb{E}[f(x^{T-1}) - f(x^*)] \\
&\quad + 2\sum_{t=1}^{T-2}(\lambda_t - \lambda_{t+1})\,\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1\,\mathbb{E}[f(x^0) - f(x^*)] \\
&\geq 2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)], \qquad (20)
\end{aligned}
$$

利用 (λ_t) 是递减的这一事实，以及 $\mathbb{E}[f(x^t) - f(x^*)] \geq 0$。结合等式（19）和等式（20），我们得到

$$
2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \leq \mathbb{E}\left[\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\mathbb{E}\,\eta_t\|\nabla f_{S_t}(x^t)\|^2\right]. \qquad (21)
$$

现在考虑等式（21）的右侧，我们有

$$
\begin{aligned}
\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 &\overset{(3)}{\leq} \frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^t) - \ell_{S_t}^*] \\
&= \frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^t) - f_{S_t}(x^*)] + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^*) - \ell_{S_t}^*]
\end{aligned}
$$

$$\leq \frac{c_{T-1}D_z^2}{\alpha} + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^t) - f_{S_t}(x^*)] + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^*) - \ell_{S_t}^*], \tag{22}$$

where the last inequality follows from the fact that η_t is decreasing and $\eta_t \geq \frac{\alpha}{c_t}$ with $\alpha = \min\{\frac{1}{2L_{\max}}, \eta_b c_0\}$, from Lemma C.4. Thus, combining eq. (21) and eq. (22) we have

$$2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \leq \frac{c_{T-1}D_z^2}{\alpha} + \sum_{t=0}^{T-1}\frac{1}{c_t}\mathbb{E}[f(x^t) - f(x^*)] + \sum_{t=0}^{T-1}\frac{\sigma^2}{c_t}. \tag{23}$$

Rearranging eq. (23) we get

$$\sum_{t=0}^{T-1}\left(2 - \frac{1}{c_t}\right)\mathbb{E}[f(x^t) - f(x^*)] \leq 2\lambda_1[f(x^0) - f(x^*)] + \frac{c_{T-1}D_z^2}{\alpha} + \sigma^2\sum_{t=0}^{T-1}\frac{1}{c_t} \tag{24}$$

and using the fact that $2 - \frac{1}{c_t} \geq 1 > 0$ eq. (24) reduces to

$$\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \leq 2\lambda_1[f(x^0) - f(x^*)] + \frac{c_{T-1}D_z^2}{\alpha} + \sigma^2\sum_{t=0}^{T-1}\frac{1}{c_t} \tag{25}$$

Dividing eq. (25) by T and using Jensen's inequality we get

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)]$$
$$\leq \frac{2\lambda_1[f(x^0) - f(x^*)]}{T} + \frac{c_{T-1}D_z^2}{T\alpha} + \frac{\sigma^2}{T}\sum_{t=0}^{T-1}\frac{1}{c_t},$$

as wanted. □

If $\sigma^2 > 0$ like in the standard SGD analysis under decreasing step-sizes, the choice $c_t = \sqrt{t+1}$ leads to the asymptotic rate $O(1/\sqrt{t})$.

Corollary D.3. *Under the setting of Theorem D.2, for $c_t = \sqrt{t+1}$ ($c_{-1} = c_0$) we have*

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{2\lambda_1[f(x^0) - f(x^*)]}{T} + \frac{D_z^2}{\alpha\sqrt{T}} + \frac{2\sigma^2}{\sqrt{T}}.$$

Proof. It follows from the well known inequality

$$\sum_{t=0}^{T-1}\frac{1}{\sqrt{t+1}} \leq 2\sqrt{T}.$$

□

Now the proof of Theorem 3.6 follows immediately from Corollary D.3 setting $\lambda_t = \lambda = \frac{\beta}{1-\beta}$ (from Proposition 2.1). Moreover we have

$$\begin{aligned}\|z^t - x^*\|^2 &\overset{(11)}{=} \|(1+\lambda)x^t - \lambda x^{t-1} - x^*\|^2\\ &\leq (1+\lambda)\|x^t - x^*\|^2 + \lambda\|x^{t-1} - x^*\|^2\\ &\leq (1+2\lambda)\max_t\|x^t - x^*\|^2,\end{aligned} \tag{26}$$

thus $D_z^2 \leq (1+2\lambda)\max_t\|x^t - x^*\|^2 = \frac{1+\beta}{1-\beta}D^2$, where $D^2 = \max_{t\in[T]}\|x^t - x^*\|$.

$$\leq \frac{c_{T-1}D_z^2}{\alpha} + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^t) - f_{S_t}(x^*)] + \sum_{t=0}^{T-1}\frac{1}{c_t}[f_{S_t}(x^*) - \ell_{S_t}^*], \tag{22}$$

其中最后一个不等式源于 η_t 是递减的，并且 $\eta_t \geq \frac{\alpha}{c_t}$ 与 $\alpha = \min\{\frac{1}{2L_{\max}}, \eta_b c_0\}$，根据引理C.4。因此，结合等式（21）和等式（22），我们得到

$$2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \leq \frac{c_{T-1}D_z^2}{\alpha} + \sum_{t=0}^{T-1}\frac{1}{c_t}\mathbb{E}[f(x^t) - f(x^*)] + \sum_{t=0}^{T-1}\frac{\sigma^2}{c_t}. \tag{23}$$

重新排列等式（23）我们得到

$$\sum_{t=0}^{T-1}\left(2 - \frac{1}{c_t}\right)\mathbb{E}[f(x^t) - f(x^*)] \leq 2\lambda_1[f(x^0) - f(x^*)] + \frac{c_{T-1}D_z^2}{\alpha} + \sigma^2\sum_{t=0}^{T-1}\frac{1}{c_t} \tag{24}$$

并利用事实 $2 - \frac{1}{c_t} \geq 1 > 0$ 等式（24）简化为

$$\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \leq 2\lambda_1[f(x^0) - f(x^*)] + \frac{c_{T-1}D_z^2}{\alpha} + \sigma^2\sum_{t=0}^{T-1}\frac{1}{c_t} \tag{25}$$

将等式（25）除以 T 并使用Jensen不等式我们得到

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)]$$
$$\leq \frac{2\lambda_1[f(x^0) - f(x^*)]}{T} + \frac{c_{T-1}D_z^2}{T\alpha} + \frac{\sigma^2}{T}\sum_{t=0}^{T-1}\frac{1}{c_t},$$

如预期所示。 □

如果 $\sigma^2 > 0$ 像在标准随机梯度下降（SGD）分析中那样采用递减步长，那么选择 $c_t = \sqrt{t+1}$会导致渐近收敛速度 $O(1/\sqrt{t})$。

推论 D.3。 在定理 D.2的设定下，对于 $c_t = \sqrt{t+1}$ ($c_{-1} = c_0$) 我们有

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{2\lambda_1[f(x^0) - f(x^*)]}{T} + \frac{D_z^2}{\alpha\sqrt{T}} + \frac{2\sigma^2}{\sqrt{T}}.$$

证明。 这源于众所周知的不等式

$$\sum_{t=0}^{T-1}\frac{1}{\sqrt{t+1}} \leq 2\sqrt{T}.$$

□

现在，定理3.6 的证明直接来自推论 D.3 的设定 $\lambda_t = \lambda = \frac{\beta}{1-\beta}$(来自命题 2.1)。此外，我们还知道

$$\begin{aligned}\|z^t - x^*\|^2 &\overset{(11)}{=} \|(1+\lambda)x^t - \lambda x^{t-1} - x^*\|^2\\ &\leq (1+\lambda)\|x^t - x^*\|^2 + \lambda\|x^{t-1} - x^*\|^2\\ &\leq (1+2\lambda)\max_t\|x^t - x^*\|^2,\end{aligned} \tag{26}$$

因此 $D_z^2 \leq (1+2\lambda)$ 最大$_t\|x^t - x^*\|^2 = \frac{1+\beta}{1-\beta}D^2$，其中 $D^2 =$最大$_{t\in[T]}\|x^t - x^*\|$。

D.3 PROOF OF THEOREM 3.7

Here we state and prove the corresponding theorem for AdaSPS in the 2.3 setting.

Theorem D.4. *Assume that each f_i is convex and L_i-smooth. Then, the iterates of 2.3 with*

$$\eta_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}\frac{1}{\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \eta_{t-1}\right\} \text{ and } \lambda_t \geq 0,$$

where $\eta_{-1} = +\infty$, $c > 0$ and (λ_t) a decreasing sequence, converge as

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\tau = \left(\lambda_1\sqrt{f(x^0) - f(x^*)} + \frac{cLD_z^2}{2} + \frac{1}{2c}\right)$ and $D_z^2 = \max_t \|z^t - x^*\|$.

Proof. Using Equation (21) from the proof of Theorem 3.6, we have

$$2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \leq \mathbb{E}\left[\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\mathbb{E}\,\eta_t\|\nabla f_{S_t}(x^t)\|^2\right]. \tag{27}$$

By Lemma C.7 we have

$$\frac{D_z^2}{\eta_{T-1}} \leq cLD_z^2\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}, \tag{28}$$

and by Lemma C.5 we have

$$\sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \leq \sum_{t=0}^{T-1}\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{2c\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}} \leq \frac{1}{c}\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}. \tag{29}$$

Now combining Equations (27) to (29) and using Jensen's inequality we get

$$\begin{aligned}
&2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \\
&\leq \mathbb{E}\left[cLD_z^2\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}} + \frac{1}{c}\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}\right] \\
&= \left(cLD_z^2 + \frac{1}{c}\right)\mathbb{E}\left[\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}\right] \\
&= \left(cLD_z^2 + \frac{1}{c}\right)\mathbb{E}\left[\sqrt{\sum_{s=0}^{T-1}(f_{S_s}(x^s) - f_{S_s}(x^*) + f_{S_s}(x^*) - \ell^*_{S_s})}\right] \\
&\leq \left(cLD_z^2 + \frac{1}{c}\right)\sqrt{\sum_{s=0}^{T-1}\mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}.
\end{aligned} \tag{30}$$

Rearranging eq. (30) we have

$$2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \leq 2\lambda_1[f(x^0) - f(x^*)] + \left(cLD_z^2 + \frac{1}{c}\right)\sqrt{\sum_{s=0}^{T-1}\mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}. \tag{31}$$

D.3 P证明T定理 3.7

这里我们陈述并证明 AdaSPS 在 2.3 设置下的相应定理。

定理D.4. 假设每个 f_i 是凸的且 L_i-光滑的。那么，*2.3* 的迭代序列，满足

$$\eta_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}\frac{1}{\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \eta_{t-1}\right\} \text{ and } \lambda_t \geq 0,$$

其中 $\eta_{-1} = +\infty$, $c > 0$ 和 (λ_t) 是递减序列，收敛为

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\tau = \left(\lambda_1\sqrt{f(x^0) - f(x^*)} + \frac{cLD_z^2}{2} + \frac{1}{2c}\right)$ and $D_z^2 = \max_t \|z^t - x^*\|$.

证明。 根据定理3.6的证明中的公式(21)，我们有

$$2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \leq \mathbb{E}\left[\frac{D_z^2}{\eta_{T-1}} + \sum_{t=0}^{T-1}\mathbb{E}\,\eta_t\|\nabla f_{S_t}(x^t)\|^2\right]. \tag{27}$$

根据引理 C.7，我们有

$$\frac{D_z^2}{\eta_{T-1}} \leq cLD_z^2\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}, \tag{28}$$

并且根据引理C.5 我们有

$$\sum_{t=0}^{T-1}\eta_t\|\nabla f_{S_t}(x^t)\|^2 \leq \sum_{t=0}^{T-1}\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{2c\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}} \leq \frac{1}{c}\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}. \tag{29}$$

现在结合公式（27）到（29）并使用Jensen不等式，我们得到

$$\begin{aligned}
&2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] - 2\lambda_1[f(x^0) - f(x^*)] \\
&\leq \mathbb{E}\left[cLD_z^2\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}} + \frac{1}{c}\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}\right] \\
&= \left(cLD_z^2 + \frac{1}{c}\right)\mathbb{E}\left[\sqrt{\sum_{s=0}^{T-1} f_{S_s}(x^s) - \ell^*_{S_s}}\right] \\
&= \left(cLD_z^2 + \frac{1}{c}\right)\mathbb{E}\left[\sqrt{\sum_{s=0}^{T-1}(f_{S_s}(x^s) - f_{S_s}(x^*) + f_{S_s}(x^*) - \ell^*_{S_s})}\right] \\
&\leq \left(cLD_z^2 + \frac{1}{c}\right)\sqrt{\sum_{s=0}^{T-1}\mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}.
\end{aligned} \tag{30}$$

将等式（30）重排后得到

$$2\sum_{t=0}^{T-1}\mathbb{E}[f(x^t) - f(x^*)] \leq 2\lambda_1[f(x^0) - f(x^*)] + \left(cLD_z^2 + \frac{1}{c}\right)\sqrt{\sum_{s=0}^{T-1}\mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}. \tag{31}$$

Now let us choose $c_q = \sqrt{f(x^0) - f(x^*)}$. Then

$$2\lambda_1[f(x^0) - f(x^*)] = 2\lambda_1 c_q \sqrt{f(x^0) - f(x^*)} \leq 2\lambda_1 c_q \sqrt{\sum_{s=0}^{T-1} \mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}, \tag{32}$$

Combining eq. (31) and eq. (32) we get

$$\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)] \leq \left(\lambda_1 c_q + \frac{cLD_z^2}{2} + \frac{1}{2c}\right) \sqrt{\sum_{s=0}^{T-1} \mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}. \tag{33}$$

Squaring both sides of eq. (33) we have

$$\left(\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)]\right)^2 \leq \tau^2 \left(\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)] + T\sigma^2\right), \tag{34}$$

where $\tau = \left(\lambda_1 c_q + \frac{cLD_z^2}{2} + \frac{1}{2c}\right)$. Now we use Lemma C.6 in eq. (33) to get

$$\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)] \leq \tau^2 + \tau\sigma\sqrt{T}. \tag{35}$$

Finally by Jensen's inequality and eq. (35) we have

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

as wanted. □

Now the proof of Theorem 3.7 follows immediately from Theorem D.4 setting $\lambda = \frac{\beta}{1-\beta}$ (from Proposition 2.1). Moreover, using eq. (26) we have $D_z^2 \leq \frac{1+\beta}{1-\beta} D^2$.

现在我们选择 $c_q = \sqrt{f(x^0) - f(x^*)}$。然后

$$2\lambda_1[f(x^0) - f(x^*)] = 2\lambda_1 c_q \sqrt{f(x^0) - f(x^*)} \leq 2\lambda_1 c_q \sqrt{\sum_{s=0}^{T-1} \mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}, \tag{32}$$

结合等式(31)和等式(32)我们得到

$$\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)] \leq \left(\lambda_1 c_q + \frac{cLD_z^2}{2} + \frac{1}{2c}\right) \sqrt{\sum_{s=0}^{T-1} \mathbb{E}[f(x^s) - f(x^*)] + \sigma^2}. \tag{33}$$

对等式(33)两边同时平方我们有

$$\left(\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)]\right)^2 \leq \tau^2 \left(\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)] + T\sigma^2\right), \tag{34}$$

其中 $\tau = \left(\lambda_1 c_q + \frac{cLD_z^2}{2} + \frac{1}{2c}\right)$。现在我们在等式(33)中使用引理C.6得到

$$\sum_{t=0}^{T-1} \mathbb{E}[f(x^t) - f(x^*)] \leq \tau^2 + \tau\sigma\sqrt{T}. \tag{35}$$

最后根据Jensen不等式和等式(35)我们有

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

如所求。 □

现在定理3.7 的证明直接来自定理D.4 ，设置 $\lambda = \frac{\beta}{1-\beta}$ (来自命题2.1)。此外，使用等式(26)我们有 $D_z^2 \leq \frac{1+\beta}{1-\beta} D^2$。

E BEYOND THE BOUNDED ITERATES ASSUMPTION

The two main theorems on the convergence of MomDecSPS/MomAdaSPS require the bounded iterates assumption. As mentioned in the main paper, this is a standard assumption for several adaptive step-sizes, see: (Reddi et al., 2018; Ward et al., 2020; Orvieto et al., 2022; Jiang and Stich, 2023). However, one can artificially remove this assumption by adding a projection step in the update rule of IMA onto a compact and convex subset $\mathcal{D} \subseteq \mathbb{R}^d$ as described in this section.

Consider the *constrained* finite-sum optimization problem,

$$\min_{x \in \mathcal{D}} \left[f(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x) \right], \tag{36}$$

where each $f_i : \mathbb{R}^d \to \mathbb{R}$ is convex, smooth, and lower bounded by ℓ_i^* and $\mathcal{D} \subseteq \mathbb{R}^d$ is a compact and convex subset. Let $X_{\mathcal{D}}^* \subseteq \mathcal{D}$ be the set of minimizers of (36). We assume that $X_{\mathcal{D}}^* \neq \emptyset$ and we fix $x^* \in X_{\mathcal{D}}^*$.

The new update rule of IMA, takes the following form (Projected IMA):

$$z^{t+1} = \text{proj}_{\mathcal{D}} \left[z^t - \eta_t \nabla f_{S_t}(x^t) \right] \tag{37}$$

$$x^{t+1} = \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1}, \tag{38}$$

with $x^0 = z^0 \in \mathcal{D}$, where $\text{proj}_{\mathcal{D}}(x) \in \arg\min_{d \in \mathcal{D}} \|d - x\|^2$. Note that the projection step is only needed for the update of z^{t+1} because x^{t+1} is already a convex combination of elements in the convex set $\mathcal{D}$, namely z^{t+1} and x^t by induction. We highlight the fact that Projected IMA is not necessarily equivalent to Projected SHB. Now the proofs of Theorems D.2 and D.4 will go through using the non-expansiveness of the projection operator, as explained in the following lemma:

Lemma E.1 (Non-expansiveness)**.** *For all $x, y \in \mathbb{R}^d$ it holds*

$$\|\text{proj}_{\mathcal{D}}(x) - \text{proj}_{\mathcal{D}}(y)\| \leq \|x - y\|$$

Proof. See Example 8.14 and Lemma 8.16 in Garrigos and Gower (2023). □

Now we have:

Theorem E.2 (Pojected IMA version of Theorem D.2)**.** *Assume that each f_i is convex and L_i-smooth. Then, the iterates of Projected IMA with*

$$\eta_t = \frac{1}{c_t} \min \left\{ \frac{f_{S_t}(x^t) - \ell_{S_t}^*}{\|\nabla f_{S_t}(x^t)\|^2}, c_{t-1}\eta_{t-1} \right\} \text{ and } \lambda_t \geq 0,$$

where (c_t) is an increasing sequences with $c_t \geq 1$, $c_{-1} = c_0$, $\eta_{-1} = \eta_b$ and (λ_t) is a decreasing sequence, converge as

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{2\lambda_1[f(x^0) - f(x^*)]}{T} + \frac{c_{T-1}D^2}{T\alpha} + \frac{\sigma^2}{T} \sum_{t=0}^{T-1} \frac{1}{c_t},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\alpha = \min\left\{\frac{1}{2L}, \eta_b c_0\right\}$ and $D = \text{diam}\, \mathcal{D}$.

Proof. We have

$$\begin{aligned} \|z^{t+1} - x^*\|^2 &= \|\text{proj}_{\mathcal{D}}(z^t - \eta_t \nabla f_{S_t}(x^t)) - \text{proj}_{\mathcal{D}}(x^*)\| \\ &\overset{\text{Lem. E.1}}{\leq} \|z^t - \eta_t \nabla f_{S_t}(x^t) - x^*\| \\ &= \|z^t - x^*\|^2 - 2\eta_t \langle \nabla f_{S_t}(x^t), z^t - x^* \rangle + \eta_t^2 \|\nabla f_{S_t}(x^t)\|^2. \end{aligned}$$

Now we continue exactly like the rest of the proof of Theorem D.2. □

超越有界迭代假设

关于MomDecSPS/MomAdaSPS收敛性的两个主要定理需要满足有界迭代假设。正如在正文中提到的，这是几种自适应步长的一个标准假设，参见：（Reddi等人，2018； Ward等人,2020； Orvieto等人，2022；Jiang和Stich，2023)。然而，可以通过在IMA的更新规则中添加一个投影步骤，人为地移除这个假设，投影到本节描述的紧凸子集 $\mathcal{D} \subseteq \mathbb{R}^d$ 上。

考虑约束有限和优化问题，

$$\min_{x \in \mathcal{D}} \left[f(x) = \frac{1}{n} \sum_{i=1}^{n} f_i(x) \right], \tag{36}$$

其中每个 f_i： $\mathbb{R}^d \to \mathbb{R}$ 是凸的、光滑的，并由 ℓ_i^*和 $\mathcal{D} \subseteq \mathbb{R}^d$ 下界约束的紧凸子集。设 $X_{\mathcal{D}}^* \subseteq \mathcal{D}$ 是（36）的最小化器集合。我们假设 $X_{\mathcal{D}}^* \neq \emptyset$ 并固定$x^* \in X_{\mathcal{D}}^*$。

IMA的新更新规则，取以下形式（投影IMA）：

$$z^{t+1} = \text{proj}_{\mathcal{D}} \left[z^t - \eta_t \nabla f_{S_t}(x^t) \right] \tag{37}$$

$$x^{t+1} = \frac{\lambda_{t+1}}{\lambda_{t+1}+1} x^t + \frac{1}{\lambda_{t+1}+1} z^{t+1}, \tag{38}$$

其中 $x^0 = z^0 \in \mathcal{D}$，其中$\text{proj}_{\mathcal{D}}(x) \in \text{argmin}_{d \in \mathcal{D}} \|d - x\|^2$。请注意，投影步骤仅用于 z^{t+1} 的更新，因为 x^{t+1} 已经是凸集 $\mathcal{D}$中元素的凸组合，即 z^{t+1} 和 x^t ，通过归纳法可知。我们强调投影IMA不一定等同于投影SHB。现在我们将使用投影算子的非扩张性，如以下引理所述，来证明定理D.2和D.4：

引理 E.1 (非扩张性)**.** 对于所有 $x, y \in \mathbb{R}^d$ 都成立

$$\|\text{proj}_{\mathcal{D}}(x) - \text{proj}_{\mathcal{D}}(y)\| \leq \|x - y\|$$

证明。 参见 Garrigos 和 Gower 中的例 8.14 和引理 8.16(2023). □

现在我们有：

定理 E.2 (投影IMA版本的定理D.2)**.** 假设每个 f_i 是凸的并且 L_i- 光滑的。那么，具有

$$\eta_t = \frac{1}{c_t} \min \left\{ \frac{f_{S_t}(x^t) - \ell_{S_t}^*}{\|\nabla f_{S_t}(x^t)\|^2}, c_{t-1}\eta_{t-1} \right\} \text{ and } \lambda_t \geq 0,$$

其中 (c_t) 是一个递增序列，包含 $c_t \geq 1$, $c_{-1} = c_0$, $\eta_{-1} = \eta_b$ ，并且 (λ_t) 是一个递减序列，收敛为

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{2\lambda_1[f(x^0) - f(x^*)]}{T} + \frac{c_{T-1}D^2}{T\alpha} + \frac{\sigma^2}{T} \sum_{t=0}^{T-1} \frac{1}{c_t},$$

其中 $\underline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\alpha = \min\left\{\frac{1}{2L}, \eta_b c_0\right\}$ 并且 $D = \text{diam}\, \mathcal{D}$。

证明。我们已经

$$\begin{aligned} \|z^{t+1} - x^*\|^2 &= \|\text{proj}_{\mathcal{D}}(z^t - \eta_t \nabla f_{S_t}(x^t)) - \text{proj}_{\mathcal{D}}(x^*)\| \\ &\overset{\text{Lem. E.1}}{\leq} \|z^t - \eta_t \nabla f_{S_t}(x^t) - x^*\| \\ &= \|z^t - x^*\|^2 - 2\eta_t \langle \nabla f_{S_t}(x^t), z^t - x^* \rangle + \eta_t^2 \|\nabla f_{S_t}(x^t)\|^2. \end{aligned}$$

现在我们继续按照定理D.2的其余证明进行。 □

Theorem E.3 (Pojected IMA version of Theorem D.4)**.** *Assume that each f_i is convex and L_i-smooth. Then, the iterates of Projected IMA with*

$$\eta_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}\frac{1}{\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \eta_{t-1}\right\} \text{ and } \lambda_t \geq 0,$$

where $\eta_{-1} = +\infty$, $c > 0$ and (λ_t) a decreasing sequence, converge as

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

where $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\tau = \left(\lambda_1\sqrt{f(x^0) - f(x^*)} + \frac{cLD^2}{2} + \frac{1}{2c}\right)$ and $D = \operatorname{diam}\mathcal{D}$.

Proof. Same as above. □

定理 E.3 (投影IMA版本的定理D.4)。 假设每个 f_i 是凸的并且 L_i- 光滑的。那么，投影*IMA*的迭代

$$\eta_t = \min\left\{\frac{f_{S_t}(x^t) - \ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}\frac{1}{\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \eta_{t-1}\right\} \text{ and } \lambda_t \geq 0,$$

其中 $\eta_{-1} = +\infty$, $c > 0$ 和 (λ_t) a是递减序列，收敛为

$$\mathbb{E}[f(\overline{x}^T) - f(x^*)] \leq \frac{\tau^2}{T} + \frac{\tau\sigma}{\sqrt{T}},$$

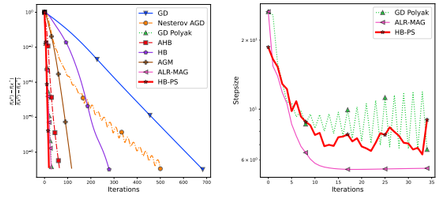
其中 $\overline{x}^T = \frac{1}{T}\sum_{t=0}^{T-1} x^t$, $\tau = \left(\lambda_1\sqrt{f(x^0) - f(x^*)} + \frac{cLD^2}{2} + \frac{1}{2c}\right)$ 和 $D = \operatorname{diam}\mathcal{D}$。

证明。 与上述相同。 □

F ADDITIONAL EXPERIMENTS

F.1 EXTRA DETERMINISTIC EXPERIMENT

Here we have included an extra experiment in the deterministic setting, this one for logistic regression. We have performed a grid search to find the best β for MomPS$_{\max}$ since no optimal choices for β are known for the general convex case. We use the same β for HB and ALR-MAG for direct comparison. For the step-size of HB we used $\gamma = 2(1-\beta)/L$ as recomended in Ghadimi et al. (2015). The results are presented in Figure 7. In this problem we observe that our step-size (MomPS$_{\max}$) is the fastest, having similar performace with ALR-MAG and GD Polyak. For the deterministic setting we have used a 12 core AMD Ryzen 5 5600H CPU to run the experiments.



(a) Relative Error Plot (b) Step-size Plot

Figure 7: Comparison of various deterministic algorithms for the logistic regression problem on synthetic data.

F.2 STOCHASTIC HEAVY BALL WITH CONSTANT STEP-SIZE

As we saw in Corollary 3.5, when $\gamma_b \leq \frac{1}{2L_{\max}}$ then we have new theoretical guarantees for SHB with the constant step-size $\gamma = \frac{1-\beta}{2L_{\max}}$. The most recent analysis of constant SHB is from Liu et al. (2020) with step-size $\gamma = \frac{(1-\beta)^2}{L} \min\left\{\frac{1}{4-\beta+\beta^2}, \frac{1}{2\sqrt{2\beta+2\beta^2}}\right\}$. As mentioned in the main text our step-size is larger when $\beta \geq \sqrt{5}-2 \approx 0.236$. In Figure 8 we provide a numerical comparison of these results for logistic regression with synthetic data. We observe that SHB with the Liu et al. (2020) step-size has faster convergence for the first iterations but it reaches a plateau much earlier. Moreover, note that as $\beta \to 1$ both step-sizes have similar performance.

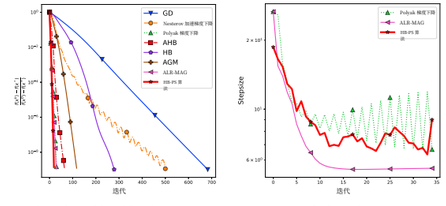
F.3 OTHER CHOICES

In this paper, we have provided convergence guarantees for SHB update rule $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$ with the MomSPS$_{\max}$ step-size given by $\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t)-\ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2}, \gamma_b\right\}$. However, as discussed in the main paper a more natural step-size would be to choose the well known SPS$_{\max}$ on the SHB rule. We call this new update rule SPS$_{\max}$ with naive momentum. Here we numerically compare these two updates. As we see in Figure 9, when β is "small" and close to 0 then both SPS$_{\max}$ with naive momentum and MomSPS$_{\max}$ are close in performance. Of course, when $\beta = 0$ they are both equal to standard SPS$_{\max}$. However, as β gets larger the performance of SPS$_{\max}$ with naive momentum gets worse and less stable and at some point it diverges. For $\beta \geq 0.7$ there is no SPS$_{\max}$ with naive momentum, since it cannot be numerically computed.

F 额外实验

F.1 超确定性实验

在这里，我们在确定性设置中包含了一个额外的实验，这次是用于逻辑回归的。我们进行了网格搜索以找到最佳的 β 对于 MomPS$_{\max}$，因为对于通用凸情况，没有已知的最佳选择。我们使用相同的 β 对于 HB 和 ALR-MAG 进行直接比较。对于 HB 的步长，我们使用了 $\gamma = 2(1-\beta)/L$，如 Ghadimi 等人 (2015) 中推荐的那样。结果展示在图 7。在这个问题中，我们观察到我们的步长（MomPS$_{\max}$）是最快的，其性能与 ALR-MAG 和 GD Polyak 相似。对于确定性设置，我们使用了一台 12 核 AMD Ryzen 5 5600H CPU 来运行实验。



(a) 相对误差图 (b) 步长图

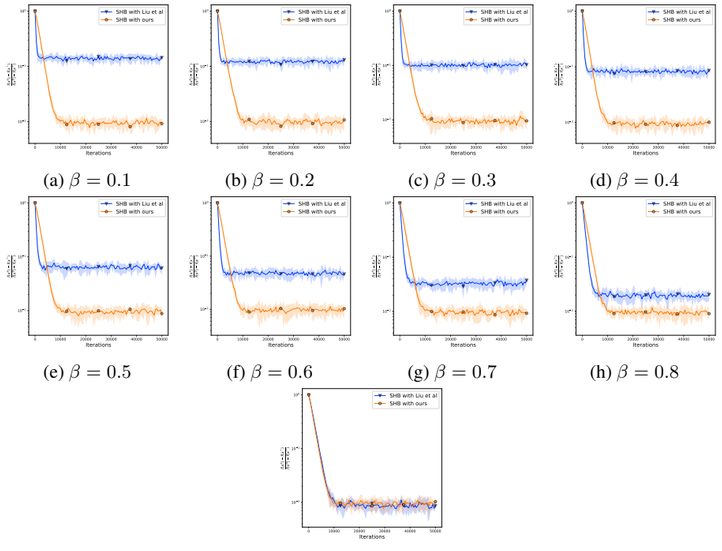
图 7：比较在不同确定性算法下，逻辑回归问题在合成数据上的表现。

F.2 带固定步长的随机重球方法

如我们在引理 3.5中所见，当 $\gamma_b \leq \frac{1}{2L_{\max}}$ 则我们为 SHB提供了新的理论保证，其步长为常数 $\gamma = \frac{1-\beta}{2L_{\max}}$。最近对固定步长 SHB 的分析来自 刘等人(2020)，其步长为 $\gamma = \frac{(1-\beta)^2}{L}\min\left\{\frac{1}{4-\beta+\beta^2}, \frac{1}{2\sqrt{2\beta+2\beta^2}}\right\}$。正如正文中所提，当 $\beta \geq \sqrt{5}-2 \approx 0.236$时，我们的步长更大。在图 8 中，我们提供了这些结果在逻辑回归问题上的数值比较，使用合成数据。我们观察到SHB 在步长为 刘等人(2020) 的设置下，在初始迭代时收敛速度更快，但更早达到平台期。此外，请注意，由于 $\beta \to 1$ 两种步长表现相似。

F.3 其他选择

在本文中，我们为SHB更新规则 $x^{t+1} = x^t - \gamma_t \nabla f_{S_t}(x^t) + \beta(x^t - x^{t-1})$与MomSPS$_{\max}$ 步长 $\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t - \ell^*_{S_t})}{c\|\nabla f_{S_t} x^t\|^2}, \gamma_b\right\}$提供了收敛保证。然而，正如在正文中讨论的那样，一个更自然的步长选择是在SHB规则上使用众所周知的SPS$_{\max}$。我们称这个新的更新规则为SPS$_{\max}$（带朴素动量）。在这里，我们数值比较了这两个更新。如图9所示，当β "较小"且接近0时，带朴素动量的SPS$_{\max}$ 和MomSPS$_{\max}$ 在性能上非常接近。当然，当 $\beta = 0$ 它们都等于标准SPS$_{\max}$时，它们是相等的。然而，随着β 的增大，带朴素动量的SPS$_{\max}$ 的性能变差且稳定性降低，在某个时刻它会发散。对于 $\beta \geq 0.7$ 没有带朴素动量的SPS$_{\max}$，因为它无法进行数值计算。

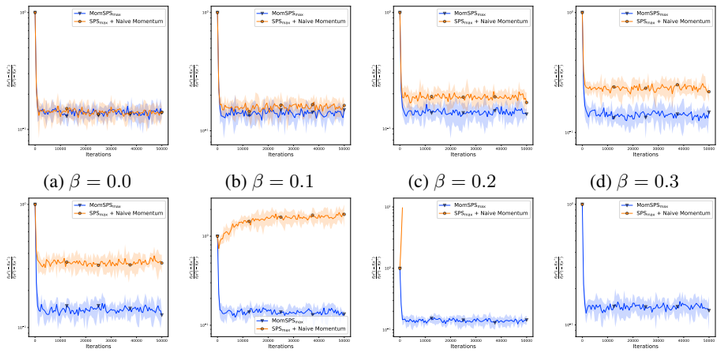


(a) $\beta = 0.1$ (b) $\beta = 0.2$ (c) $\beta = 0.3$ (d) $\beta = 0.4$

(e) $\beta = 0.5$ (f) $\beta = 0.6$ (g) $\beta = 0.7$ (h) $\beta = 0.8$

(i) $\beta = 0.9$

Figure 8: Comparison of SHB with constant step-size. (Liu et al., 2020) vs Corollary 3.5 for various momentum coefficients β on logistic regression with synthetic data.



(a) $\beta = 0.0$ (b) $\beta = 0.1$ (c) $\beta = 0.2$ (d) $\beta = 0.3$

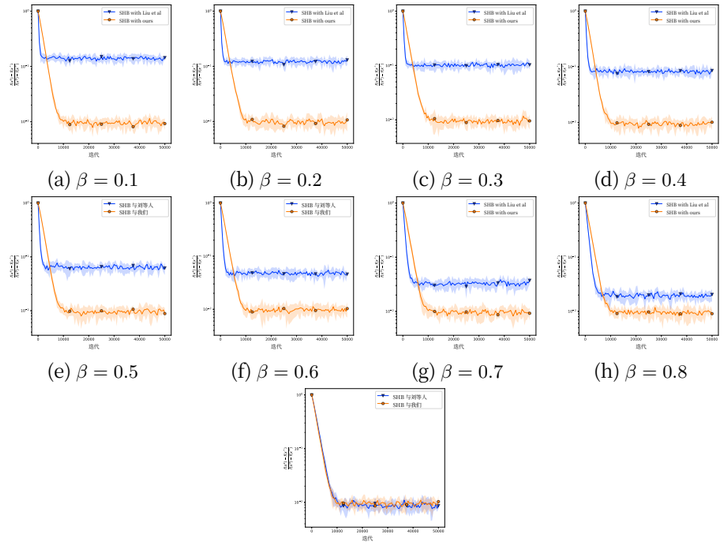
(e) $\beta = 0.4$ (f) $\beta = 0.5$ (g) $\beta = 0.6$ (h) $\beta = 0.7$

Figure 9: Comparison of MomSPS$_{\max}$ versus SPS$_{\max}$ with naive momentum for various momentum coefficients β on logistic regression with synthetic data.

Furthermore, recall that our decreasing variants MomDecSPS and MomAdaSPS are respectively given by

$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\sqrt{t+1}\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}\sqrt{t}}{\sqrt{t+1}}\right\}, \qquad \text{(MomDecSPS)}$$

$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}. \qquad \text{(MomAdaSPS)}$$

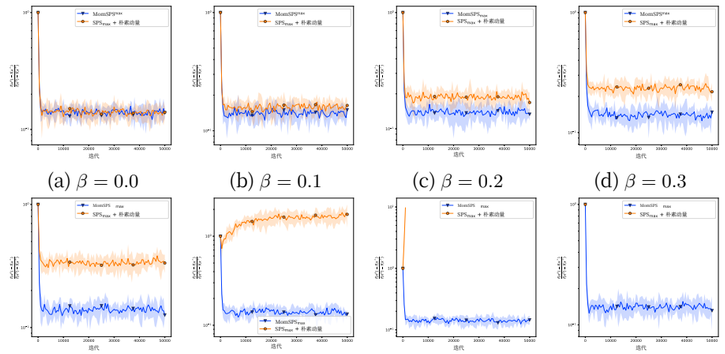


(a) $\beta = 0.1$ (b) $\beta = 0.2$ (c) $\beta = 0.3$ (d) $\beta = 0.4$

(e) $\beta = 0.5$ (f) $\beta = 0.6$ (g) $\beta = 0.7$ (h) $\beta = 0.8$

(i) $\beta = 0.9$

图8：与 SHB 固定步长的比较。 (刘等人， 2020) vs 定理 3.5 对于不同的动量系数 β 在逻辑回归合成数据上的表现。



(a) $\beta = 0.0$ (b) $\beta = 0.1$ (c) $\beta = 0.2$ (d) $\beta = 0.3$

(e) $\beta = 0.4$ (f) $\beta = 0.5$ (g) $\beta = 0.6$ (h) $\beta = 0.7$

图 9：比较MomSPS$_{\max}$ 与 SPS$_{\max}$ 在不同动量系数下使用朴素动量的对比 β 在逻辑回归合成数据上的表现。

此外，请回忆我们的减少变体 MomDecSPS和 MomAdaSPS 分别是给出

$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\sqrt{t+1}\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}\sqrt{t}}{\sqrt{t+1}}\right\}, \qquad \text{(MomDecSPS)}$$

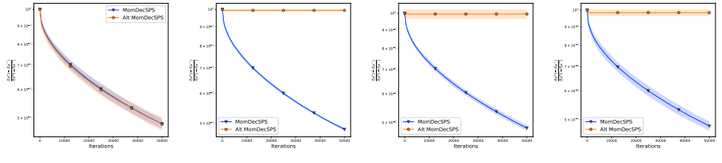
$$\gamma_t = \min\left\{\frac{(1-\beta)[f_{S_t}(x^t) - \ell^*_{S_t}]}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s) - \ell^*_{S_s}}}, \gamma_{t-1}\right\}. \qquad \text{(MomAdaSPS)}$$

The above step-sizes are theoretically inspired by IMA, as explained in Appendix C.3, but again a natural question is what happens if we take the "correcting factor" $1-\beta$ outside the minimum. We call these step-sizes Alternative MomDecSPS and Alternative MomAdaSPS and are given by

$$\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t)-\ell^*_{S_t}}{c\sqrt{t+1}\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}\sqrt{t}}{\sqrt{t+1}}\right\}, \tag{Alt MomDecSPS}$$

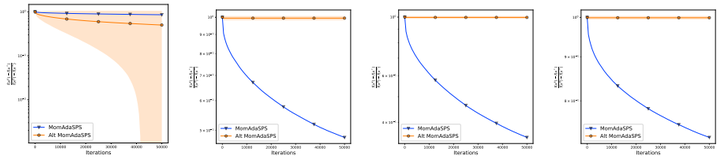
$$\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t)-\ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s)-\ell^*_{S_s}}}, \gamma_{t-1}\right\}. \tag{Alt MomAdaSPS}$$

In Figure 10 and Figure 11 we see that when $\beta = 0$ we have similiar performances, however when $\beta > 0$ then our proposed step-sizes are significantly better. A possible reason is that for both Alt MomDecSPS and Alt MomadaSPS it holds $\gamma_t \leq (1-\beta)\gamma_{t-1}$ for all t. Thus inductively we get $\gamma_t \leq (1-\beta)^t\gamma_0$. So when $\beta > 0$, $(1-\beta)^t \to 0$ as $t \to \infty$ which means that the step-size is too small and it barely updates x^t.



(a) $\beta = 0.0$ (b) $\beta = 0.3$ (c) $\beta = 0.6$ (d) $\beta = 0.9$

Figure 10: Comparison of MomDecSPS versus Alternative MomDecSPS for various momentum coefficients β on logistic regression with synthetic data.



(a) $\beta = 0.0$ (b) $\beta = 0.3$ (c) $\beta = 0.6$ (d) $\beta = 0.9$

Figure 11: Comparison of MomAdaSPS versus Alternative MomAdaSPS for various momentum coefficients β on logistic regression with synthetic data.

F.4 Is γ_b needed?

Recall the definition of MomPS$_{\max}$, $\gamma_t = (1-\beta)\min\left\{\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\}$. The more natural choice would be without the upper bound γ_b. Moreover, as in Appendix F.3, we can have the naive version of both of the above step-sizes, i.e. with no correcting factor $1-\beta$. We compare the following step-sizes on both the least squares problem, in Figure 12, and logistic regression problem, in Figure 13, in the deterministic setting on synthetic data. The method is SHB with $\beta = 0.97$ for the least squares and $\beta = 0.3$ for the logistic regression.

$$\gamma_t = (1-\beta)\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2} \tag{MomPS}$$

$$\gamma_t = \frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2} \tag{Alt MomPS}$$

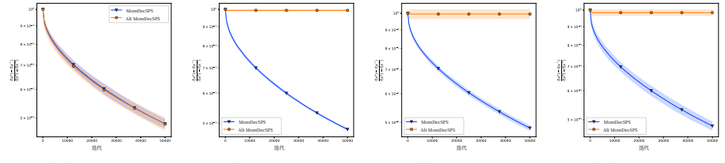
$$\gamma_t = (1-\beta)\min\left\{\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\} \tag{MomPS$_{\max}$}$$

上述步长在理论上受IMA启发，如附录C.3所述，但再次一个自然的问题是，如果我们把"校正因子" $1-\beta$ 放在最小值之外会发生什么。我们称这些步长为Alternative MomDecSPS和AlternativeMomAdaSPS，其表达式为

$$\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t)-\ell^*_{S_t}}{c\sqrt{t+1}\|\nabla f_{S_t}(x^t)\|^2}, \frac{\gamma_{t-1}\sqrt{t}}{\sqrt{t+1}}\right\}, \tag{Alt MomDecSPS}$$

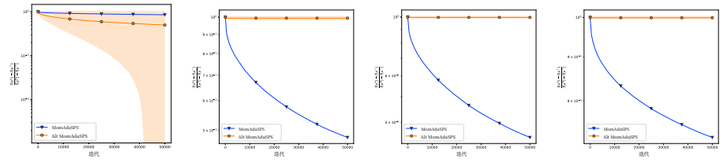
$$\gamma_t = (1-\beta)\min\left\{\frac{f_{S_t}(x^t)-\ell^*_{S_t}}{c\|\nabla f_{S_t}(x^t)\|^2\sqrt{\sum_{s=0}^t f_{S_s}(x^s)-\ell^*_{S_s}}}, \gamma_{t-1}\right\}. \tag{Alt MomAdaSPS}$$

在图10 和图11 中，我们看到当 $\beta = 0$ 我们获得相似性能时，但当$\beta > 0$ 时，我们提出的步长则显著更优。一个可能的原因是，对于Alt MomDecSPS和Alt MomadaSPS，均成立 $\gamma_t \leq (1-\beta)\gamma_{t-1}$ 对所有 t。因此，我们归纳得到$\gamma_t \leq (1-\beta)^t\gamma_0$。所以当 $\beta > 0$时，$(1-\beta)^t \to 0$ 作为 $t \to \infty$ 这意味着步长太小，几乎不进行更新 x^t。



(a) $\beta = 0.0$ (b) $\beta = 0.3$ (c) $\beta = 0.6$ (d) $\beta = 0.9$

图 10：比较MomDecSPS与 Alternative MomDecSPS 在不同动量系数下的表现 β 在逻辑回归合成数据上的结果。



(a) $\beta = 0.0$ (b) $\beta = 0.3$ (c) $\beta = 0.6$ (d) $\beta = 0.9$

图11：比较MomAdaSPS与 Alternative MomAdaSPS 在不同动量系数下的表现 β 在逻辑回归合成数据上的结果。

F.4 是否需要 γ_b ？

回忆 MomPS$_{\max}$ 的定义 $\gamma_t = (1-\beta)$ 最小 $\left\{\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\}$。更自然的选择则不包括上界 γ_b。此外，正如附录 F.3中所展示的，我们都可以获得上述两种步长的朴素版本，即不带校正因子 $1-\beta$。我们在图 12所示的确定性设置下合成数据的最小二乘问题，以及图 13所示的逻辑回归问题上，比较了以下步长。该方法是 SHB，对于最小二乘问题使用 $\beta = 0.97$，对于逻辑回归问题使用 $\beta = 0.3$。

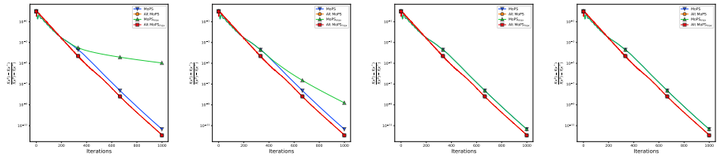
$$\gamma_t = (1-\beta)\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2} \tag{MomPS}$$

$$\gamma_t = \frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2} \tag{Alt MomPS}$$

$$\gamma_t = (1-\beta)\min\left\{\frac{f(x^t)-f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\} \tag{MomPS$_{\max}$}$$

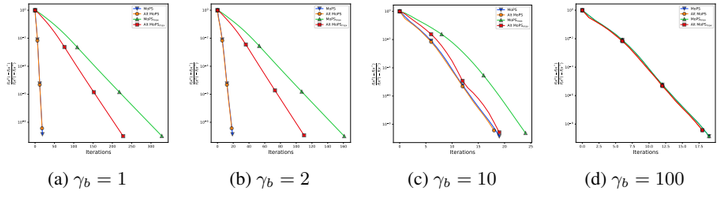
$$\gamma_t = \min\left\{\frac{f(x^t) - f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\}. \qquad \text{(Alt MomPS}_{\max}\text{)}$$

We see that if γ_b is chosen large enough, all these step-sizes have similar performance. However, we highlight the fact that convergence guarantees are known only for MomPS$_{\max}$.



(a) $\gamma_b = 0.1$ (b) $\gamma_b = 0.5$ (c) $\gamma_b = 1$ (d) $\gamma_b = 10$

Figure 12: Comparison of various upper bounds for MomPS$_{\max}$ and other alternatives on least squares with synthetic data.



(a) $\gamma_b = 1$ (b) $\gamma_b = 2$ (c) $\gamma_b = 10$ (d) $\gamma_b = 100$

Figure 13: Comparison of various upper bounds for MomPS$_{\max}$ and other alternatives on logistic regression with synthetic data.

F.5 PLOTS FOR C_1 AND C_2

Below it a plot of the constants C_1 and C_2 as functions of β with $\gamma_b = 2$ and $\alpha = 1$ (recall $\gamma_b \geq \alpha$) from Theorem 3.2. Both functions are hyperbolas and the vertical line is the constant $\beta = \frac{\alpha}{2\gamma_b - \alpha}$. We see that in the interval $\beta \in \left[0, \frac{\alpha}{2\gamma_b - \alpha}\right)$ both functions are increasing.

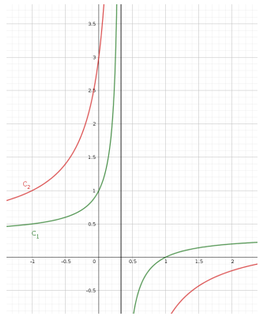
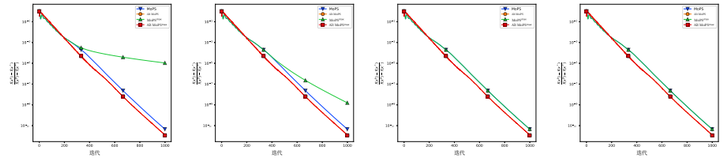


Figure 14: Plots for the constants C_1 and C_2 as functions of β. Here we have chosen $\gamma_b = 2$ and $\alpha = 1$.

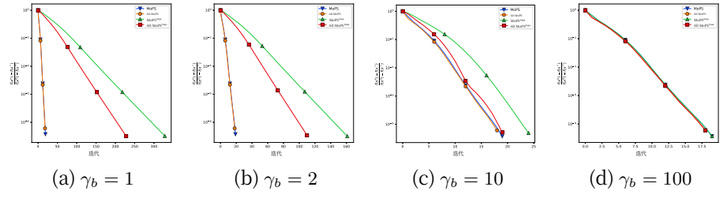
$$\gamma_t = \min\left\{\frac{f(x^t) - f(x^*)}{\|\nabla f(x^t)\|^2}, \gamma_b\right\}. \qquad \text{(Alt MomPS}_{\max}\text{)}$$

我们看到，如果 γ_b 选择得足够大，所有这些步长都具有相似的性能。然而，我们强调的是，已知收敛保证仅适用于 MomPS$_{\max}$。



(a) $\gamma_b = 0.1$ (b) $\gamma_b = 0.5$ (c) $\gamma_b = 1$ (d) $\gamma_b = 10$

图12：比较各种上界MomPS$_{\max}$ 及其他替代方案在合成数据上的最小二乘法比较。



(a) $\gamma_b = 1$ (b) $\gamma_b = 2$ (c) $\gamma_b = 10$ (d) $\gamma_b = 100$

图13：比较各种上界MomPS$_{\max}$ 在逻辑回归合成数据上的其他替代方案。

F.5 C_1 和 C_2的图

下面是常数 C_1 和 C_2 作为 β 的函数 $\gamma_b = 2$ 和 $\alpha = 1$ (回顾 $\gamma_b \geq \alpha$)来自定理3.2。这两个函数都是双曲线，垂直线是常数 $\beta = \frac{\alpha}{2\gamma_b - \alpha}$。我们看到在区间 $\beta \in \left[0, \frac{\alpha}{2\gamma_b - \alpha}\right)$ 内这两个函数都是递增的。

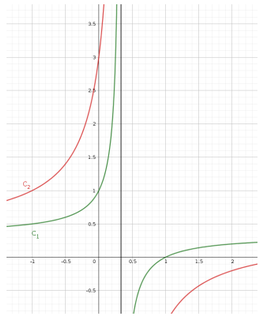


图14：常数 C_1 和 C_2 随 β变化的曲线图。这里我们选择了 $\gamma_b = 2$ 和$\alpha = 1$。

F.6 MORE CONVEX EXPERIMENTS AND PARAMETER SETTINGS

In this section, we list the parameters, architectures and hardware that we used for the deep learning experiments. The information is collected in Table 2. We also include some extra experiments in Figures 15 to 18.

Hyper-parameter	Value
Architecture	Logistic Regression
GPUs	1x NVIDIA GeForce RTX 3050
Batch-size	See caption of each plot
Epochs	100
Trials	5
Weight Decay	0.0

Table 2: Logistic regression experiment

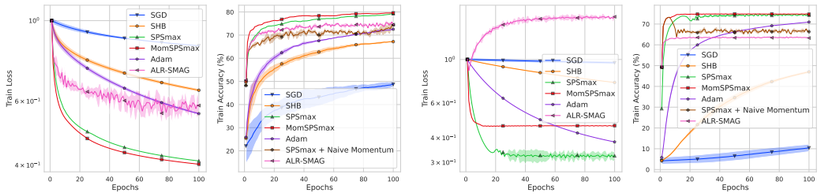


Figure 15: LibSVM dataset: vehicle, Batch size: 16

Figure 16: LibSVM dataset: letter, Batch size: 256

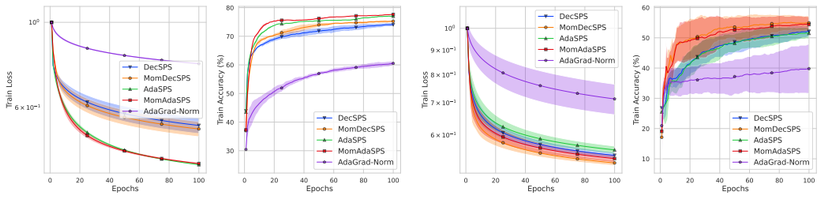


Figure 17: LibSVM dataset: vehicle, Batch size: 85

Figure 18: LibSVM dataset: glass, Batch size: 32

F.7 MORE DEEP LEARNING EXPERIMENTS AND PARAMETER SETTINGS

In this section, we list the parameters, architectures and hardware that we used for the deep learning experiments. The information is collected in Table 3. We also include some extra experiments in Figures 19 and 20. For ALR-SMAG we use $c = 0.1$ in the DL experiments.

Hyper-parameter	Value
Architecture	ResNet 10/34 (He et al., 2016)
GPUs	1x Nvidia RTX 6000 Ada Generation
Batch-size	256
Epochs	100
Trials	5
Weight Decay	0.0

Table 3: CIFAR10 experiment

F.6 更多凸实验和参数设置

在本节中，我们列出了用于深度学习实验的参数、架构和硬件。信息收集在表2中。我们还包含一些额外的实验在图15 到18中。

超参数	值
架构	逻辑回归
GPUs	1x NVIDIA GeForce RTX 3050
批大小	参见每个图的图例
周期	100
试验	5
权重衰减	0.0

表2：逻辑回归实验

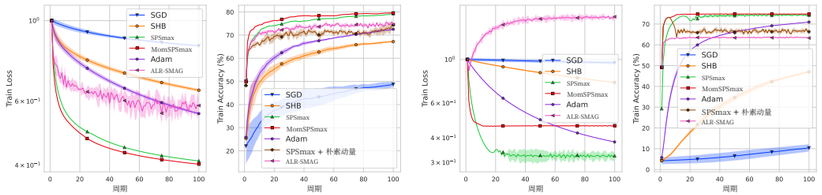


图15：Libsvm数据集：车辆，批大小：16

图16：Libsvm数据集：字母，批大小：256

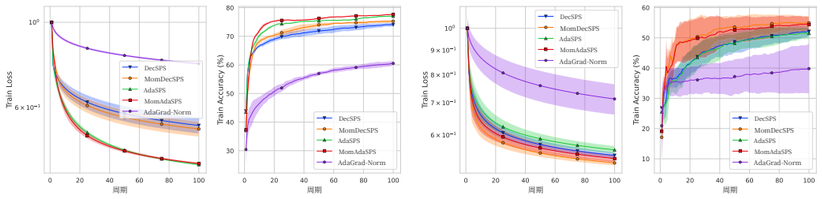


图17：Libsvm数据集：车辆，批大小：85

图18：Libsvm数据集：玻璃，批大小：32

F.7 更多深度学习实验和参数设置

在本节中，我们列出了深度学习实验中使用的参数、架构和硬件。这些信息收集在表3中。我们还包含一些额外的实验在图19 和20中。对于ALR-SMAG，我们在深度学习实验中使用 $c = 0.1$ 。

超参数	值
建筑结构	ResNet 10/34 (He等人,2016)
GPUs	1x Nvidia RTX 6000 Ada Generation
批大小	256
周期	100
试验	5
权重衰减	0.0

表3：CIFAR10实验

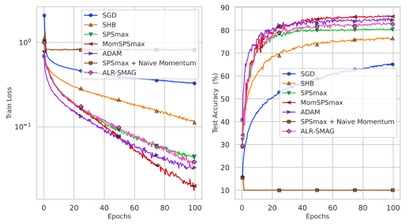
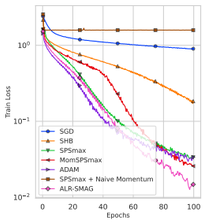


Figure 19: Resnet 18 on CIFAR 10



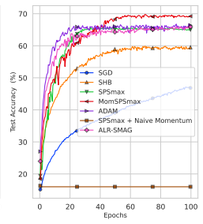


Figure 20: Resnet 18 on CIFAR 100

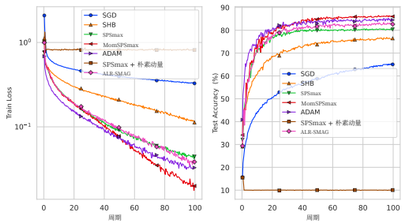
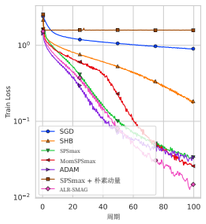


图19：Resnet 18在CIFAR 10上的表现



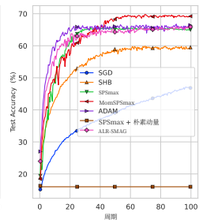


图20：Resnet 18在CIFAR 100上的表现